

COMMON COMMODITIES
AND INDUSTRIES

THE FISHING
INDUSTRY

W. E. Gibbs

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Title: The Fishing Industry

Author: William E. Gibbs

Release date: November 4, 2016 [eBook #53447]

Most recently updated: October 23, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/53447

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HAULING THE TRAWL

Frontispiece.

PITMAN'S COMMON COMMODITIES
AND INDUSTRIES

THE
FISHING INDUSTRY

BY

W. E. GIBBS, D.Sc.



LONDON
SIR ISAAC PITMAN & SONS, LTD.
PARKER STREET, KINGSWAY, W.C.2
BATH, MELBOURNE, TORONTO, NEW YORK
1922

PRINTED BY
SIR ISAAC PITMAN & SONS, LTD.
BATH, ENGLAND

PREFACE

IN this little book I have tried to describe concisely, yet clearly and comprehensively, the great work of our sea fisheries. It is notoriously difficult to write a small book on a large subject, and I expect there are many who will detect sins of omission.

The book is chiefly concerned with fisheries for edible fish. I have included a chapter on whale fisheries, since whale oil is now used largely in the manufacture of such food substances as lard substitute and margarine. No account of seal "fishing" is included, as seals are not fished but are generally hunted on shore. I have not included fisheries for pearls, sponges or seaweed. To its cost the nation knows little of the methods and organization and achievements of the Fishing Industry. I sincerely hope that this little book may do something to stimulate a wider and deeper interest in this vitally important British industry.

My cordial thanks are due to Mr. J. A. Robertson, O.B.E., of Fleetwood, and to Mr. W. T. Sinderson, of Grimsby, who have very kindly read through the manuscript and given me the benefit of their valuable experience and advice.

I am indebted to Prof. James Johnstone, of Liverpool University, for much of the information contained in Chapters I and II, and also for permission to use the illustrations on pages 17 and 29.

For other illustrations I make grateful acknowledgement as follows: for Nos. 7, 8, 9, 10, 15, 16, 17 and 19, from *The Sea Fisheries*, to the author, Dr. J. Travis Jenkins, and the publishers, Messrs. Constable; for the frontispiece and No 18, to the Grimsby Coal, Salt and Tanning Co; for Nos. 11, 14 and 20 to Mr. Walter Wood, of the Mission to Deep Sea Fishermen.

Mr. R. A. Fleming, of Liverpool University, very kindly copied Nos. 2, 3, 4 and 5 for me from Day's *British Fishes*.

Chapter V is based upon Bitting's monograph on "the preparation of the cod and other salt fish for the market." (U.S. Dept. Agric. Bur. of Chem. Bull. No. 133).

W. E. G.

RUNCORN, 1922.



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THE
FISHING INDUSTRY

CHAPTER I

INTRODUCTION

IN its essential features the story of the gradual rise and development of the fishing industry closely resembles that of its sister industry, agriculture. In both cases man became skilled in harvesting long before he understood anything of the art of cultivation. Primitive man roamed from place to place in the wake of the annual wave of harvest, gathering wild crops of grain, berries and fruits. Ultimately he became alive to the significance of seed, and the nomad settled down to raise crops year after year in the same place. Gradually he acquired a knowledge of the conditions of temperature, moisture, and quality of soil that favoured the growth of his plants. Finally, he discovered the principle of the rotation of crops, and, by this, not only increased the productivity of his land but also laid the foundations of a systematic agriculture. Of recent years agriculture has been rapidly developing into a science. Chemistry, physics, botany, plant physiology, and bacteriology, all contribute increasingly to a full understanding of the inner processes of the growing plant, and indicate more and more clearly the exact relations that exist between the conditions of growth and the character and amount of the resulting product.

The art of fishing is one of the oldest in the world, yet even to this day the fisherman is simply a hunter, gathering where he has not sown, and differing little, save in mechanical efficiency, from his primitive ancestor fishing with spear and trap.

Only in recent years has any systematic attempt been made to understand something of the forces that produce the annual harvest of the sea. We know very little about the habits of the various fishes that constitute this harvest—their food, their migrations, their reproductive processes, and, in general, the conditions upon which their healthy life and development depend. We have developed highly efficient fishing implements, but we have yet to learn to use them wisely and not too well; to increase the fertility of the various fishing grounds rather than depopulate them by over-fishing and the destruction of immature fish.

The fisherman's harvest differs from that of the farmer in one important respect. Fishes grow for three or four, or more, years before they are mature. Now, only mature fish as a rule have any considerable commercial value, and only mature fish are able to reproduce their kind and so maintain the existence of the fishery. On the fishing grounds, both mature and immature fish are mingled together, and in capturing the one it is practically impossible to avoid netting the other. To some extent the capture of immature fish is avoided by making the mesh of the net of such a size that the smaller fish can escape. With drift nets only mature fish are caught, the small ones escaping; but with trawl nets it is otherwise. The trawl net is essentially a large string bag that is drawn open-mouthed along the sea bottom, scooping up wholesale all bottom-living fish, such as cod, haddock, sole and plaice. All go into the net, both large and small, and, although the young fish ultimately escape through the meshes, many of them are damaged in so doing, while many young, flat fish, lying on the sea bottom, are damaged by the foot rope of the net, as it passes over them. Certain fishing grounds, such as the Dogger Bank, were almost depopulated of flat fish in the years just previous to the war.

Fortunately for the future of the fisheries, the trawl, can only be worked on smooth ground, and at depths not exceeding two hundred and fifty fathoms, so that only a small percentage of the actual fishing grounds is affected by it. Also, when a fishing ground shows signs of becoming exhausted by over-fishing, it is less frequented by fishermen, owing to the reduced catches that can be obtained, and thus it tends automatically to recover. Nevertheless, it is desirable that fishing should be so organized and restrained, that the fertility of the fishing grounds is not imperilled. In the distant future it may become possible to re-stock partially exhausted grounds with young fish, artificially reared in a hatchery.

Oceanography—the study of the ocean and its inhabitants—is one of the youngest of sciences. Yet, to an island people such as we are, it should be one of the most important, for it is only by the study of oceanography that we can hope to found a systematic, organized aquiculture.

The beginning of a simple aquiculture is to be seen in the cultivation of shellfish, such as oysters and mussels, by the inshore fishermen.

Of recent years, experiments have been carried out by the Fishery Boards of England, Scotland, Germany, and the United States of America, with the object of increasing the productivity of certain fishing grounds by adding large numbers of artificially hatched, young fish. For some years the Fishery Board for Scotland added annually about twenty million plaice larvae to certain confined sea areas (Upper Loch Fyne), and found, as a result, that the number of young plaice on the shallow beaches was doubled.

In some cases a new species of fish has been introduced into a particular fishing ground, with marked success. Thus the U.S.A. fisheries collected and hatched the eggs of the shad on the Atlantic coast and introduced the larvae into the Pacific, with the result that a profitable shad fishery has now been established on the Californian coast.

The application of science to the fishing industry is not restricted to biological investigations of the food, habits and development of living fishes. It is developing new processes for the better preservation of edible fish for food purposes, so that the large quantities of fish caught periodically—for example, in the summer herring fishery—may be stored up for gradual consumption during the winter. It has shown that fish waste can be manufactured into glue, cattle food, and fertilizers. It has developed into a profitable industry the extraction of oils from both edible and inedible fish, and the conversion of these oils into hard fats, suitable for the manufacture of soap and margarine. It has demonstrated that the skins of certain fish, notably the shark, can be tanned to make excellent leather.

With the exception of these pioneer experiments and investigations, however, the fishing industry of to-day is simply an organized art—the art of catching wild fish. The story of the industry is essentially a description of the methods that are used for capturing the various species of fish that are of commercial importance, and for handling, curing, and disposing of the catch.

Great Britain is situated in the midst of the greatest fishing grounds of the world. The British fishing industry is the most efficient and the most highly developed of any. Consequently, since fishing methods are essentially the same everywhere, it will be sufficient for us to consider, with few exceptions, the methods and equipment that are used by our own fishermen around our own shores.

There is direct evidence that, as early as the third century, A.D., fish were caught in considerable quantities round the coast of Britain by the natives and used as food. Little is known about the early development of a fishing industry in this country. We know that in the fourteenth and fifteenth centuries, fish was in demand throughout the country, partly because of the religious observance of fast days, and partly, no doubt, because it afforded a welcome change in the regular winter diet of salted meat. In those days there was no winter root crop, so that cattle were killed in autumn and salted down for consumption during the winter.

In disposing of their catch, the fishermen were handicapped by the almost complete lack of transport facilities from the coast inland. Their produce would be distributed by pack-horse, so that fresh fish would be practically unknown beyond a distance of a few miles from the coast. Consequently, all fish for inland markets were salted. The fish were pickled in brine, as the art of dry-salting was then unknown in this country.

To develop a successful fishing industry, it was necessary, then, as it is to-day, either to dispose of the catch quickly on the spot, or to preserve the fish so that it could be transported to distant markets. In 1347, a Dutchman, William Beukels, of Biervelt, invented an improved means of curing and pickling herring, which was essentially the modern process of gutting the fish and packing them in dry salt. At this time the Baltic herring fishery, carried on by the Hanseatic League, dominated the markets of Europe. But the new method of curing, exploited by the Dutch, improved the quality and keeping powers of the fish to such an extent that, by the end of the fifteenth century, the Dutch fishing industry was supreme, and had become a powerful and valuable national enterprise. In the sixteenth century, as many as two thousand Dutch herring “busses” (as the boats were called) would gather on St. John’s day at Brassa Sound, in the Shetlands, to begin the summer herring fishery. The fish were caught with drift nets, were salted and packed in barrels, and carried home by the fast-sailing, attendant “yaggers.” Ashore they were repacked in fresh salt in new barrels. Over a million barrels were packed in a year. When caught, the fish would be worth about a million pounds, and when retailed about two million pounds. Contemporary illustrations of the methods of curing and salting then in use reveal the astonishing fact that even to the smallest detail the methods that

were employed in Holland in Elizabeth's day are identical with those that are employed at Yarmouth to-day.

As a direct result of the great development of their trade in salted herrings, the Dutch gradually gained a naval and maritime supremacy in Europe which they maintained until it was wrested from them by the English.

English sea-power in the early years of the sixteenth century was in a decadent condition. The ports and harbours had been neglected, and had become silted up, so that the condition of the shipping industry in general, and of the Navy in particular, had reached a very low ebb. In 1561, Mr. Secretary Cecil, alarmed by the growing menace of the Dutch naval ascendancy, proposed three remedies for restoring the strength and importance of the navy. He proposed:

(1) That the fishing industry be promoted, as it provided a valuable recruiting ground for the navy;

(2) That merchandise be extended, and so provide increased employment for the shipping industry;

(3) That piracy be encouraged, privately-owned privateers forming valuable auxiliaries in time of war.

He thought that the fishing industry could be stimulated immediately by renewing the fast days, which had fallen into disuse since the abolition of the monasteries.

He suggested that two days a week—Wednesday and Friday—should be meatless days.

In 1563, he tried a measure of Protection, a Navigation Act being passed, making it illegal to buy or sell foreign-caught fish, and attempts were made to prevent Dutch and other foreigners from fishing in English waters. These measures, although passed by Parliament, do not appear to have been enforced.

James I issued two proclamations, imposing licences and dues upon foreign fishing vessels fishing in British waters. No attention was paid to these, and it was left to Charles I, some years later, to enforce them. Other steps taken by both Charles I and Charles II consisted mainly in the

formation of Royal Fishery Companies. Various fishery companies and societies succeeded one another up to the end of the eighteenth century. They do not appear to have been successful in establishing a flourishing fishing industry, and in 1718 (George I) an act was passed by which fishermen were to be rewarded for their catch by a bounty. Bounties were to be paid for several kinds of fish: thus, for every barrel of white herrings of 32 gallons, exported beyond the seas, the bounty was 2s. 8d.; for full red herrings, 1s. 9d. per barrel; for empty red herrings, 1s. per barrel.

The conditions upon which the bounty was to be paid were fully set forth in a later act in 1750 (George II). The construction of herring vessels was encouraged by a bounty of 30s. per ton, paid out of the Customs, for decked fishing vessels of from twenty to eighty tons.

The time and place of fishing were stipulated, as well as rules for the proper management and prosecution of the fishery. Each vessel was to have on board twelve Winchester bushels of salt for every last of fish such vessel was capable of holding, the salt to be contained in new barrels.

In 1757, the bounty was increased to 50s. per ton, but was reduced to 30s. again in 1771. It was further reduced to 20s. in 1787, and an additional bounty of 4s. per barrel added. This was made proportional to the tonnage, so that no vessel could claim more than 30s. per ton—unless the vessel caught over three barrels per ton, in which case a bounty of 1s. per barrel was granted upon the surplus quantity.

While the bounty often undoubtedly encouraged the development of the fishery, the development was not so rapid or so extensive as it would otherwise have been, owing to the duty on imported salt. The weight of the duty was such that the fishermen threw fish overboard rather than cure it, only landing that which could be brought in fresh.

In 1808 the bounty was raised to £3 per ton on every British built and British owned fishing boat of not less than sixty tons burden, properly manned, registered, and navigated and employed in herring fishing. The maximum tonnage on which the bounty was payable was one hundred tons. Two shillings per barrel was paid on properly cured and packed herrings.

After the peace of 1815, the naval wars and the press gangs had reduced the sea fisheries to negligible proportions, but the existing bounties were

continued until 1829, and encouraged the rapid revival of the industry. By 1829, the fishing industry was well established, and thereafter steadily developed in value and importance.

The modern organization and development of the fishing industry began between 1870 and 1880, following the introduction of steam fishing vessels. The old sailing smacks and drifters were necessarily limited in their scope and capacity. They could only fish in certain weathers; they required skilled handling; their effective area of operation was restricted by the necessity for bringing the catch ashore as fresh as possible; their trawling power depended upon the wind.

A sail boat was generally the property of a small family group of fishermen, who worked the boat and fished, while one of their number—the ship's husband—stayed ashore to purchase stores and tackle, and dispose of the catch. The proceeds of the boat were shared among the owners. These privately owned sail boats were to be found in every little harbour on every coast of Britain. The fishermen themselves were a fine, sturdy, independent class of men, skilful seamen, and all-round fishermen, able to turn their hands to any form of fishing, whether lining, trawling, or drifting.

The introduction of steam trawlers and drifters has completely changed the character and organization of the fishing industry. Instead of being individualistic, it has become collective, and instead of being the common industry of every seaside village, it has become controlled by large limited liability companies, and centralized in a few large ports.

Steamers were first used in 1870, to collect the catch from the sail boats on the fishing grounds, bringing it home with all speed while the fishing boats remained at sea. This naturally enabled the fishing boats to catch more fish, and also made possible the use of larger boats fishing further afield. A logical development of this step was the construction of actual steam-driven fishing boats—trawlers and drifters. These steamers soon proved to be superior to the sail boats. They were able to fish in all weathers, even in a calm. Owing to their greater power, also, they were able to use much larger nets and fish in deeper waters.

Steam trawlers and drifters are much more expensive than smacks or sailing drifters. They can only be berthed and handled satisfactorily in harbours that are equipped for the unloading and dispatching of large

quantities of fish. From the very beginning these steamers were owned by large limited companies rather than by individuals, and the industry has tended to become more and more centralized at certain large ports, for example, Aberdeen, Hull, Grimsby, Yarmouth, Lowestoft, Milford Haven, and Fleetwood. The rise and development of many of these ports, for example, Aberdeen and Fleetwood, has been in direct response to the demands made upon them by the new steam fishing industry.

The introduction of steam fishing made longer voyages possible, and led to the development of new fishing grounds. Steam trawlers from British ports now fish as far north as Iceland and the White Sea, as far west as Newfoundland, and as far south as Morocco, making voyages of many week's duration.

The re-organization of the fishing industry led to specialization amongst the fishermen themselves. The old sailing fisherman was essentially an all-round man. He was equally expert at lining, drifting and trawling. The skipper of a steamer, however, is a specialist; he is either a liner-, a drifter-, or a trawler-man. Generally, also, he keeps to a given region—Iceland, the White Sea, the North of Scotland, the North Sea, or the Bay of Biscay.

In the three years preceding the war (1911 to 1914) the development of the steam fishing industry had become almost stationary. This was probably due in part to over-capitalization, resulting in lower profits. It was feared also that the greatly increased efficiency of the steam trawlers tended to produce a condition of over-fishing in certain areas, with the result that catches obtained in those areas progressively diminished; for example, the average catch per boat per day in the North Sea during three successive periods was as follows—

1903 to 1906	17·2 cwts.
1907 to 1910	16·7 „
1911 to 1913	15·3 „

The fishermen became alarmed and development was arrested. This tendency to over-fish certain grounds has been effectively checked during the war by the almost complete cessation of offshore fishing. There is thus every probability that such grounds have now recovered, and further that, in

many cases, grounds such as the Dogger Bank, that had become almost depopulated, will have become restocked.

The successful development of steam fishing has necessarily reacted upon the prosperity of the individual fishermen in the various fishing villages, with their smaller, privately-owned sail boats. They were faced with two alternatives: either to combine together to acquire steamers, and so maintain their position in the offshore fisheries, or to devote their attention to the development of inshore fishing. Many of the larger sailing drifters have now been fitted with petrol engines, which make it possible for them to compete with the steam drifters for herring and mackerel.

Generally speaking, however, the outlook for the small fishermen of the English and Scottish coast villages—the real fisher folk—is discouraging. The tendency of legislation, however, just before the war was to encourage this class of fishermen by restricting the operations of the steam trawlers in certain localities. In 1910-1914, with the object of protecting the inshore fishermen, the Fishery Board of Scotland prohibited trawling in the Moray Firth area, only drifting and lining being permitted. Since this prohibition only applied to British subjects, certain East Coast fishing companies evaded it by transferring their vessels to foreign flags, registering them in a foreign port and employing a foreigner as a dummy skipper. The Board secured convictions against these offenders in the Sheriff's Court, but the convictions were upset subsequently by the Foreign Office. The original prohibition was then strengthened by a new law which made it illegal to land fish in Scotland, if caught by vessels registered in a foreign port.

During the war, the inshore fisherman found himself in a comparatively advantageous position, as the high price of coal made steam fishing less profitable. Further, the offshore trawling grounds were mostly closed, and the majority of the steam trawlers and drifters were on war-service. For the time being, therefore, inshore fishing with smacks was placed at an advantage.

A number of fishermen's co-operative societies were formed to organize the sale and distribution of the produce of these inshore fisheries. This also tended to make the position of the inshore fisherman more secure.

The old order changeth, and although there is that connected with this transformation in the fishing industry which is to be regretted, yet, on the

whole, the developments of the past forty years have undoubtedly transformed the fishing industry into a very efficient and valuable national asset. Individually, the present-day steam fisherman is very much inferior to his sailing predecessor. The centralization of the industry in a few big ports, although undoubtedly making for much greater efficiency, bears hardly on the type of the old class of expert fishermen; but these are the almost inevitable consequences of such a transition.

But what is the present condition of the industry, and what is its future likely to be? The prosperity of the inshore fisherman, as well as that of his offshore rival, is vitally important to the welfare of this country; there should be room and opportunity enough for both. The inshore fisherman, protected by legislation and secured by well-organized co-operation, can increase very considerably the amount of our available home-grown food supply. The superior power and equipment of the big steam trawlers and drifters, properly utilized and encouraged, should be one of the most valuable industrial assets of the State. We are not a great food-producing nation; on the contrary, in the years before the war, we actually imported more than 40 per cent of our total food requirements. We are surrounded by seas that teem with every form of edible fish. British enterprise has built up a fishing industry which is the greatest and most efficient in the world. In 1914, our fishing boats were practically equal in numbers and equipment to those of all the other countries in North-West Europe put together. Nearly 70 per cent of the fishing boats in the North Sea were British. The total produce of our sea fisheries has nearly doubled since the beginning of the century. The annual catch in the last few years before the war averaged over a million tons. It was worth about fifteen million pounds when landed, and may be valued at nearly fifty million pounds by the time it reached the consumers. Of all this splendid food that is obtained at our very doors by our own people, less than half is retained for consumption in this country. Out of 600,000 tons of herrings landed annually in this country before the war, over 500,000 were exported, chiefly to European countries. Herrings have a high food value, and contain a large amount of easily digested fat, and if all the herrings landed in this country were consumed at home, it would only allow two herrings a week to each adult individual in all the population. An increased home consumption of fish, would effect a corresponding saving in imported meat.

Owing to this remarkably small home demand for fish, the fisherman has had to depend upon foreign markets, chiefly Germany, Poland, Russia and the Levant. The present adverse rate of exchange with these countries, and the increased cost of fishing operations, make it impossible for the foreign importer to take our fish, except on terms which our fishermen cannot consider. These markets are therefore closed, and unless other outlets are found for its produce, the industry will be threatened with ruin.

In 1920, the Government guaranteed the cure of herrings up to 880,000 barrels; unfortunately, they were only able to dispose of them in European markets at a great loss. The Government, therefore, have decided this year (1921) to withdraw their guarantee.

It would seem that, in view of the present failure of the foreign markets, vigorous steps should be taken to encourage the consumption of fish in this country, and so preserve this valuable industry from ruin. A national scheme of development should be inaugurated, having for its objects, (1) the systematic exploitation of local and periodic coastal fisheries; (2) the discovery of methods of preserving for future consumption fish that cannot be disposed of just when it is caught; (3) the education of the public to use more freely the large supplies of excellent fish food that are available at our very doors.

CHAPTER II

CHARACTERISTICS AND HABITS OF FISHES

FISHES are the most primitive vertebrate, i.e. backboneed, creatures known. All reptiles, birds, and animals have gradually evolved from fish-like ancestors by a series of age-long processes, the stages of which are recorded in fossilized remains that are found in various rock strata throughout the world. A fish lives exclusively in water. It has no lungs, but extracts oxygen from the water as it passes over the surface of its gills. Instead of limbs, it has fins, with which it balances itself and propels itself through the water. Its skin is either bare, e.g. the cat fish, or is covered with scales, e.g. the herring, or with bony plates, e.g. the sturgeon. The skin of certain sharks is studded with minute teeth and produces, when cured, the well-known shagreen leather. In nearly all cases the skin of fishes is liberally supplied with small glands which constantly produce a lubricating mucus. This mucus greatly reduces friction between the fish and the water through which it moves.

The body of a fish is adapted to move swiftly and smoothly through the water; it is shaped more or less like a torpedo, but this form is greatly modified in different species. Certain species of fish living at the bottom of the sea, for example skates and rays, have become flattened, as though by a pressure applied vertically downwards. Others, for example plaice, flounder, sole, appear to have been flattened sideways. In the various members of the eel family, the body is greatly elongated.

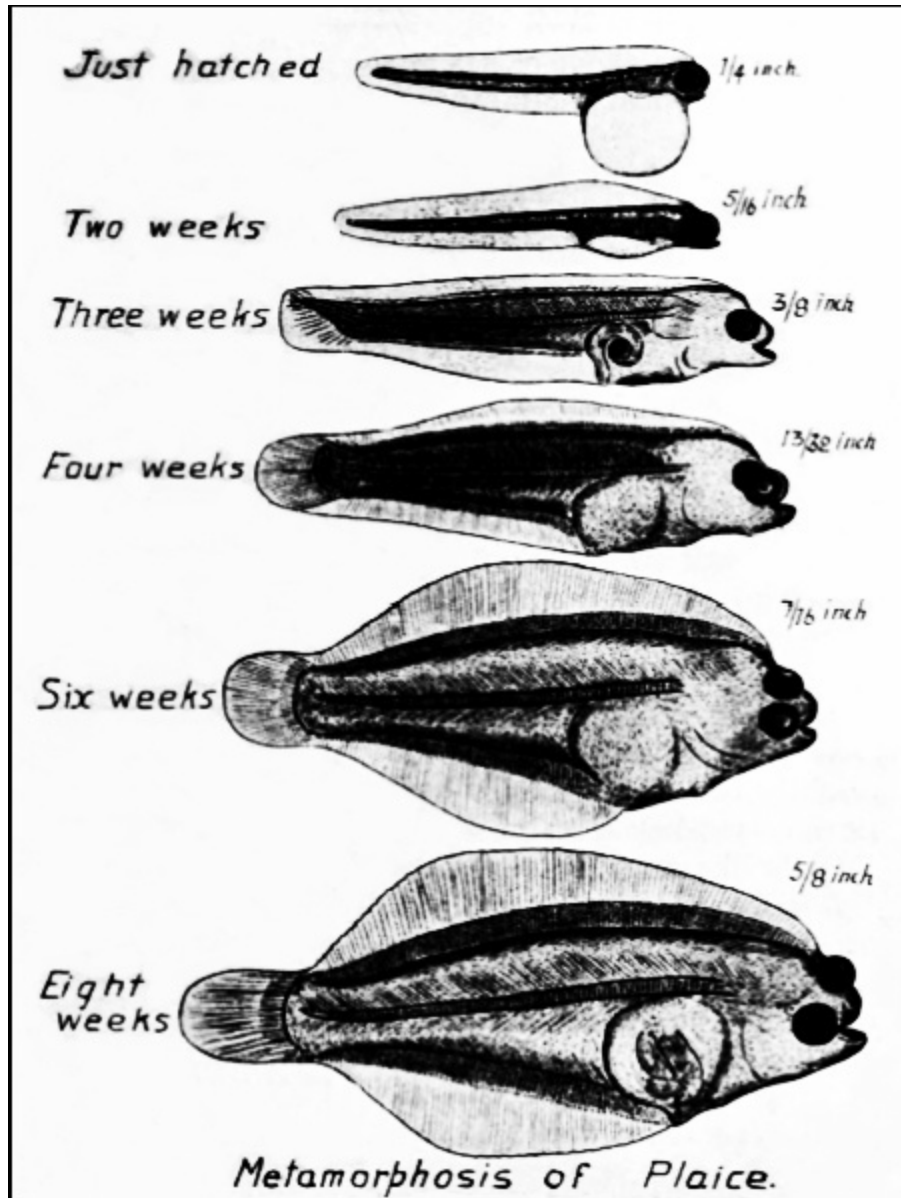


FIG. 1

The body of a fish is generally coloured and marked in such a way that it becomes practically invisible when seen from above or below, the under-surface being silvery white, and the upper surface generally olive or blackish-green. Sometimes, as in the mackerel, the upper surface is mottled, resembling rippled water.

Most small fish in ponds and streams reflect their surroundings so well, and are coloured and marked in such a way, that they are almost invisible to

the large fish, for example pike, that prey upon them. Generally, they reveal their presence by the flash of light reflected from above by their scales, as they turn suddenly to snap at a morsel of food. In the same way, many predatory fish, e.g. the angler fish, resemble their surroundings so closely that the fish for which they are lying in wait swim within easy reach of them without perceiving their danger. Many fishes, particularly in tropical waters, are remarkable for their bright and gorgeous colouring. It is impossible to preserve these colours in their natural brightness after the fish have been taken from the water, but amongst the brightly coloured corals, and anemones and seaweeds, in the crystal clear water of their natural environment, they flit like gorgeous tropical birds in a tropical forest.

Distribution. Fishes are found in practically every ocean, lake and river in the world, with a few notable exceptions, such as the Dead Sea, in which the concentration of salt is too high. They appear to exist at all depths of water, and have been found in the sea as deep down as 2,720 fathoms. Fish living at this depth generally possess enormous mouths, long, attenuated, soft bodies, and are equipped with highly developed phosphorescent organs.

The distribution of a particular species appears to depend upon the salinity of the water, the temperature of the water, the kind and quantity of food available and the prevailing intensity of sunlight. It is possible to divide fish into four well-defined groups, according to the salinity of the water in which they are found: (1) Marine fish: those that live always in the sea, for example herring, haddock, shark. (2) Fresh-water fish: those that live always in fresh water, for example carp, trout, pike. (3) Many fish live in brackish water, and appear to be able to accommodate themselves easily to considerable changes in salinity, e.g. sticklebacks, gobies, grey mullets and blennies. Such species naturally are widely distributed; thus, a particular kind of grey mullet (*Mugil capito*) is found without any appreciable difference in form on nearly every coast of the Atlantic Ocean. (4) The fourth group of fish are migratory. Some species, for example salmon and shad, live and develop in salt water, but ascend rivers to spawn, i.e. to lay their eggs, in fresh water. Others, such as eels and certain pleuronectids, for example the flounder, live and develop in fresh water, and descend rivers to the sea to spawn. Many fresh water fish, e.g. trout, forsake the large streams in the spring and ascend small brooks, where the young can be reared in greater safety.

Of these different groups or species, the marine fishes are industrially by far the most important, for at least two-thirds of all the fish in the world live in the sea, and the capture of these sea-fish in enormous quantities constitutes the fishing industry, with which we are concerned.

The different species of marine fishes can be divided into three well-marked groups, according to their habits and habitats.

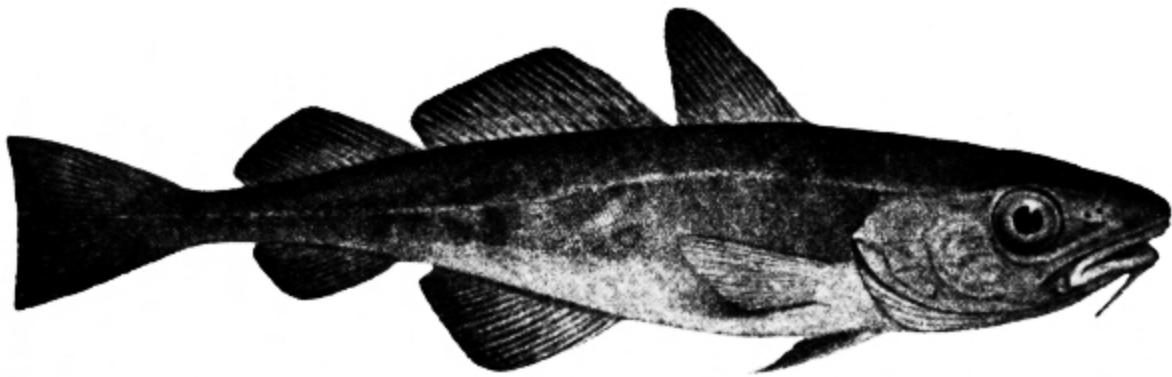


FIG. 2
COD (*Gadus morrhua*)

Length up to 5 ft.; usually caught at about 3 ft.

Food.—Small crustaceans, molluscs, and young fish.

Range.—North of Norway and Iceland to the Bay of Biscay, and from Greenland to New York.

(1) There are the true deep-sea fishes that live at the bottom of the sea, for example cod, haddock, plaice, sole. These are called “demersal” fish. Fish, like birds, inhabit a medium that is continuous throughout the world. A glance at the map of the world will show that the three great oceans—Atlantic, Indian and Pacific—are united in the southern hemisphere. In Tertiary times, it is practically certain that the Pacific and the Atlantic oceans were also united at Darien, and that the Mediterranean was united with the Red Sea. Apart, therefore, from differences in local conditions, for example of temperature and food supply, there is practically no obstacle to the world-wide distribution of any particular species of fish. At the bottom of the sea, the temperature, the food supply, and the general conditions of

life are singularly uniform all over the world, consequently there are no barriers at all to the dispersion of demersal fish, and we find various species widely distributed in all seas. Demersal fish, on the whole, are more primitive in type than those that live nearer the surface. They have well-developed senses of touch and smell by means of which they hunt for their food. They differ markedly in structure and shape from surface or shallow-water fish, their bodies being designed to resist the greater pressure of deep water. The body is generally lean and is enclosed by a wall of muscular fibre. Shallow-water fish, if introduced into deep water, would be crushed inward by the pressure. Similarly, the deep-living, demersal fish are unable to accommodate themselves to shallow water and, if placed in it, soon become unhealthy. A cod floats helplessly on its side when placed in shallow water, owing to the dilatation of its swimming bladder. If the bladder is pricked it collapses, and the fish is able to regain an upright position. This is done when cod and other similar demersal fish are kept alive in sea-water tanks on board ship, to be delivered to the markets alive. In Denmark, fish are delivered alive to the shops. When fishes from great depths are brought to the surface, their bodies break into pieces owing to the reduced external pressure, the scales start from their skin and the eyes from their sockets.

There are two distinct types of demersal fish: the “round” and the “flat.” The body of a round fish is more or less circular in cross-section, for example cod, while that of a flat fish is flattened, for example sole, ray.

The most important edible demersal fish can be classified as follows—

(a) The *Gadidae*—related to the cod.

Cod—inhabits northern waters, notably the North of Britain, Iceland and Newfoundland.

Ling—inhabits northern waters: West of Scotland and Ireland, and North towards Iceland and Newfoundland.

Haddock—inhabits northern waters. Nearly half the total catch is obtained in the North Sea, from the White Sea to the Bay of Biscay.

Whiting—found in great numbers in the North Sea. It is more coastal than the cod or haddock.

Hake—found from Norway to the Mediterranean. The greater part of the catch is obtained off the south-west of Ireland. Hake is also caught off Morocco and in the Bay of Biscay.

(b) The *Pleuronectidae*—related to the plaice and sole.

Sole—a shallow-water fish, common in the Irish Sea, and particularly abundant in southern waters down to Morocco.

Plaice—inhabits northern waters—all round Britain and Iceland.

Flounder—inhabits estuaries, for example, of the North Sea and the Baltic.

Halibut—inhabits northern waters. It attains a large size, six feet or more.

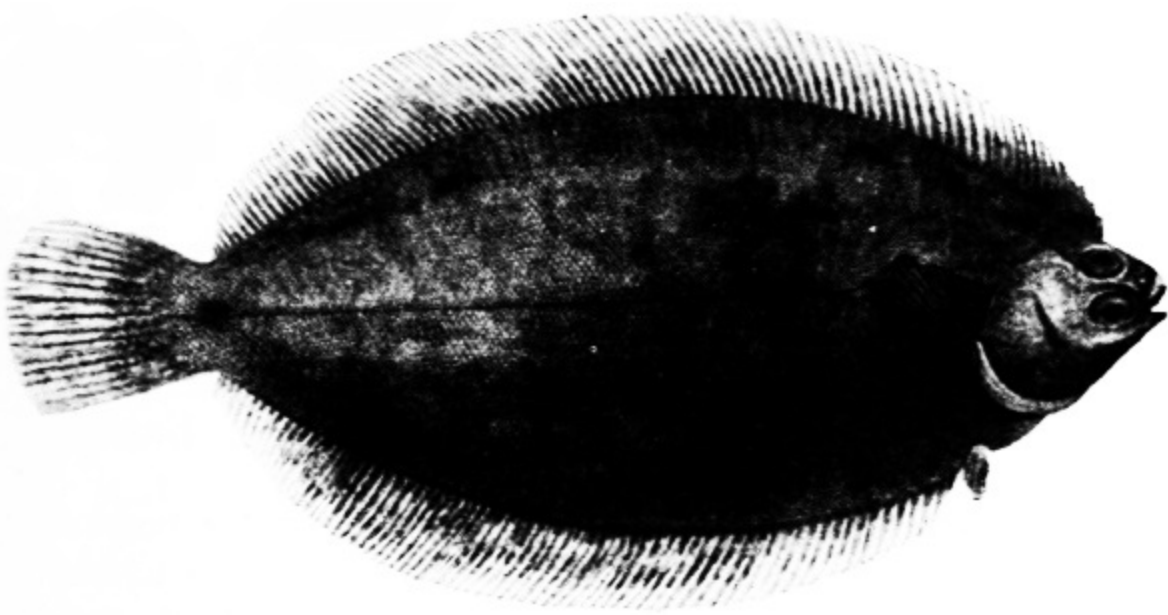


FIG. 3

LEMON SOLE (*Pleuronectes microcephalus*)

Length up to 16 ins.

Food.—Small crustaceans and worms.

Range.—From North of Europe to the Bay of Biscay.

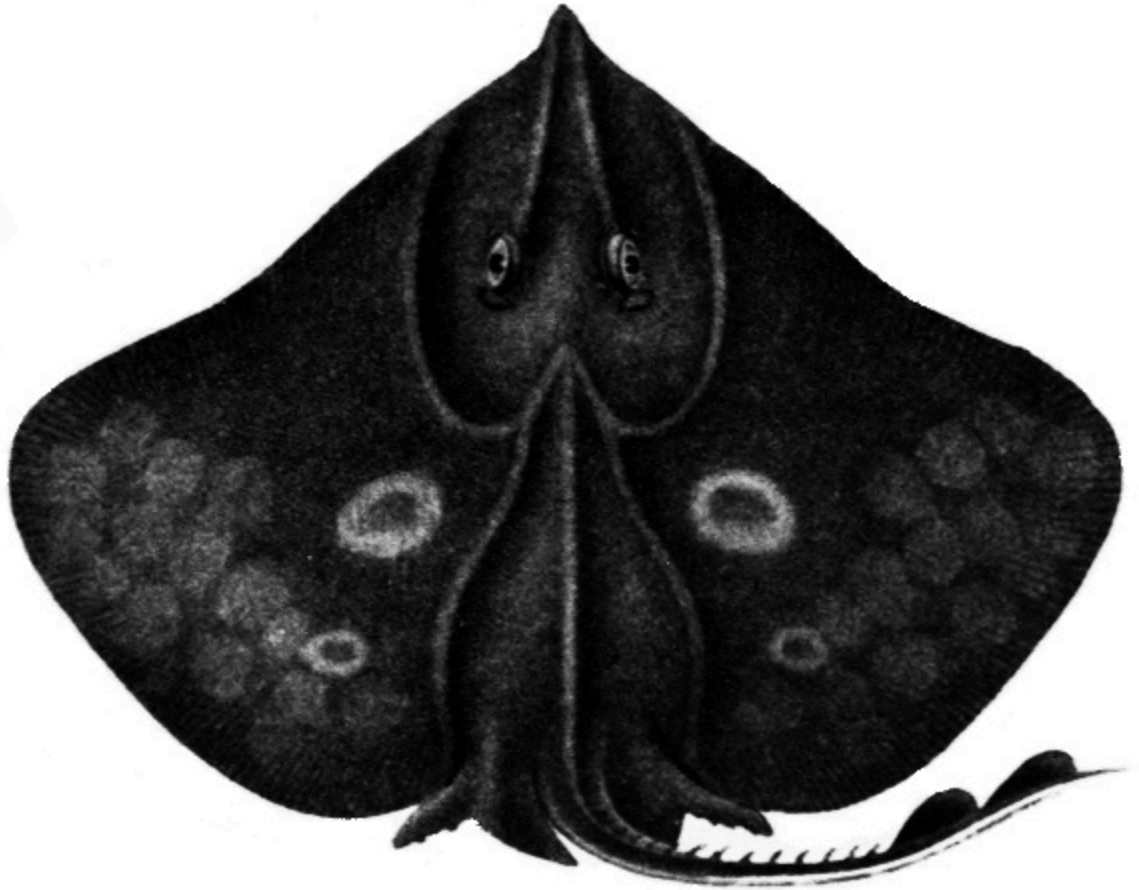


FIG. 4
SKATE (*Raia batis*)

Length up to 7 ft.

Food.—Crustaceans and molluscs, and fish.

Range.—Round the British Isles and along the coast of Western Europe.

Turbot—not very abundant. It inhabits the deeper parts of the North Sea.

Brill—inhabits southern waters, and is fairly abundant.

(c) The *Raïdac*.

Skates and rays—found all round Britain, more particularly the Western area of the English channel.

(2) The various species of fish that inhabit the surface waters of the sea are called “pelagic.” They include the herring, mackerel, tunny, flying fish, sword fish, and many sharks, also various marine mammals, such as whales, grampuses, porpoises, dolphins. Amongst pelagic fish are included some of the smallest (plankton) as well as some of the largest (whales) of all living creatures. Pelagic fish pass their whole life swimming at or near the surface. They enter the shallow water offshore only for prey or, in some cases, periodically to spawn. The majority spawn in the open sea, far from land. Unlike demersal fishes, the distribution of the different species of pelagic fishes depends very much upon local conditions of light, water temperature, and the character and quantity of food available. They do not hunt their food individually to the same extent as demersal fishes, but generally filter it from the water as it passes through their gill-openings. Although not so widely dispersed as demersal fish, they are, in favourable circumstances, dispersed over large areas by swimming and by ocean currents.

All pelagic fish are “round.” With the exception of the mackerel, the important edible pelagic fishes belong to the herring family, and are known as the Clupeidae. They include—

Herring—found from the White Sea to the Bay of Biscay. It is the most abundant of all food fishes.

Sprat—found from the North of Europe to the Mediterranean.

Pilchard—ranges from the English Channel to Madeira and the Mediterranean. Skipper “sardines” are young herring, pilchard, and brisling.

There is also—

Mackerel—found from the North Sea to Madeira and the Mediterranean.

(3) The shallow-water of the seashore is inhabited by certain animals (shellfish) not found elsewhere, including various mollusca, e.g. mussel, cockle, oyster and periwinkle, and crustacea, e.g. lobster, crab, prawn, shrimp. In addition to these, there are various species of immature offshore fish, e.g. plaice and dabs. The inhabitants of this shallow, coastal water are called “littoral” fish. The distribution of such littoral fish depends not only

upon the water temperature and the amount of light, but also upon the character of the shore—whether it is rocky, or soft and sandy—and more especially upon the animal and vegetable products of the adjacent land, e.g. plants, seaweed, worms. Littoral fish do not swim very far, but become scattered inadvertently over considerable distances by currents and other mechanical means.

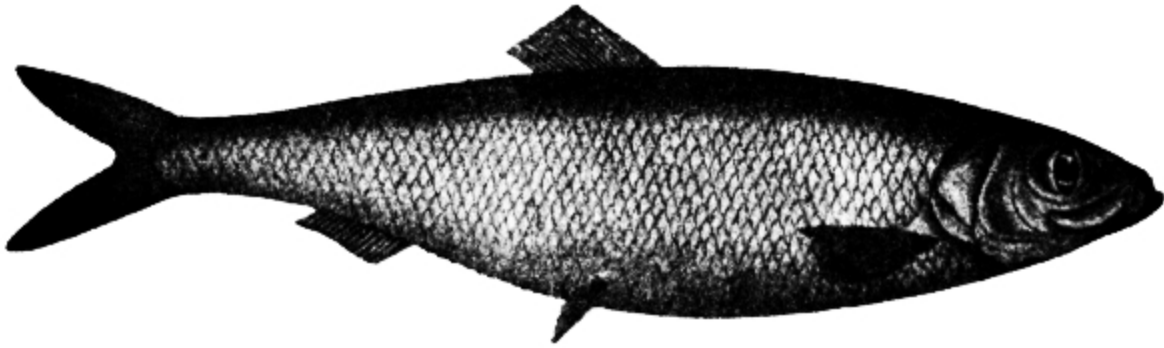


FIG. 5
HERRING (*Clupea harengus*)

Length slightly above 12 ins.

Food.—Plankton (*copepoda*).

Range.—From the White Sea to the Bay of Biscay.

Certain kinds of shellfish, for example oysters, mussels, cockles, live in the sand or attached to the stones or seaweed on the seashore, generally between high and low watermarks. They obtain their food from the water as it streams over their gills. They require adequate room for growth and development, and constant irrigation by water containing sufficient floating food. When mussel beds or oyster beds become overcrowded, the fish are ill-nourished, their health is impaired and their growth is arrested. It has been shown that, if they are transferred to new beds, their condition rapidly improves and ultimately they increase considerably in size. All edible shellfish need systematic care and attention. Their cultivation by man affords the simplest instance of an attempt at a systematic aquiculture.

Food. The surface water of the sea abounds in minute forms of vegetable and animal life. This vast floating population of microscopic organisms is

called the “plankton.” Just as man and all land animals depend ultimately for their food supply upon grass and other green-leaved plants which, under the influence of sunlight, are able to transform the inorganic constituents of the atmosphere and the soil into organic foodstuffs—albumen, fat, carbohydrates—so the minute unicellular marine plants of the plankton are able, under the influence of sunlight, to convert the inorganic constituents of their environment into fat, albumen and carbohydrate. Upon these minute organisms, therefore, directly or indirectly, all marine life depends.

In addition to these minute plants, the plankton contains nearly all forms of marine life at some stage or other of their life history. Fish are only found in it as eggs, or larvae. Crustacea of all kinds are present, and form one of its most important constituents. Crabs and lobsters spend their larval, free-swimming career among the plankton, until they reach the adult stage and settle down to the bottom. Various minute crustacea, known as “Copepoda” (lit., oar-footed) spend the whole of their lives drifting about in the surface water. They occur in incredibly large numbers, and are the most abundant of all forms of marine life. These copepoda form the main source of the food of pelagic fish, such as the herring, mackerel and sprat.

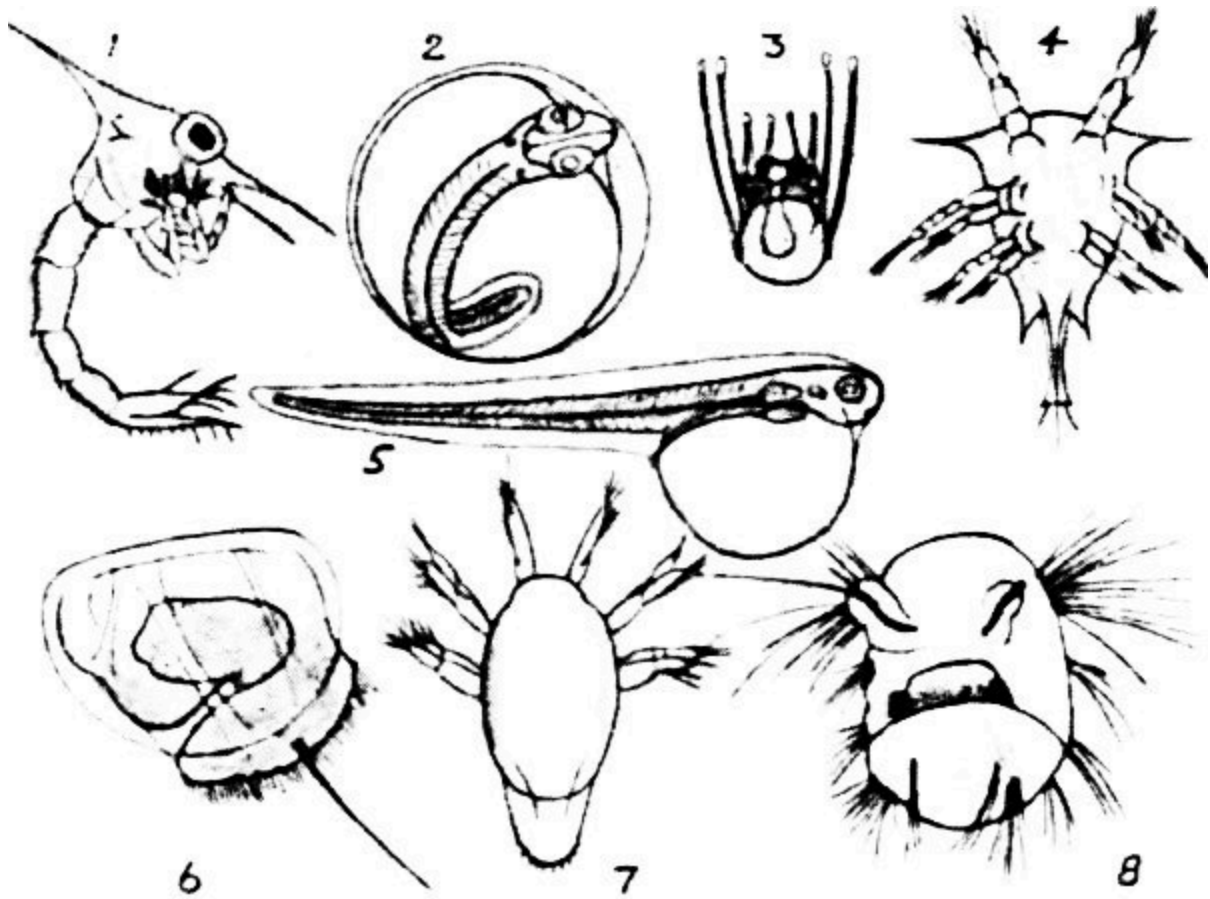
The larvae of the edible molluscs, oyster, mussel, cockle, develop in the warm surface water until they settle to the bottom and begin their adult life.

There are also many larval forms of marine worms and jellyfish, and many kinds of microscopic, unicellular organisms, some of which are vegetable and others are clearly animal. The chief animal forms belong either to the Infusoria, the Foraminifera or the Radiolaria. The shells of the two latter forms accumulate at the bottom of the sea, producing the deposits known as the Globigerina and Radiolarian oozes. In this way, chalk deposits were formed in primitive times.

The most important vegetable planktonic organisms are the Diatoms. Their accumulated shells form important deep-sea deposits.

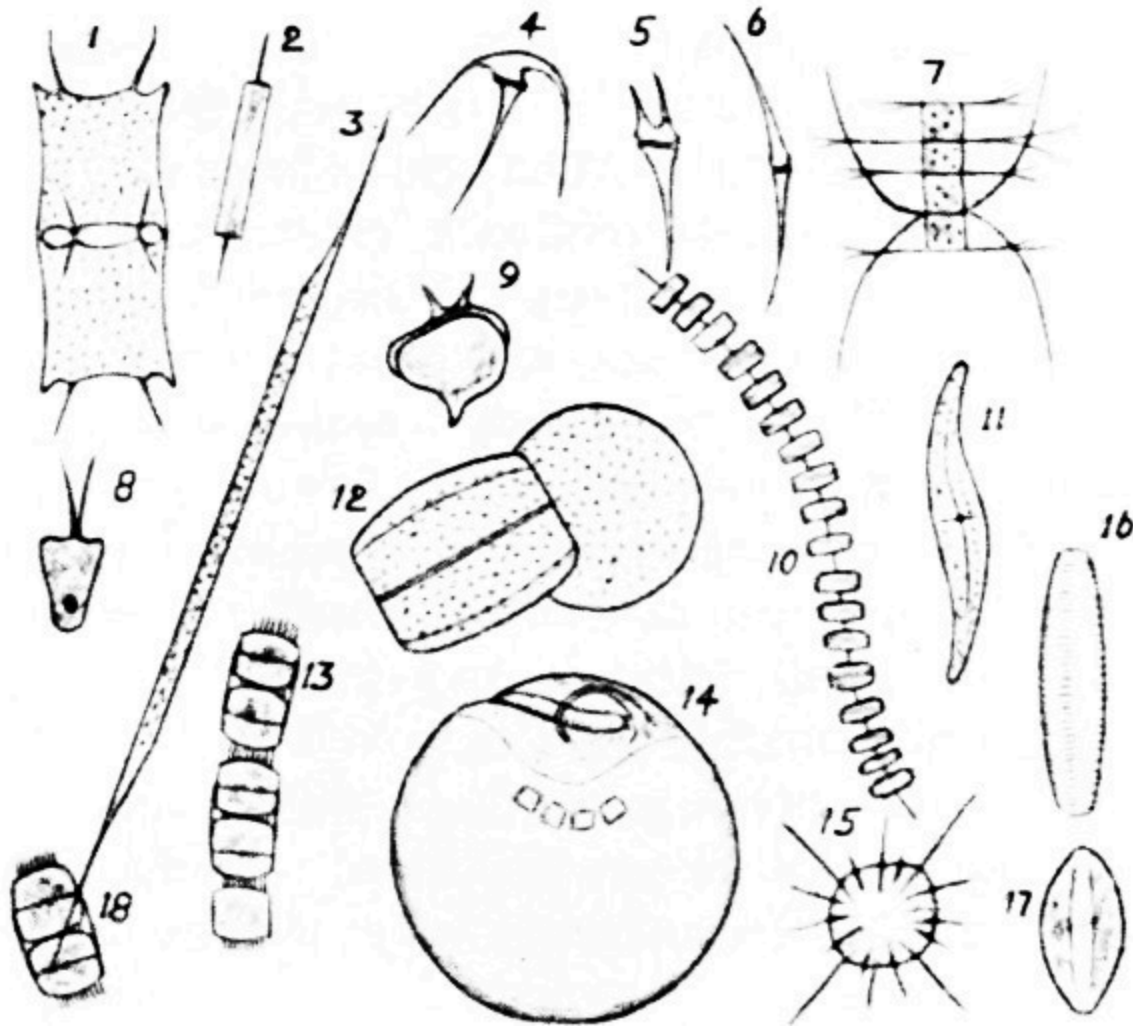
The numerous varieties of planktonic life can thus be divided into two groups: those minute animal and vegetable organisms that pass the whole of their existence at the surface of the sea—the true constituents of plankton all the year round—and the eggs and larvae of many species of fish that are found among the plankton only at certain times of the year—notably in spring and summer.

The quantity of organic food substances such as albumen, fat and carbohydrate, that is contained in the plankton produced annually by a given area of the sea, has been compared with the quantity of such substances produced by a similar area of land in crops such as pasture, hay, lupine and peas. In this way, it has been estimated that the productivity of the sea is about 20 per cent less than that of average land.



PLANKTON: LARVAE

1. Crab zoea; 2. Fish egg; 3. Sea Urchin pluteus; 4. Barnacle nauplius; 5. Fish larva; 6. Mussel larva; 7. Copepod nauplius; 8. Worm larva.



PLANKTON: UNICELLULAR ORGANISMS

1, 2, 3, 7, 10, 11, 12, 13, 16, 17, 18. Diatoms; 4, 5, 7, 9, Peridinians 8. An Algal spore; 14. Noctiluca; 15. A Radiolarian.

FIG. 6

Unlike that of the land, the productivity of the sea is greater in colder latitudes than in the tropics. This somewhat unexpected fact is attributable to the action of denitrifying bacteria which, flourishing more readily in warm, tropical waters, effectively reduce the amount of available nitrogen compounds in the water. In colder waters, denitrifying bacteria are less active, and nitrates and nitrites are available in larger quantities for the nourishment of the plankton.

All the great fisheries of the world are prosecuted in cold or temperate seas; as examples of this we have the Banks of Newfoundland, the cod fisheries of Norway, and the great trawling grounds of the North Sea and the North Atlantic.

All fish, during the larval stage of their development, feed first upon the contents of the yolk sac which, when they are hatched, is attached to their ventral surface. When the yolk is absorbed, the larvae feed upon the microscopic plankton that abound in the water on every side. The surface water, with its warm temperature, high plankton content and sunlight, forms an ideal nursery for the very young fish of all species. Demersal fish, as they complete the larval stage of their development and descend into deeper water, have to rely for their food either upon the various species of young shellfish and crustacea that drop from the surface water as they develop, or hunt for their food amongst the small fish, mollusca, crustacea, worms and seaweeds of the sea-bottom. Plaice feed chiefly upon cockles and other mollusca, which in their turn feed upon diatoms. The cod is almost omnivorous, greedily devouring small fish, crustacea, worms or mollusca; its favourite food, however, is shrimps and prawns. These, in their turn, feed upon smaller invertebrates, for example small jellyfish and larval molluscs, and these upon microscopic plankton.

Pelagic fish, herrings and mackerel, feed almost entirely upon the larger plankton, mainly copepoda (small, shrimp-like crustacea). These may be present in the surface water in enormous quantities at certain times. In many cases, shoals of herring or mackerel probably follow special swarms of copepoda. Mackerel also feed upon young fish, hermit crabs, and prawns.

With a few notable exceptions, the various species of demersal fish feed upon smaller fish. Thus—

The hake, normally a deep-water fish, ventures inshore in pursuit of herrings, pilchards, mackerel.

The ling, turbot, brill, dog fish live entirely upon small fish. The dog fish swarms on certain fishing grounds and is often a serious pest to the drift-net fishermen, destroying their nets as well as the fish that are attached to them.

The whiting, like the cod, feeds upon small fish, and upon crustacea and mollusca.

The food of the haddock consists of mollusca, crustacea and marine worms, etc.

The sole lives on small crustacea, for example shrimps, and marine worms.

Skates and rays feed upon mollusca and crustacea.

Most shellfish live in shallow water and feed upon the plankton.

The methods by which fish obtain their food differ greatly according to the species of the fish. Pelagic fish, e.g. herring and mackerel, sprat and pilchard, obtain their food almost automatically as they swim open-mouthed through the water in which it abounds. These direct plankton-feeders possess comb-like structures—the gill-rakers—attached behind the gill openings, and as the food-bearing water streams through the mouth and gill openings of the fish, these structures strain the food from it. The fish licks the plankton from its gill-rakers with its tongue and swallows it.

Many pelagic fish, e.g. carp, trout, salmon, look for their food while swimming through the well-lighted surface water.

Demersal fish—flat fish, cod, haddock, etc.—seek their food by scent and touch. The cod possesses a barbel attached to its chin, by means of which it feels for its food.

The Angler or Devil fish is a curious creature, from three to four feet long, and appearing to consist almost entirely of head. It has a large mouth, and teeth that are hinged so as to admit food, but prevent it from escaping. The devil fish has a long feeler on the top of its head, terminating in a tassel which, moved by the water, attracts the attention of small fish and lures them to their fate. This tassel is a sensory organ and, when it is touched by the small fish, the angler fish snaps upwards with unerring aim at a point immediately in advance of the tassel.

The dog fish seeks its food exclusively by scent. If its sense of smell be destroyed, it ceases to feed spontaneously.

The sole also seeks its food by smell. It is quite unable to recognize a worm by sight or touch, even when hung just above its head, but feels aimlessly over the ground seeking it by smell.

Reproduction. Fish are male and female and, with few exceptions, reproduce their kind by laying eggs. The number of eggs laid by an individual female fish during a single spawning varies greatly, according to the species. The average number of eggs spawned by a single female fish in the course of one season, is—

Ling	18,500,000
Turbot	8,600,000
Cod	4,500,000
Flounder	1,000,000
Sole	570,000
Haddock	450,000
Plaice	300,000
Herring	32,000
Shark	A few—not more than a dozen.
Dog fish	
Skate	

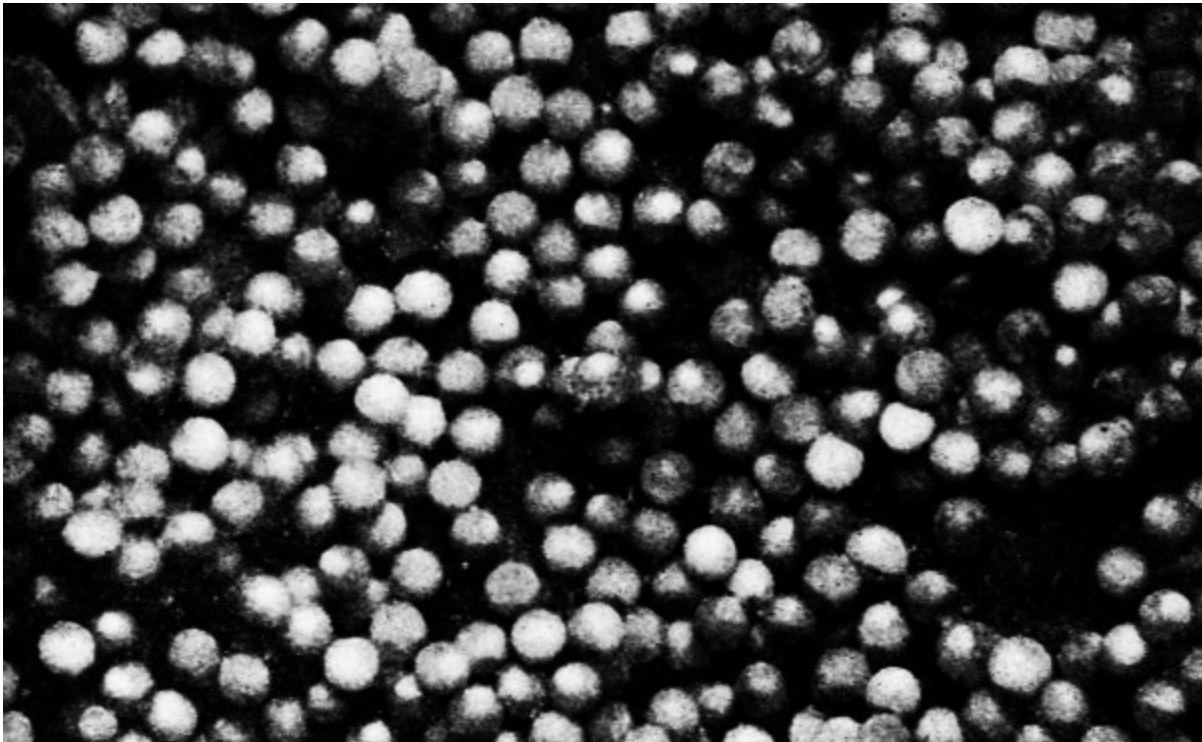


FIG. 7
HERRING EGGS—×5

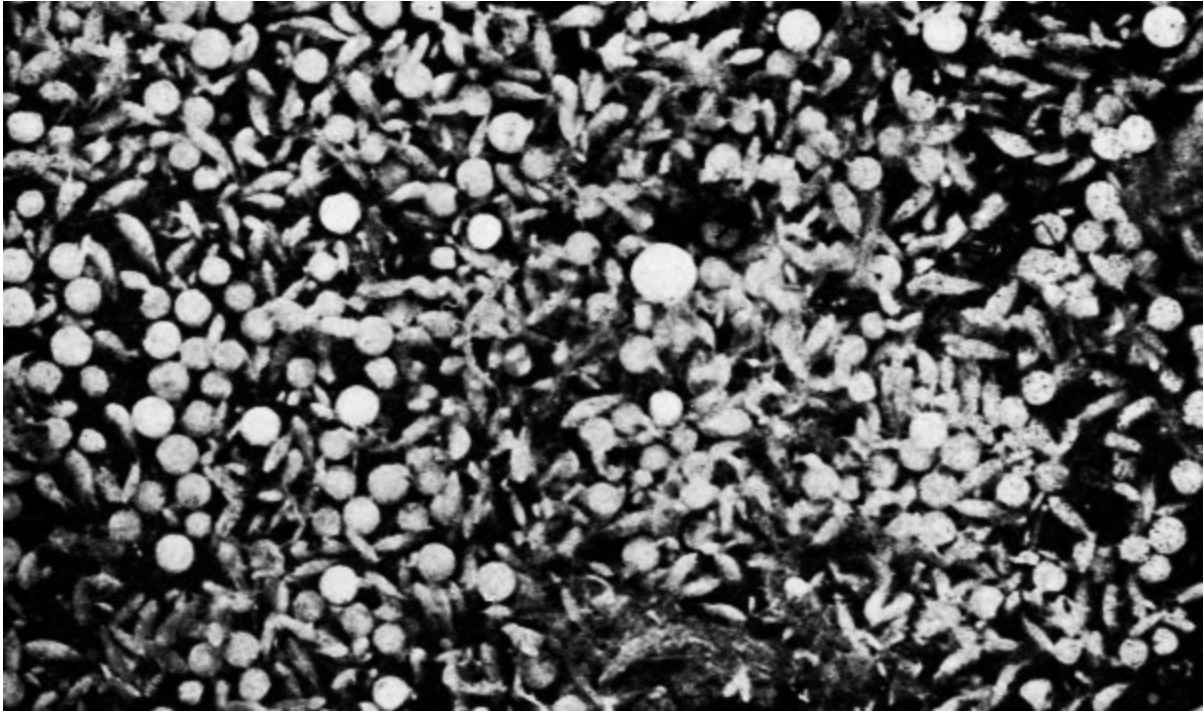


FIG. 8
PLANKTON CONTAINING FISH EGGS—×3

The large egg is that of a plaice: the smaller ones are cod and whiting.
The copepod is a calanus.

The eggs of the cod, whiting, haddock, fluke, plaice, etc., are relatively small, varying from $\frac{1}{6}$ of an inch in the case of a halibut, to $\frac{1}{25}$ of an inch in a flounder. The eggs are discharged into the water by the female. This process takes place gradually, and generally occupies many weeks. A few of the eggs come to maturity at a time, and are extruded. They are fertilized in the water by the spermatazoa of the male, which are discharged into the water at the same time as the eggs. The fish, both male and female, are closely crowded together on the spawning grounds, so that the fertilization of the eggs is fairly complete. With few exceptions, the eggs of most species are buoyant and float to the surface, where they drift in the warm surface water until, happily, they hatch. Unhappily, however, a very large

proportion of them never reach maturity, for, either as eggs, embryos or larvae, or post larval young fishes, they soon fall a prey to marauding fish. It is estimated that, of the thirty-two thousand eggs laid annually by each female herring, not more than two reach maturity.

The spawning grounds of the herring are not definitely known. Research is being carried out at present with a view to solving this question. Haddock are to be caught in various likely parts of the sea, marked with the place of capture, and their interiors examined for herring spawn.

Certain demersal fish, notably shark, dog fish and skate, deposit a few large, demersal eggs—about a dozen in the year—in a carefully selected spot. The incubation period of these eggs is unusually long, being from six months to over a year, according to the species and the temperature of the water.

Parental care is exhibited by very few fishes in this part of the world, although many foreign fish build nests and care for their young, often carrying them in their mouths. Certain kinds of dog fish and angel fish keep their young inside their oviducts until they are completely formed. The only notable example of a fish common to British waters that exercises parental care is the stickleback. Spawn is deposited by a number of different females in a nest constructed of stones and weed, and is guarded by a male until all the eggs are hatched.

The eggs of the crustacea, for example the lobster, are found attached in large numbers to the swimmerets—feathery processes that are situated underneath the tail. When in this condition, the lobster is known as “berried,” and, if captured, should be returned to the sea. The eggs are sticky and are laid while the lobster lies on her back, and so become attached to the hairs of these feathery processes. Berried crabs, prawns and shrimps may also be observed on the seashore in the spring and early summer.

The mollusca, e.g. mussels, periwinkles, oysters, deposit their eggs in the sea-water. The eggs float to the surface, hatch out, and drift about with the other constituents of the plankton. The fully developed larvae fall to the sea bottom and become attached to seaweed and stones.

The period of incubation of fish eggs varies according to the species of fish, and for the same species is prolonged by a low temperature. Plaice eggs, fertilized in January, hatched in eighteen days; others, fertilized in April, were hatched in nine days.

All fish, on emerging from the egg, enter upon a larval stage in which they resemble each other very closely (see [Fig. 1](#)). (Thus, the larvae of plaice are quite symmetrical, like those of the cod or other round fish.) The newly hatched larvae drift helpless in the water for two or three weeks, during which time they subsist upon the contents of the yolk sac, which they carry attached to their ventral surface. When this is exhausted, they feed upon the microscopic plankton which abound in the surrounding water.

The characteristic forms of the different species of flat fish are gradually assumed by the young fish during the period of their larval development. The appearance of a newly-hatched young plaice exhibits little change during the first week or so, other than that due to the gradual disappearance of the yolk sac. The young fish grows very slowly, and, twenty-one days after hatching, is only $\frac{3}{8}$ of an inch in length. For thirty days the development of the young fish is entirely symmetrical. During the succeeding fifteen days, the shape and appearance of the fish become profoundly modified. The left eye gradually moves upwards and forwards, until it attains its final position above and in front of the right eye. At the same time, the fish gradually acquires a new swimming position, finally swimming on what is really its left side. This left side becomes colourless. With these changes in form and habit, there proceeds a transformation in the diet of the fish. At twenty-one days it feeds upon the young stages of various crustacea. Gradually it acquires a taste for copepoda and the larvae of mollusca and crustacea. After its metamorphosis is complete, it feeds upon various worms, small shrimps and small, bottom-living crustacea. The adult plaice feeds upon mollusca of the cockle and mussel families.

The Migration of Fishes. Fishes, like birds, migrate over great distances at certain seasons of the year. In most cases, this migration occurs just before spawning, and is evidently connected directly with the spawning instinct. True marine fishes, such as the herring, haddock, plaice, cod, associate in vast numbers at spawning time, choosing a locality in which the temperature and food supply will be favourable to the development of the young larvae. Generally, the spawning ground is in deep water. The eggs

are buoyant, and drift up to the warm surface water and hatch out amongst the plankton. The herring differs from most other pelagic fish in laying its eggs in relatively shallow water, over a rocky bottom covered with seaweed. The eggs are denser than sea-water and are covered with an adhesive substance, so that they sink to the bottom and become attached to the stones and seaweed.

It is at the time of this annual migration to the spawning grounds that the fish are most profitably caught, for not only are they gathered together in large numbers, but, just before spawning, the fat content and general condition of the fish, and therefore its food value, reach a maximum. After spawning, the food value of the fish is at a minimum, and remains comparatively low until a few months before the next spawning.

The plaice migrates in the autumn from the feeding grounds in various parts of the North Sea to the spawning grounds near the Straits of Dover. Spawning takes place between December and March. In the spring and summer it returns northwards to the feeding grounds in the centre of the North Sea.

In the Irish Sea, there are two distinct annual migrations of plaice. The first occurs in summer (from June to September), the larger plaice moving from the warmer, shallow water inshore to the deeper, cooler waters offshore. In winter and spring, (from October to May), the mature plaice migrate from Morecambe and Liverpool bays to the spawning ground in deep water to the North-East of Douglas (Isle of Man).

In winter, also (from November to January), a large number of plaice gather in Red Wharf Bay, off the north coast of Anglesey, probably because it is sheltered from the prevailing south-east winds. In February they commence their spawning migration round the coast of Anglesey to Cardigan Bay.

Certain species of fish, instead of migrating from one part of the sea to another, migrate from the sea to rivers (anadromous), or from rivers to the sea (katadromous).

Thus, in the spring or autumn, according to species, the anadromous salmon and shad ascend rivers to spawn. The eggs are deposited on clean gravel in clean water, where they are likely to remain undisturbed. The

salmon does not feed when in the river, and after spawning, becomes very thin and in poor condition.

The Alaskan salmon, from which the bulk of American canned salmon comes, exists in five species. It has a similar spawning habit to the British salmon, except that the same species always tends to use the same rivers. Once having spawned, the fish dies, so that the parents never see their offspring. The young larvae hatch out in the fresh water and make their way to the sea, where they pass the whole of their lives until they are mature, some years later, and then, in their turn, ascend the rivers to spawn.

Eels are normally fresh-water fish. After living for six or seven years in rivers and ponds and streams, they become mature and migrate to the sea to spawn. This spawning always takes place in deep water (over five hundred fathoms), the particular region chosen depending upon the species. Eels from the British Isles and North-West Europe spawn in deep Atlantic, some hundreds of miles west of Ireland. In the autumn, the mature eels move down the rivers to the sea. When approaching maturity, the yellowish coat of the eel changes to silver. These “silver” eels pass into the sea and are never seen again. It is probable that the eel only spawns once in its life and then dies. The spawn floats to the surface and hatches out into curious little transparent, leaf-shaped larvae. These larvae develop rapidly into elvers and commence the return journey to the shores and rivers. In the spring, the young eels ascend the rivers in enormous swarms. Many of them leave the rivers and travel over damp ground and grass to isolated pools and lakes. It is probable that the eels that are found in the Thames travelled overland from the Severn.

The Baltic flounder migrates in winter from rivers and estuaries to the open sea, and spawns in spring in deep water. It returns in the summer when the spawning is over. By observing the movements of marked fish, it has been shown that the fish move at an average rate of from three to four miles per day. During its seaward migration, the flounder takes no food, but uses the material stored up in its tissues for the development of its reproductive organs.

In addition to these spawning migrations, there are migrations that are prompted by a search for food, or for warmer or colder water.

In northern and temperate seas, the surface water grows warmer with the spring. This warming influence spreads northwards from the equator, producing what is known as the annual wave of sea temperature. A direct result of the rise of temperature and the increased sunshine is a rapid increase in the amount and quality of the plankton. It is not surprising, therefore, that fish migrate in the wake of this annual wave of sea temperature, attracted by the increased food supply, and possibly, also, by the warmer water.

The mackerel is a southern fish, and prefers the warm water of the Mediterranean and West African coast. In spring, as the wave of rising sea temperature travels northwards, it migrates to the English Channel and the North Sea. This migration is often directly associated with the presence, in large quantities at that season, of a particular kind of copepod in the surface water of the English Channel.

Phosphorescence. Many marine creatures, ranging from deep-sea fish living in the dark abysses of the ocean to various species of the minute plankton drifting in the surface water, possess phosphorescent organs, which emit light of low intensity similar to that of a glow-worm and firefly. In many cases the light appears to possess some important function, and highly specialized organs are developed. In such cases the light is only emitted in response to some stimulus—thus, the phosphorescence of the surface water of the sea, when disturbed by the blade of an oar, is due to the disturbance of myriads of minute planktonic organisms, equipped with phosphorescent organs, either protozoa or protophyta; many pelagic copepods are phosphorescent. In other cases, phosphorescence appears to be a more or less accidental by-product of some other process, and of little or no significance. The substance which produces the glow is contained in the slimy secretion produced by the epidermal glands of the fish, and, as phosphorescence can only occur in the presence of oxygen, it is evident that the light is produced by the slow oxidation of this substance. The colour of the light emitted by marine organisms is generally blue or light green, but red and lilac also have been observed. The distribution and colour of the light or lights produced by individual fish vary with the different species. In many cases it would appear that these points of light provide the means by which fish recognize each other in the dark depths of the ocean. Some fishes possess highly developed phosphorescent organs known as

photophores, consisting essentially of a group of gland cells that secrete the phosphorescent fluid. These organs are generally distributed in rows along the sides and ventral surface of the fish. Some fishes possess more complex and highly developed organs containing, in addition to the gland cells, a system of blood vessels and nerves, a transparent, protecting membrane and reflector, an iris-like diaphragm and a lens. These more complex organs are generally larger and less numerous than the simpler ones. Possibly they are used to search for, or to attract, prey.

The phosphorescence of decaying fish and meat is due to the presence on the fish or meat of certain bacteria of putrefaction, which are themselves phosphorescent. When seen under the microscope, the individual bacteria appear as shining points of light.



CHAPTER III

METHODS OF FISHING

FISH may be captured with spear, trap, line or net. Which of these methods is employed necessarily depends very much upon the size and habits of the fish, and upon the skill and available equipment of the fishermen.

Spears and traps were used in prehistoric times and survive to this day in various forms, e.g. harpoons, lobster pots, hedge baulks, fishing weirs and the various ingenious traps and entanglements that are used by primitive races in all parts of the world. The logical development of the spear and the trap into the line and the net was made possible by the invention of string.

To design and construct a trap, it is generally necessary to know something of the habits of the fish to be caught. Hedge-baulks and fishing weirs are fairly extensive enclosures made of brushwood, basket work, stakes or stones, constructed on the foreshore in such a way that at high tide the sea carries the fish into the enclosure and leaves them there when it recedes. These fishing weirs are probably the primitive origin of most forms of fishing nets.

The crab or lobster pot or creel is constructed of basket-work, in shape somewhat like a safety inkpot, so that the lobster or crab can easily enter it, but, once in, is unable to escape. Lobster pots, suitably baited with fish and weighted, are distributed over the fishing ground—a rocky bottom full of crevices—from small, open boats, and are gathered the next day.

Fishing with hook and line is also a very ancient method. Before the discovery of metals, the hooks were made of bone. Some people—notably the Chinese—frequently use unbaited hooks, and rely upon the jerk of the hook at the right moment to secure the fish. Generally, however, the hook is suitably baited, the method being used chiefly for fish that seek their food by scent or sight, e.g. cod and shark. Line fishing for cod is still employed on a large scale off the North of Scotland and the coast of Newfoundland.

In lining, the fish are caught individually. A “line” may be as much as seven miles long. Short pieces of line from two to three feet long are attached to it at regular intervals. These lines are called the “snoods,” and

carry the hooks. The line is usually shot at night, and fished in the morning. In most cases line fishing is rapidly being superseded by trawling.

The invention of netting marked a notable advance in the primitive development of the fishing industry. The net in all its various forms and applications is the characteristic and all-important implement of the fishing industry. A net may be used either to surround a fish and drag it out of the water, as in seining or trawling, or it may be used to enmesh the fish, as in drift netting. The rise and development of the sea fishing industry has been due very largely to the gradually improved efficiency of the net.

Nets were originally used on the shore. A long strip of netting was attached to upright stakes, to form an enclosure with an opening towards the sea, constructed like a fishing weir in such a way that the fish enter the enclosure at high tide and are unable to escape. Such devices constructed on shore are known as “fixed engines”; they include stake nets, poke nets, stream nets and purse nets. The net may simply form the wall of an enclosure (stake net). This enclosure may be furnished with a pocket at one corner (poke net). It may consist essentially of one long, deep pocket kept open by rings or stakes at intervals (purse and hose nets). It may be simply a wall of netting into which the fish thrust their heads; owing to their gill openings they are unable to withdraw and so become entangled (stream net).

The first development of a movable net was the seine or drag net. The seine is a semi-circular drag net, which is shot in shallow water so as to enclose an area of water close to the shore. It is then hauled ashore, and gathers up the fish that are in the enclosed area of water. Such a net is limited to inshore use. Generally, a line is attached to each end of the net. The free end of one of these lines is made fast to the shore by a stake, and the net is paid out from a small boat. When the whole of the net has been paid out, the boat travels round until the net forms a semi-circle of which the diameter is parallel to the shore; the net is then hauled in.

The seine net was used in ancient times by Phoenicians, Greeks, and other Mediterranean peoples. Various types of seines are in common use today. In Denmark a seine net is employed to catch eels and plaice. On the Cornish coast pilchards are caught with a large seine up to two hundred fathoms long and eight fathoms deep. In the United States a seine is used in

water of any depth to catch mackerel. Rings are attached to the foot-rope of the net, and by passing a line through these rings and drawing it tight, the net is transformed into a bowl of netting. This is called the purse seine.



FIG. 9
TRAWLING (*circa 1750*)

The seine was first improved by the addition of a pocket at its centre. Then the sides or wings were gradually lengthened, until finally it developed into a deep, conical, bag-shaped net, furnished with long arms or wings. This was dragged along the bottom, behind a boat in full sail. The net was weighted and its mouth kept open by attaching its upper edge to a beam of wood (beam trawl). When the net was full of fish, it was run ashore. Ultimately, instead of drawing the net ashore, the fishermen remained at sea and hauled the net on board with a winch. In this way the seine net gradually developed into the trawl net. The trawl net marked a big improvement, for it could be fished in deeper water further from shore, and thus greatly increased the scope of fishing operations, and led to the rapid growth and improvement of demersal fishing.

Trawling is said to have been invented at the end of the seventeenth century by the Brixham fishermen. The first trawlers were quite small

vessels, and were followed towards the end of the eighteenth century by the smack. The smack reached its maximum size and efficiency at about the middle of the nineteenth century. Some of the smacks that are still fishing from Brixham—durable, seaworthy, and with beautiful lines—are probably a hundred years old.

In 1870, there were a thousand first-class smacks in the North Sea, three hundred in the English Channel, and over a hundred in the Irish Sea.

The smacks were fitted with a tank in the well of the ship, in which the fish were kept in sea-water and brought in alive. In Denmark to-day, plaice are brought ashore and sold alive.

The subsequent development of trawl fishing has been in the construction of larger nets, worked by more powerful trawling vessels driven by steam.

The size of beam trawl that can be worked by a large sailing smack is limited by the trawling power of the vessel, and also by the difficulty of constructing and handling very long beams. The maximum length of beam in general use by sailing smacks is fifty feet. The length of the net, from its mouth to the narrow of “cod” end, rarely exceeds a hundred feet. To each end of the beam is attached a triangular trawl-head of iron, which moves along the ground and serves to keep the beam about three and a half feet above the ground. These trawl crossheads are attached to the ship by bridles and warp.

The upper edge of the net is attached to the beam, the lower edge being attached to a stout rope—the foot-rope—the ends of which are made fast to the crossheads. This foot-rope, being considerably longer than the beam, sweeps along the ground abaft of the beam, to form a deep curve known as the “bosom” of the net. The result is that, when the foot-rope disturbs the fish so that they leap to avoid it, the beam has passed on overhead and they leap into the net.

Pockets are formed in the sides of the net by lacing the top and bottom together for about two-thirds of the distance from the mouth of the net towards the cod end. The mouth of a pocket is at the cod end of the net, so that fish reaching the cod end and attempting to return to the mouth of the net, generally enter the pockets. A flap of netting suspended some distance

inside the mouth of the net serves as a valve. It is easily lifted by the incoming fish, but tends to prevent their escape.

The netting is of hemp, the mesh gradually increasing from one inch at the cod end to about two inches near the mouth, and is preserved with tar.

When fishing, the vessel moves ahead at a steady, slow rate of from two to three miles per hour, dragging the trawl behind it. Smacks always trawl with the tide. If they trawl against the tide, the net is lifted from the ground.

During fishing the cod end is closed by the cod line, but at the conclusion of the trawl the net is hoisted aboard, mouth upwards, and the contents are discharged upon the deck by drawing the cod line.

The otter trawl that is used by modern steam trawlers is from seventy to one hundred and twenty feet wide across the mouth, according to the character of the fishing, and a hundred and ten feet long from the mouth to the cod end. The otter trawl is shown in [Fig. 17](#). It differs from the beam trawl in that its mouth is kept open, not by being attached to a beam, but by otter boards, which are attached one to each side of the mouth of the net. These are attached to the net and to the warps by which the net is towed in such a way that the pressure of the water upon them causes them to diverge, thus keeping the mouth of the net open. The size of a beam trawl is necessarily limited by the length of beam obtainable. The size of the otter trawl, however, is obviously only limited by the power of the steam trawler. The otter boards measure 11 ft. by 4 ft. 6 ins., are shod with iron, and weigh 15 cwts. each. The warps, as the ropes are called which attach the otter boards to the ship, are from three hundred to a thousand fathoms long—generally a little over three times as long as the depth of the water in which the trawl is to be used. Each board is attached to the steamer by a separate warp. The upper edge of the mouth of the net is attached to a strong rope, called the “head” rope. The lower edge of the mouth of the net is also attached to a strong rope, called the “foot” rope. As in the beam trawl, the foot rope is considerably longer than the head line, and forms a bosom. Traps and pockets also are inserted in the sides of the net. When trawling on rough ground, the foot rope is furnished with large, heavy, wooden rollers, called the “bobbins.”

Trawl fishing, until quite recently, was almost entirely confined to demersal fish, such as cod, plaice, haddock and halibut. In recent years,

however, considerable quantities of herring have been caught by trawlers.

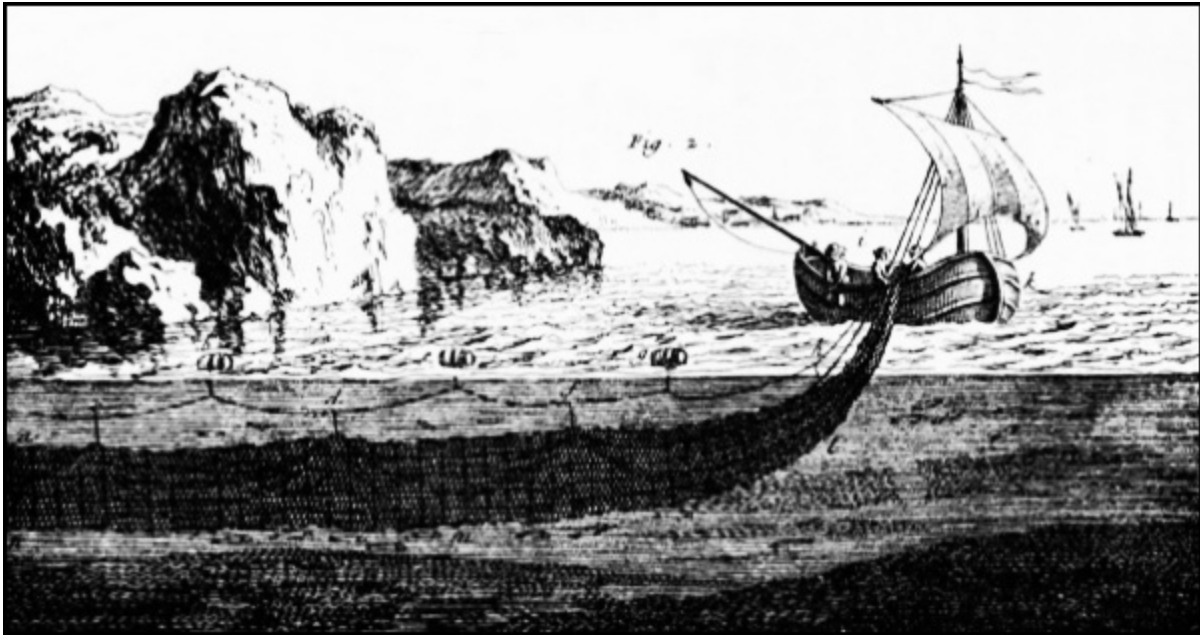


FIG. 10
DRIFTING (*circa* 1750)

Drifting. The drift net is essentially a completely submerged, vertical curtain of netting, one end of which is attached to a boat called a drifter. The net extends in a straight line from the boat, and may be as much as three miles long. Unlike the trawl net, the drift net generally catches one kind of fish only—either herring or mackerel—drift net fishing being carried on at a time when these fish come together in shoals near the surface for the purpose of spawning. The trawl obviously only captures fish living at the bottom. At the same time, of course, it captures all the fish at the bottom, whether immature, or useless star fish, etc. The drift net, on the other hand, is generally used for a particular kind of fish—herring, mackerel, sprat—and only catches fish above a certain size.

A drifter may be as much as 90 ft. long, with 20 ft. beam and 10 ft. draught. Its foremast is so constructed that it may be lowered when the vessel is steaming against a head wind, or when it is fishing. The ordinary sailing drifter is rapidly being superseded by the steam drifter, partly because the greater power of the steam driven boat increases its capacity

and scope, and, further, owing to the centralization of the industry at a few big ports at certain times of the year, these harbours are so crowded that it is almost impossible to handle a sailing drifter in them. Many of the larger sailing drifters have been equipped with petrol engines which largely discount this disadvantage. A steam drifter can travel at from 11 to 12 knots, and both steamers and sailers carry a fishing crew of seven men and a boy.



FIG. 11
A SINGLE-BOATER AT FOLKESTONE

Inshore Fisheries. The development of steam fishing—trawling and drifting—has resulted in the re-grouping of the fishing industry into two well-marked divisions. Fisheries, whether trawling, drifting or lining, that are carried on in deep water far from shore in large steamers, for the most part owned by limited liability companies, are known as offshore fisheries. The fisheries of the seashore, carried on by small, privately-owned, sailing smacks and cutters within territorial waters, are distinguished by the term “inshore fisheries.” The inshore fisheries are mainly for shellfish, crabs, lobsters, shrimps and immature deep sea fish such as plaice, soles, flounders, dabs, codling and sprats.

Shrimps and whiting are caught with trawl nets of 25 ft. beam or less, and of about $\frac{1}{4}$ in. mesh. The net is generally drawn behind a small cutter, but frequently it is used in shallow water with a horse and cart. These nets are generally made of flax or cotton, and are either tanned or tarred, in order to preserve them.

Smaller, fine-meshed, trawl nets are used for catching shrimps and also immature plaice, soles and dabs. These shrimp nets are either attached to a long handle and pushed through the water in front of the fisherman (push nets), or drawn behind a small boat or a horse and cart (trawl nets).

Larger fish are sometimes caught in shallow water by casting a net over the fish so as to enclose it (cast nets). The fisherman of the Eastern Mediterranean uses a cast net with conspicuous skill. The net is essentially a circular disc of netting, to the circumference of which small weights are attached at regular intervals. A cord is attached to the centre of the net, and the fisherman, standing knee-deep in the water, grasps the net by its centre, swinging it round his head, and casts it so that as it approaches the water it opens out, and with a soft splash sinks through the water until it lies outstretched over the fish. It is then drawn up by the string attached to its centre, and the weighted edges fall together enclosing the fish.

Fish are often caught on shores and in rivers by causing them to pass between converging walls of stakes or basket work, until they enter an enclosure, the floor of which is covered by a net. When the fish have gathered in the enclosure, the net is pulled up.

The simplest form of inshore fishery is that for periwinkles, in which they are simply picked off the rock. Mussels live on the sea bottom, on the lower half of the foreshore. They generally attach themselves to a stone by a thread. They are usually collected at low tide by hand or, when submerged, are raked from the bottom. The rake is from 2 to 3 ft. wide, and is furnished with teeth 10 ins. long, the back of the rake being covered with netting. Sometimes the mussels are submerged even at low water and then a short rake is used.

Cockles live about an inch or so below the surface of the sand, and maintain a connection with the water above by means of small tunnels in the sand. They occur abundantly in many places between high and low watermark. When the cockles are abundant they are raked out of the sand,

the rake being from 10 ins. to 1 ft. wide, with teeth 1 in. long. The cockles are riddled, the small ones being rejected. When the cockles do not exist in such large numbers, they are obtained by means of a "jumbo." This is essentially a block of wood, 3 or 4 ft. long, and 1 ft. wide, furnished with two upright handles. The jumbo is rocked to and fro on the surface of the sand, with the result that the cockles are gradually worked up to the surface.



CHAPTER IV

THE HERRING FISHING INDUSTRY

HERRINGS abound in the waters round the coast of Great Britain. Ordinarily they are widely scattered in deep water, but at certain times of the year they come together in shoals in the warmer water near the surface for the purpose of spawning. It is at this time that they are of greatest value for food purposes and, being gathered together in shoals, are most economically caught.

The herring may spawn at any time of the year. In this respect it differs from all other British marine food fishes. Most British caught herrings spawn during September and Autumn. Very little spawning takes place during late winter and spring, i.e. just after minimum sea temperature. Each local race (or species) appears to spawn at a constant time of the year. The date of the annual spawning, and hence the herring fishing season, varies from point to point round the coast. Herrings caught at different places show well-marked differences in appearance and quality, which are evidently due to differences in species and feeding ground. The food value of the herring will depend also upon the time of the year at which spawning occurs. Thus, in the Irish Sea, there are two races of herrings—the Manx and the Welsh. The Manx herring spawns in summer (September), and is rich in fat; the Welsh herring spawns in winter (November and December), and is poor in fat. Herrings are first caught off the West coast of Scotland in the waters round the Hebrides. This fishing begins in the middle of May, its chief centre being Stornoway. In early June herrings are caught in the waters round the Orkneys and Shetlands, and then in succession off Wick, Fraserburgh and Peterhead, and the Northumberland coast (Eyemouth, Berwick and Sea Houses). About the middle of July the herring fishery season begins at Blyth and Shields, and at Scarborough and Grimsby towards the end of July. At Yarmouth and Lowestoft it begins early in October. The last herrings to be caught in British waters are caught round Devon and Cornwall in December.

Of the various kinds of herring obtained at different places, the largest and finest fish are those caught in Downings Bay off the North of Ireland,

Castle Bay off the Island of Barra in the South Hebrides, and off the Shetlands. Herrings differ very much in their suitability for handling, keeping and curing. Most herrings have a small gut which is easily removed without seriously damaging the body of the fish. Blyth and Shields herrings, however, are very rich and fat, and have a specially big, distended gut. Such herrings are difficult to clean because, when this large gut is removed, the belly of the fish is so tender that it is often broken. Herrings caught off these ports are fat and oily, so that many are landed in a broken condition. The Yarmouth herring is firm and hard, and is the best adapted for handling and curing.

Unlike that of the cod, the flesh of the herring is very rich in oil and fat. The body flesh of the herring consists essentially of two well-developed layers of adipose tissue, alternating with two layers of muscular tissue. The fat in this adipose tissue is very liquid and oily, and tends to make the fish tender. The actual amount of body fat varies widely throughout the year. It gradually rises to a maximum before spawning takes place, and diminishes slightly before spawning and afterwards rapidly to a minimum. Thus, the fat content of Manx summer herrings is about 2 per cent during the winter, and rises rapidly in June and July, until in August, just before spawning, it is over 30 per cent. The herring has a small liver which also contains some oil.

Fishing is carried out with drifters. Practically all drifters to-day are steam-driven, although recently a number of motor-driven drifters have come into use. Motor-driven drifters are mostly sailing boats converted. Each drifter carries a crew of seven men, including the skipper and engineer. The boats are largely privately owned and the crew work on a share basis. A number of boats are owned by companies.

The boats from the various fishing ports work round the coast, following the fishing from port to port. At Yarmouth during the fishery season there are about 1,200 drifters from nearly all the fishing ports round the coast. Stornoway, Wick, Fraserburgh, Peterhead, Aberdeen, Berwick, Whitby, and Yarmouth are all well represented.

Each boat carries from 70 to 80 nets. The nets are approximately 1 in. mesh. Each net is essentially a long rectangular curtain, hanging vertically in the water. Its upper edge, which is about 55 yds. long, is buoyed up by about 80 to 84 corks distributed equidistantly along it from end to end. The

net is about 6 yds. wide. Each net hangs with its upper edge about 2 fathoms below the surface of the water, being attached at each corner to two pellets or bladders, resembling large footballs, and serving as floats.

Fishing nets and sails are often coated with warm gelatine, and then immersed in a strong solution of tannin. This renders the gelatine insoluble and preserves the nets against the attacks of destructive organisms.



FIG. 12
HERRING DRIFTER

When fishing, the boat takes up a position stern on to the tide. The nets are paid out over the bow and connected up in line, and carried by the tide till they form one long line, one end of which is attached to the drifter. The position of the nets is indicated by the line of bladder floats.

The fish swim against the nets, push their heads through, and then, owing to their gill openings, find that they cannot withdraw their heads, and in this way are caught in enormous numbers. Generally, fishing goes on all night, and in the morning the nets are hauled in, and, together with the attached fish, are thrown into the hold situated amidships. The drifters then return

with all possible speed to the fish wharf. While the boats are returning to port, the men draw the nets from the hold and shake them free from any entangled fish. When the drifter reaches port, she moors alongside the fish wharf, bow on, and unloads her cargo of fish, using her derrick mast. The fish are unloaded in a round basket which is stamped by the Fishery Board's officer as holding a quarter of a "cran." The word "cran" is derived from the "crown" branded by the Fishery Board's officer on each of the two wooden shafts in the basket.

The cran is the measure which is universally used in the trade. At Yarmouth and Lowestoft originally herrings were counted out and sold by the "last." A cran averages from 900 to 1,000 herrings and weighs approximately 3 cwts. A "last" equals ten crans, and originally consisted of 13,200 herrings, counted out. This method, of course, was too slow and has now been abandoned.



FIG. 13
CURING YARD AT YARMOUTH

The herrings, as they are removed from the ship, are put into special baskets called "swills," each swill holding half a cran. The swills containing

the day's catch are arranged in rows on the fish wharf, opposite each drifter. It is a great sight to see about four or five hundred drifters lying, bow on, alongside the fish wharf for about 2½ miles, all unloading fish as fast as they can.

A good day's catch would consist of about 90 crans. A good catch, therefore, would average about 100,000 herrings, and would weigh about 13 tons. Some boats come in with as many as 160 crans of fish, and the total "cranage" for a day may exceed 30,000. The total catch for Yarmouth on a good day would be about 30,000,000 herrings, weighing about 4,000 tons.

Sometimes when the catch has been poor, the drifters remain out on the fishing grounds for another day, rather than come home with a small catch. In this case, the two catches are kept separate, the first catch being called "overdays." Overdays are worth about half the price of fresh fish and are, of course, less suitable for high grade curing.

After it has been purchased by the curer, the fresh herring may develop into a salted herring, a red herring, a bloater, or a kipper, depending upon the degree of salting and smoking to which it is subjected. Herrings are sometimes put into cold storage, to be withdrawn subsequently as occasion demands, either to be salted or, more frequently, to be consumed fresh. Cold storage affords a convenient method of preserving herrings when there is a glut, for at such times it is often impossible to deal with the herrings adequately in the ordinary curing yards.

Salted Herrings. The fresh herrings are delivered to the curer's yards. Here, the fish are emptied into broad, shallow troughs, which generally run from end to end of the yard. The troughs are about 4 ft. wide, and are generally made of wood and arranged at a convenient working height. Usually, the trough is situated just inside the boundary wall, and the fish are delivered into it through large openings in the wall.



FIG. 14
SCOTTISH FISHER GIRLS

The fish are gutted and salted by Scottish girls—many of them from the Hebrides—who come to Yarmouth and other places in the season for this purpose. These girls are all brought up in Scottish villages, and are extraordinarily expert in all the operations connected with the cleaning and salting of the fish. They work in crews of three, and take very good care that each member of the crew is a good worker, as they are paid according to the amount of work they do.

Each girl receives 25s. a week as a kind of subsistence allowance, and is paid 1s. a barrel for the work she does.

As the fish are delivered into the gutting trough, they are liberally sprinkled with salt, thus enabling the women to grasp the fish easily, as otherwise the fish are too slippery for quick handling.

The women work standing in a row beside the trough. They pick up a fish, gut it by inserting a sharp knife just below and behind the gills, and with a quick, upward cut, bring away the gut. The guts drop into small tubs placed in front of each worker, and are collected periodically and sold to manufacturers of manure. Behind each woman are three shallow tubs or baskets, and after she has gutted a fish, she throws it behind her into one of the three tubs, according to its quality and size. In this way, the two operations of gutting and selecting the fish are combined. As the tubs of gutted fish become filled, they are taken away by other girls to the barrel packers, and are packed in separate barrels, according to quality or size. The barrels are arranged in long rows, generally parallel to, and at some distance behind, the gutting trough. A girl will pack about three barrels in an hour.

The gutted fish are first of all emptied into large, shallow tubs called "rousing tubs," placed just behind the row of barrels, and are again sprinkled with salt.

The packer takes an armful of fish from the rousing tub and drops them into the barrel. Each time the fish are taken from the rousing tub the contents of the tub are well stirred up. The fish are then packed in the barrel in layers, bellies upward, and each layer is liberally sprinkled with salt. In this way each individual fish is first of all thickly coated with salt in the rousing tub, and adjacent layers of fish in the barrel are also separated by a layer of salt. In this packing process, it is important that the fishery salt used should be coarse, reasonably hard, slow in dissolving and present in considerable excess. It should be coarse enough to prevent the fish from touching each other, thus enabling the brine to penetrate to every part. It should be hard enough to withstand the pressure of the fish in the barrel. It should dissolve slowly, so that the salting process takes place gradually, enough salt remaining undissolved throughout the process to keep the fish from touching. Altogether, about 1 cwt. of salt is used for each barrel of herrings cured.

The barrel, when fully packed, is covered over and left for about eight days. During this time, the salt extracts water from the fish and dissolves in

it to form a saturated brine. The efficiency of this salting process necessarily depends upon the salt being present in considerable excess, so that the brine formed is kept saturated, and consequently continues to withdraw water from the fish.

At the end of eight days, the barrels are opened, an inch hole is drilled in the side at the bilge, and the pickle allowed to run out. It is found that, owing to the withdrawal of water from them, the herrings have shrunk considerably, and some more salted herrings are added to the barrel, until it is full again. It is then fastened down permanently, turned over on its side and filled with brine pickle, and corked up.

The brine pickle which is formed during the eight days is not allowed to run to waste, but is used for filling up the barrels after they have been repacked. This brine pickle contains amino bases, together with small quantities of coagulable proteids, and is of distinct nutritive value. The Poles and Russians, who are great consumers of these salted herrings, actually use the pickle as a kind of sauce or gravy, dipping their bread in it. This, together with the general demand for salted herrings in these two countries, may very largely be due to the comparative scarcity and high price of salt there.

A cran of herrings (about 1,000 fish, weighing approximately 3 cwts.) uses up 1 cwt. of salt and, when completely salted, just fills a barrel. The curer estimates that 5 to 6 tons of salt will be sufficient for 100 crans of herrings. Herrings salted in this proportion should be exported and consumed before the warm weather comes, as they are liable to decay if the temperature rises above 70° F. The herrings that were packed for the British Government (1920-1921) were salted more heavily than usual (7 to 8 tons of salt per 100 crans), as, owing to the uncertain condition of the Russian and German markets, it was necessary to keep some of the fish in stock for a considerable time. Such a heavily-salted fish would be unpalatable to the home consumer.

In Yarmouth and Lowestoft, and also in Scotland, 100 crans of herrings should fill, when cured, from 125 to 130 barrels.

Herrings are sometimes salted at sea, 1 ton of salt being used to each last (10 crans) of herrings. Such herrings are mostly used to make "red herrings."

Red Herrings. A considerable trade in red herrings is done with the Mediterranean and the Levant. For this trade, the fish must be thoroughly smoke-cured, otherwise they will not keep in the comparatively warm climate. The fish are first of all dry-salted in concrete tanks about 10 ft. square and 6 ft. deep, arranged under the floor of the curing house. Fresh fish and salt are simply thrown in and mixed up, and left to develop their own pickle.

Generally speaking, 1 ton of salt is used to 10 crans of herrings, and each tank will hold from 20 to 30 crans of the fish. The fish should be left in these salting tanks for five days at least; sometimes, of course, they are left for months, according to the trade, in which case the tanks practically serve as storage tanks for the salted fish. The fish are removed from the tank as required, washed, and put on “speets” and smoked. A “speet” is a wooden rod about 3 ft. 6 ins. long and pointed at one end. The fish are threaded on the speet through the gill openings and mouth, each speet holding from 20 to 30 fish. The speets are then stacked horizontally on racks in the smoke house “loves” (lofts), about 6 ins. apart and about 12 ins. above each other, until the smoke house is filled from the roof to within a few feet of the floor. When the smoke-house is filled, fires are lighted on the floor. Generally, the fuel used is oak turnings, shavings, and sawdust. This material burns quickly, and gives a very resinous smoke which not only dries the fish, but also permeates it thoroughly.

The rate of curing and the character of the finished product depend upon the temperature of the smoke, and the proportion of antiseptic resinous materials in it. When the oak or other suitable hard wood fuel is in the form of turnings or dust it burns quickly, and thus produces a fairly hot smoke, containing antiseptic substances—for example, guaiacol and creosol. Such a smoke will cure the fish quickly.

If oak billets or logs are used they burn comparatively slowly. The smoke, therefore, is not so hot and, since slow combustion in this case probably means more complete combustion, the proportion of resinous constituents in the smoke is liable to be considerably diminished. When oak billets are used, therefore, curing takes place much more slowly.

The temperature in the smoke house will also depend very much upon the prevailing weather temperature outside. In cold weather it is difficult to

keep the temperature up sufficiently. The curing takes longer, and results in a hard cured product. In very warm weather, on the other hand, it is difficult to keep the temperature down, and a “fired” fish is sometimes produced, i.e. one which is half-cooked and soft. Such a fish is clearly unsuitable for packing for export.

Generally speaking, the temperature of the smoke should be such that the curing takes about 10 days.

After smoking, the fish are taken off the speets and selected according to quality. Those which are large and perfect fetch a better price, and command an entirely different market from those which are damaged or broken.

During the smoking of red herrings, the fires are lit each night, and simply allowed to burn themselves out.

Bloaters. There are two kinds of bloaters: those intended for the home trade and those intended for the Mediterranean trade. For the home trade the herring is lightly salted by immersing it in brine for two hours or less. It is then dried in the smoke-house for one night, using billets. Unlike “reds” or kippers, it is not cured by the smoke, but simply dried. The bloaters for the Mediterranean trade are salted in concrete tanks in exactly the same way as red herrings, but, instead of being smoke-cured for 10 days or so, they are simply smoke-dried for two days.

Kippers. Kippering is the only process in the herring industry in which the fish are split before curing. Fresh herrings (sometimes over-day herrings) are bought early in the morning from the drifters and taken to the curing yard. They are split down the back, close to the backbone, and gutted and thrown into large, open baskets. The basket and its contents (about 50 herrings) are then plunged into a tank of running water, and violently agitated to wash blood and slime from the fish. The fish are then thrown into brine in large tanks about 6 ft. by 5 ft. by 4 ft., until the tank is full. Salt is then sprinkled on the surface, and the fish are left from half to one hour, according to their size.

They are then hung on kipper speets. A kipper speet differs from a bloater speet. It is a square bar of wood about $3\frac{1}{2}$ ft. long, and of 1 in. square cross-section. It is supported horizontally. The split herrings are opened out

and impaled upon hooks at intervals along each side of the speet. Each speet in this way will carry about eight or nine herrings a side. The speets are then stacked on racks in the “loves” of the smoke-house, are smoked over-night, using fires of oak turnings and sawdust, and are packed the next morning in boxes.

The herring is probably the most abundant food fish known. During the autumn herring fishery of 1920, over 1,000,000 crans of herrings were landed at Yarmouth and Lowestoft. If we assume that one cran measure contains 1,000 herrings, we see that over 1,000,000,000 herrings were caught in less than 4 months, and this probably represents only a small fraction of the number present on the fishing grounds. In 1913, 11,762,748 cwts. of herrings, of value £4,412,838, were landed in Great Britain. In the same year, the exports of herrings from the British Isles were as follows—

Fresh herrings	1,464,296 cwts.	worth	£1,212,493
Cured herrings	8,797,106	„ „	5,333,113
Total	<u>10,261,402</u>	„ „	<u>£6,545,606</u>

The quantity of herrings caught by other European countries is as follows

	cwts.	£
France (1911)	7,846,503	529,739
Germany (1913) Fresh	148,354	75,738
„ „ Salted	1,030,039	563,033
Holland (1911)	1,685,751	919,973
Norway (1912)	4,404,400	580,570
Denmark (1912)	845,295	140,051
Sweden (1912)	861,420	205,555
Belgium (1911)	13,000	5,000
	<u>16,834,762</u>	<u>3,019,659</u>

CHAPTER V

THE NEWFOUNDLAND COD FISHERY

THE cod is widely distributed in the northern and temperate seas of Europe and America. It lives close to the bottom, in from 25 to 50 fathoms of water, and feeds upon fish, small crustacea, worms and mollusca. The cod spawns in the Spring. Of the 4,000,000 or so eggs that are spawned by a single female cod, comparatively few are hatched, and fewer still reach maturity. The young are about 1 in. long by the beginning of the summer, and become fit for the market at the end of the second year. Usually, the fish are mature at the end of the third year, and then measure about 3 ft. in length, and weigh from 12 to 20 lbs. They are in the finest condition in October, November and December.

In addition to its great value as a food fish, the cod, like the sturgeon, yields isinglass (a pure fish gelatine) from its swimming bladder, and oil from its liver. Cod-liver oil is largely used as a remedy for scrofulous complaints—probably owing to its content of vitamins. It is also used effectively in cases of pulmonary consumption.

Cod is fished along the coasts of Newfoundland and Labrador, and on the Banks. The Banks stretch for about 300 miles in a south-east direction from the coast of Newfoundland towards the middle of the North Atlantic. They are swept by the cold Labrador current. A branch of the Gulf Stream passes over the southern portion of the Banks. These currents bring enormous quantities of plankton and small fish, which provide excellent food for the many varieties of fish and small, invertebrate, marine animals that inhabit the Banks. These, in their turn, provide abundant food for the cod.

The cod, together with other demersal fish, including haddock, hake and pollack, is caught with baited hooks and lines. This fishery has continued with unbroken prosperity for nearly four centuries. In addition to the Newfoundland boats, a large number of American boats set out for the Banks from Gloucester (Mass.). Most of the boats are sailing boats of about 35 tons capacity, and of sturdy construction. Each boat carries eight dories

—small row-boats about 15 ft. long—amidships. The crew consists of a captain and cook, and sixteen men—two for each dory.

The “Banks” stretch for about 300 miles, by 200 miles wide, in a southeasterly direction, towards the centre of the North Atlantic. The depths in which the fishing is carried on range from 20 to 120 fathoms off the coast of Newfoundland, from 15 to 90 fathoms on the Banks, and from 100 to 135 fathoms at the edge of the Banks. The vessel starts out for the fishing grounds with about 400 hogsheads of salt, and from 15,000 to 25,000 lbs. of bait. The bait is generally frozen squid and herring. Capelan is also used as bait, but has to be obtained at Miquelon, the last port of call before putting out to the Banks. The bait must be well iced, as the cod will not bite well if the bait be tainted.

During the second trip, squid is used as bait and is caught on the fishing grounds.

As the boat approaches the fishing grounds, the dories are made ready. Each dory carries four tubs of baited lines. A tub contains nine lines, each 50 fathoms long. When fishing, these lines are all strung together, so that each dory will run a string 1,800 fathoms long—about two miles. Each line carries about 90 hooks—that is, 3,200 hooks to each dory. A vessel with eight dories will thus set about 16 miles of line, carrying about 25,000 hooks. The hooks are attached to the lines by means of shorter lines called “gangings”—in Scotland they are known as “snoods”—about 2 ft. long. The complete line, as set by a dory, is called a “trawl.”

On arriving at the fishing grounds, soundings are made to determine the depth and character of the bottom. The best fishing is obtained over a gravel bottom. The trawls are then set while the vessel is in motion (a flying set), and if the fish are found to be abundant the vessel drops anchor.

The flying set is carried out as follows: The dories are towed astern and, when the right spot has been selected, are dropped at regular intervals until all are away. Each dory as it is dropped rows off at right angles to the course of the vessel, and in the same general direction, throwing out its trawl as it proceeds until it is all set. The vessel then returns diagonally across the fishing grounds to the starting point, picking up the dories as their trawls are set. After a time, the dories are dropped again in the same order as before, and the men haul up the trawls and take the fish off. Each

dory is then picked up in succession together with her catch. If this flying set is successful, and other conditions are favourable, the vessel drops her anchor and fishing proceeds.

The manner in which the trawls are set depends upon the tide. They are always set as far as possible with the tide. Thus, the dories on the side of the vessel against which the tide is flowing row out against the tide, until they are about a trawl-length from the ship. They then set the end of the trawl at the point, and work towards the vessel. On the other side of the vessel the trawl is set from the vessel with the tide towards the dory. Each end of the trawl is attached to an anchor by a line 1 fathom in length, and to a buoy by a line 25 fathoms longer than the depth of the water at that point. Thus, the trawl is situated just above the ground. The trawls are set once a day and drawn three hours afterwards, or set in the afternoon and drawn the following morning. The shorter the time between setting and drawing, the better the condition of the fish. In hauling the trawl, one man stands in the bow of the boat and hauls in the trawl, detaching the fish, the other man receiving the trawl and coiling it. A dory carries on an average 1,000 lbs. of fish, and may sometimes make two or three trips before the line is cleared.

The fish are “gaffed” from the dories to the fishing vessel and are kept on deck, packed between division boards to prevent sliding or turning of the fish by the movements of the vessel.

When the fish are all aboard, they are split and cleaned and salted down. The crew is divided into splitting gangs, each consisting of three men—the throater, the gutter, and the splitter. The throater grasps the fish by the head with the left hand, and, holding it with its back on the edge of a tub, cuts its throat just behind the gills, and makes a slit down the belly. The head is then broken off by downward pressure against the edge of the tub, and the fish is passed on to the gutter. He opens the belly with his left hand, removes the liver for oil, and tears out the viscera. The fish then goes to the splitter, who completes the ventral splitting of the fish and removes the backbone.

After being well washed, care being taken to remove all blood, the fish are passed down a canvas chute into the hold, where they are carefully salted and piled in “kenches.” The fish are laid on their backs alternately nape and tail, salt being liberally sprinkled between the adjacent layers.

Nearly 1½ bushels of salt are used per 100 lbs. of fish. The pickle formed by the salt and the juices of the fish drains away to the bottom of the hold, from which it is pumped overboard. As the kench or pile settles, more fish are added, so as to keep the compartment full. Kenching begins in the forward compartment of the hold, and is carried on from side to side of the vessel. Each kench is about 4 ft. by 7 ft., and the full height of the hold. The refuse is thrown overboard.

In addition to the “trawl” fishing, many boats use hand-lines. For this purpose, the lines are somewhat smaller, and only 13 ft. long. About 100 barrels of bait are taken (slack-salted clams obtained on the coast of Maine), any additional bait that may be required being caught on the fishing grounds—squids, hagdens, and clams taken from the stomachs of fish.

When the vessel reaches the fishing grounds, the dories row away in all directions, each man for himself. The dory is anchored in water from 18 to 40 fathoms deep. Each fisherman uses two lines carrying two hooks a piece. The boats generally go out at sunrise and return to the fishing boat about six hours later. Two boatloads—that is, 2,000 lbs. of fish—make a good day’s work.

On returning to the vessel the fish are pitched on deck and counted, only cod of over 22 ins. length being considered. Smaller fish, and the “shack”—pollack, haddock, cusk and hake—being counted separately. The fish are then dressed and salted, as already described.

In some cases, hand-line fishing is carried on from the deck of the fishing boat itself, while the boat drifts. Each man uses one line carrying two hooks. The bait consists of iced cockles, broken with a hammer. The positions on the deck are followed by the crew in rotation, to give all an equal chance. As the fish are “landed” they are thrown on to the deck, each man keeping his count by cutting out the tongues and keeping them in a separate bucket.

On the Georges Bank, south-east of Gloucester, which is one of the favourite fishing grounds, the fish are caught by hand-line from the deck of the ship while at anchor. Frozen herring are used as bait, when possible. All the fish caught on the Georges Bank are salted, except the halibut, which is iced. Some idea of the value of these grounds is gained from the fact that a single fisherman may take 500 fish in a day. The Georges Bank area yields

about 70 per cent of the total catch, the Grand and Western Banks accounting for the remaining 30 per cent. Approximately 60 per cent of the fish are brought in iced, and 40 per cent salted.

On returning to port the fish are pitchforked on to the wharf, and sorted into snappers (less than 16 ins. from nape to tail), medium, and large (over 22 ins.) Generally, they are divided as follows: 4 per cent snappers, 41 per cent medium, and 55 per cent large. Each class is weighed separately and carefully examined for any indication of spoilage. Any suspected fish are thrown out. The fish are then washed and put with salt into butts in the store. Fish that are brought in iced whole are sorted and weighed, and then beheaded, gutted, and split and salted. About eight bushels of salt are used to each hogshead of fish. The fish are kept, salted down in hogsheads until required, care being taken that the fish are kept covered with strong brine.

After salting, the fish are dried. The salting process effects partial drying by extracting a large proportion of the flesh fluids of the fish. The extraction of water by the salt is assisted by kenching, the fish at the bottom of the kench being pressed down by the weight of those above.

The fish are taken from the butts as required, and are piled in a kench about 4 ft. high, to express and drain off the pickle. At the end of two days the fish are re-piled, the top fish becoming the bottom, and so subjected to full pressure. If the weather is unfavourable for drying, they are re-kenched every two or three days.

The fish are then dried by exposing them to wind and sun on a bed of latticework about 8 ft. wide and 30 ins. above the ground, and as long as necessary, called a "flake." The drying yard is known as the flake yard. The latticework is constructed of triangular-section, wooden laths, placed about 3 ins. apart, the fish resting on the upper edges of the laths.

In the hot weather, the fish are protected from sunburn by canvas awnings, and from rain at night by coops.

With a warm sun and a good breeze, drying will be complete in about 10 hours. Thorough drying throughout the body of the fish is accomplished by drying on the flakes until the surface is dry and crystallized. The fish is then kenched, and the dry surface salt extracts more moisture from the interior. The fish is then dried again, thus ensuring a much more complete result.

Fish are also dried in some factories in large, steam-heated shelf driers. This method is inclined to be too rapid, with the result that the fish are only surface dried instead of being uniformly dried right through.

After drying, the fish are kenched in the store until required. They are then skinned, the bones are removed, and they are moulded into blocks which are cut up into cakes for packing and export.

It is estimated that the loss in weight during the different operations is as follows—

Dressing	40 per cent
Salting (full pickle)	17 „
Drying	4 „
Skinning and boning	13 „
Total loss	<u>74 „</u>

The fresh waste, skins, bones, etc., of the fish are worked up for glue, the residue being manufactured into fertilizer. The best glue is obtained from the skins. The cod and cusk skins are superior in this to the skins of hake and haddock.

The oil is extracted from the livers. That from fresh livers is refined and used for medicinal purposes, while that from old livers is used for tanning chamois leather. The value of this oil is considerable, as much as £150 being received by a boat in one trip for the oil alone.

In 1914, Newfoundland exported 60,000 tons of cod meat, worth £1,600,000. The chief market is the Mediterranean.

CHAPTER VI

TRAWL FISHERIES

UNLIKE the drift net, which only catches fish of one species and of fairly uniform size when they are swimming near the surface, the trawl net scoops up practically all the inhabitants of the sea bottom, including round fish, e.g. cod and haddock; flat fish, e.g. sole and plaice, as well as various invertebrates (jelly fish), and marine plants and stones. The trawl is essentially a flattened, conical net that is dragged open-mouthed along the sea bottom. The two kinds of trawl in common use—the beam trawl and the otter trawl—differ in the method that is adopted for keeping open the mouth of the net. The beam trawl is used by sailing vessels, the otter trawl by steamers.

Sailing trawlers are divided into two classes: first class smacks and second class cutters. The smack is a two masted vessel with fore and aft rig, generally making a five or six day voyage, and trawling in depths of up to 40 fathoms. The cutter makes shorter voyages—20 hours—and generally keeps within territorial waters.

To work a beam trawl successfully, it is necessary to know the character of the sea bottom, whether rough or smooth, and also the time and direction of the tide. The net is trawled with the tide a little faster than it is running, so that sufficient resistance is encountered to keep the net extended. In shooting the trawl, great care must be taken to make it alight on its runners in the correct position for trawling. If the net be twisted, or if it alight upside down, it has been shot “foul,” and has to be hauled up and shot again. In preparing for a shot the net is lowered over the side by adjusting the bridle ropes, and the beam is coaxed into its proper position while the net is still near the surface. The net is then gradually lowered, the boat moving slowly forward. The trawl is generally hauled for the duration of a tide—that is, six hours—during which time it will travel about 15 miles. The net is generally hauled in by a steam capstan, driven by a small donkey engine. When the trawl comes alongside, the beam is secured and the net is gradually hauled over the side by hand until the cod end appears; this is

then made fast to a rope and tackle, and hauled above the deck. The cod line is untied and the fish are discharged upon the deck.

Since trawling is generally carried out on smooth ground, the greater proportion of the catch consists of certain kinds of demersal fishes that frequent sand and gravel. Of these, the most important are cod, haddock, whiting, ling, hake, catfish, sole, plaice, turbot, and brill. Certain of these species also frequent rocky ground, and are taken in such areas by the line fishermen.

Generally speaking, line fishermen work in deeper water than trawlers and capture larger fish, though of fewer species, e.g. cod, halibut, ling, skates and rays.

The original sailing trawlers are rapidly being superseded by steam trawlers. The first steam trawling company was formed in 1882. It had a capital of £20,000 and a fleet of four vessels. It trawled on the Dogger Bank for three years with marked success. After this the future of steam trawling was assured. The steam trawler is many times more efficient than a smack, for it can fish in nearly all weathers, including calm, and it can trawl over rough bottoms, owing to its greater power, and can go much further afield.

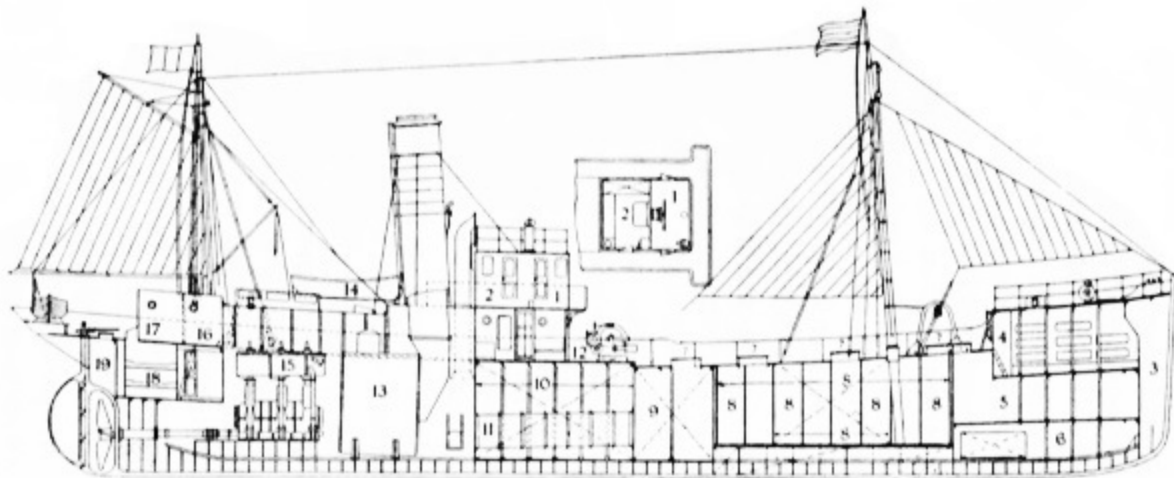


FIG. 15
MODERN STEAM TRAWLER (SECTION)

Total length, 160 ft.

Length between perpendiculars, 148·5 ft.

Greatest breadth (frame), 23 ft.

Draught, $13\frac{3}{4}$ ft.

Explanation of Section.—1. Wheelhouse. 2. Captain's cabin. 3. Collision bulkhead. 4. Crew's quarters. 5. Store for gear, nets, etc. 6. Chain locker. 7. Fish-pounds (on deck). 8. Fish-hold. 9. Cross bunker (for coal). 10. Main bunker. 11. Passage to bunker. 12. Steam-winch. 13. Stokehold. 14. Lifeboat. 15. Triple expansion engines (650 indicated h.p.). 16. Bathroom. 17. Mate's quarters. 18. Dining-room and berths for engineers. 19. Storeroom.

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Modern British steam trawlers travel as far afield as Iceland, Newfoundland and Morocco.

Steam trawling developed rapidly, and resulted in a correspondingly rapid decrease in the number of sailing trawlers. Between 1893 and 1903, the number of first class smacks in Great Britain decreased from over 2,000 with an average tonnage (net) of 57·4 to less than 900 with an average tonnage (net) of 40. From 1903 until the present day, the number had remained between 900 and 800; it would seem, therefore, that the relative numbers and importance of smacks and steam trawlers gradually attained to a condition of equilibrium. Between 1900 and 1906 the increasing importance of steam trawling received a temporary check. A steam trawler in those days would cost about £10,000 to construct and about £5,000 a year to operate; their commercial success, therefore, depended upon correspondingly large and valuable catches of fish being obtained. When first introduced on the fishing grounds round the coast their superior efficiency and speed amply compensated for their high cost. About 1900, however, the catch obtained by these vessels on the home fishing grounds began to diminish, and the fishermen became alarmed lest the greatly increased efficiency of steam trawling should prove to be its own undoing, and result in the depopulation of the fishing grounds by over-fishing. Between 1900 and 1906, the number of steam trawlers fishing from British ports only increased by 200, whereas, during the preceding 10 years, the numbers had increased from a few hundred to over 2,000.

The anticipated exhaustion of the home grounds led to the steam trawler prospecting further afield. These longer voyages, as far as Iceland and the White Sea and Morocco, were very successful. The result of this was that larger steam trawlers were built, capable of undertaking long voyages of

many weeks' duration. Between 1900 and 1906 the average net tonnage of the steam trawlers increased from 54 to 62. The steam trawlers, in opening up new and more distant fishing grounds, left the home grounds to the smacks. Consequently we find that the smacks confined their operation to the smooth ground in home waters, leaving the rough and more distant grounds to the steam trawlers. A direct result of this gradual redistribution of the fisheries between sailing smacks and steamers was the development of specialized fishing ports. Such ports as Lowestoft, Brixham and Ramsgate, off which good fish are obtainable and which are within easy access of good markets, have retained their importance as smack ports; on the other hand, the development of steam trawling has led to the rapid growth of deep water ports, such as Fleetwood, Grimsby, Hull, Aberdeen, and Milford Haven. In Grimsby, originally one of the greatest strongholds of smack fishing, smacks have been entirely displaced by steam trawlers, owing to the special facilities which the port offers in being near cheap coal, in possessing deep water, and in being in direct rail communication with large markets for trawl fish.

There is no doubt that the rapid development of steam trawling was accelerated by the invention of the otter trawl. This is not only a larger net than the beam trawl, but is for all but small, flat fish, a much more efficient instrument. From the study of market statistics between the years 1889 and 1898 Garstang has calculated that a steamer caught on the average between four and seven times as much fish in the year as a sailing smack.

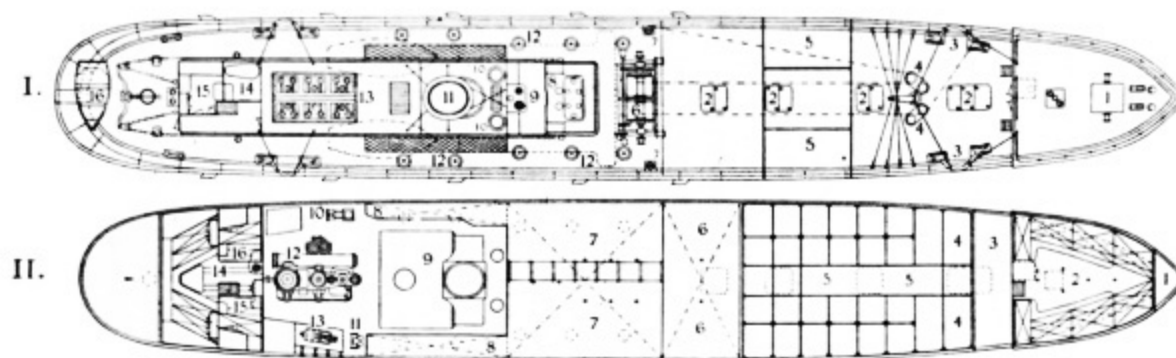


FIG. 16

I.—PLAN ON DECK.

II.—PLAN BELOW DECK.

Plan of Arrangements on and below Deck.—(I) On deck: 1. Winch. 2. Hatches. 3. Gallows. 4. Bollards. 5. Fish-pounds. 6. Steam-winch (for trawl). 7. Blocks. 8. Officers' messroom. 9. Galley. 10. Ventilators. 11. Funnel. 12. Bunker-hatches. 13. Engine-room skylight. 14. Bathroom. 15. Mate's cabin. 16. Lifeboat.

(II) Below deck: 1. Collision bulkhead. 2. Crew's quarters. 3. Storeroom. 4. Iceroom. 5. Fish-hold. 6. Reserve coal bunker. 7. Main bunker. 8. Side bunkers. 9. Stokehold. 10. Main pump. 11. Auxiliary pump. 12. Engines. 13. Dynamo. 14. Cabin. 15 and 16. Chief and second engineers' quarters.

[Larger image](#) (124 kB)

A modern steam trawler is from 150 to 160 ft. long by 25 ft. beam and 12 ft. depth, constructed with a high bow and a low, flat stern. Her net tonnage is from 60 to 200, her bunker capacity 250 tons, with storage room for up to 120 tons of fish. She is fitted with triple expansion engines of from 40 to 85 horse power. The forward part of the ship is occupied by the living quarters of the crew, rope and net store, iceroom, and fish-hold. Larger vessels, making trips to distant grounds, will take as much as 30 tons of broken ice; this ice is distributed over the fish in layers, after they have been cleaned and gutted. In practically all modern fishing ports there is a special ice factory situated near the quay, and ice is manufactured by the ammonia process, crushed, and delivered to the ships through zinc-lined chutes. The fish-hold in the forward part of the ship extends right across the ship and is from 9 to 10 ft. high, divided by a partition into two compartments, each compartment fitted with two shelves 5 ft. long, on which the fish are piled. These shelves reduce compression and facilitate the storage of the fish, the front of each compartment being closed with boards as it becomes full. She generally carries three or four trawl nets, one on her starboard and the other on her port, one or two being down below in reserve. The boat is fitted with four gallows, two forward and two aft, one on each side of the boat. These gallows are used for lifting the otter boards out of the water when the trawl is hauled in.

The ship carries nine hands, consisting of skipper, mate, boatswain, two deck hands, cook, two engineers and a fireman.

On the fishing grounds, fishing is continuous. The net is trawled for from two to four hours, although on grounds where fish is plentiful (e.g. Iceland) the trawl is frequently hauled every half-hour. It is then hauled aboard, and the cod end containing the fish is swung over the deck. The cod line is unfastened so that the cod end of the net opens, and the fish are discharged into a pound formed on the deck by horizontal 9" × 3" deal boards. The net is cleaned and shot again.

On smooth ground trawling is commercially possible at all depths down to 300 fathoms. In few cases, however, is trawling carried on at greater depths than 200 fathoms.

Owing to the large amount of stores and repairs, etc., connected with the maintenance of a fleet of steam trawlers, most large owners maintain fairly elaborate premises in the neighbourhood of the fish dock. These premises generally consist of a net-making hall in which nets are made by women working with shuttles, a large bath of tar or tanning material below in which the net is soaked, also a wood yard and blacksmith's shop, containing a steam hammer, a plumber's shop, a boat-builder's shop, a large store-room fitted with the necessary stores and spares.

During the war the steam trawlers were commandeered by the Government for use as patrol boats and mine sweepers. It is estimated that 10 per cent of our steam trawlers and drifters and their crews were lost during the war.

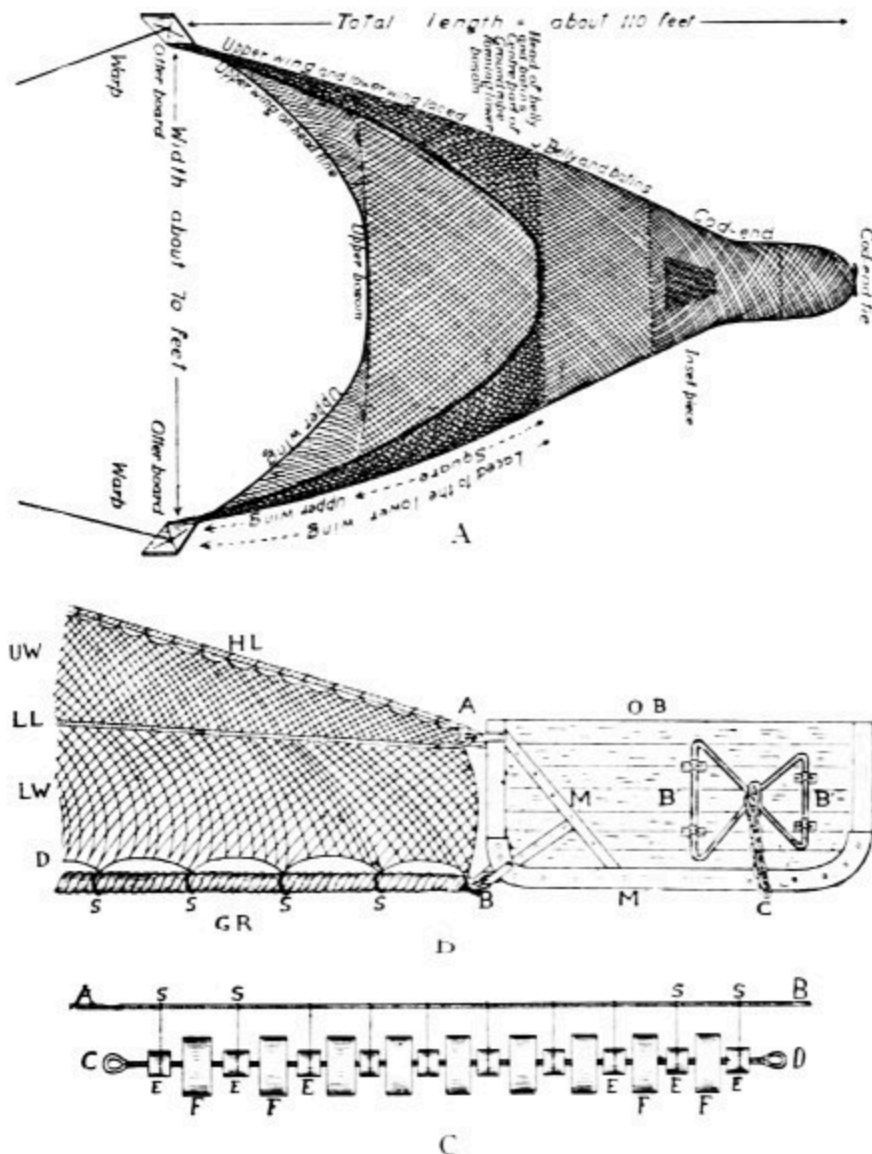


FIG. 17

A.—The otter trawl.

B.—Attachment of board to net. OB. Otter board. B. Iron brackets. C. Chain to connect with warps. M. Metal strengthening pieces. M'. Iron shoe. HL. Head line. UW. Upper wing. LW. Lower wing. LL. Lacing connecting wings. GR. Ground rope. D. Balch of lower wing. SSS. Twine settings connecting balch to ground rope. A. Headline and lacing connected to board by shackle. B. Toe of ground rope connected to board by shackle.

C.—Bosom of a bobbin foot-rope for use on rough ground. AB. Balch line on head of belly and connecting with bosom of wings. SS. Wire seizings connecting balch to small intermediate bobbins, 6" diameter (EE). Large bobbins up to 24" diameter (FF).

When steam trawling was first introduced it aroused general opposition, for there was not only the fear that their efficiency would lead to over-fishing in certain grounds, but it was said that the trawl, when dragged along the bottom, destroyed the eggs and killed the immature fish. The line fisherman found that steam trawling made it more difficult to catch demersal fish with baited hooks. He attributed this to the effect of over-fishing, but it is probable that contact with the otter trawls had made the fish rather more shy and, therefore, more difficult to catch by this method. It is unlikely that steam trawling will lead to serious over-fishing, except possibly amongst such sedentary fish as soles and plaice. It must be remembered that trawling is only commercially possible on comparatively smooth ground and down to depths of about 200 fathoms. Probably, therefore, the actual area trawled is only a small proportion of the total area that is inhabited by fish. It is possible, of course, that extensive and long continued trawling in a confined and relatively isolated area may scare the fish away; it is probable, however, that any area in which over-fishing appears to have produced temporary exhaustion will tend to recover automatically, since it would naturally be abandoned temporarily by the trawlers for more profitable fishing grounds. There is no doubt that trawling, unless the size of the mesh is carefully controlled, tends to remove large numbers of immature fish. Generally in ordinary beam trawling—cod, plaice, haddock, etc.—the mesh varies from 3 ins. diameter near the mouth of the net to about $1\frac{1}{4}$ ins. diameter at the cod end. If a much smaller mesh were used the resistance encountered by a full-sized net would be so great that it would be almost impossible to draw the net through the water. Smaller trawls of $\frac{1}{2}$ in. mesh are used in shallow coastal waters for catching shrimps, small plaice and whiting. The size of mesh largely determines the size of fish that will be retained by the net, since the smaller, immature fish readily escape through the meshes. Of recent years the various fishery boards, with a view to preventing the catching of such small, immature fish, have increased the size of mesh that is to be used—particularly when trawling within the three mile limit, where the greatest proportion of immature fish is generally encountered. For steam trawlers

working in deep water a $2\frac{1}{2}$ in. mesh is generally used, but within the three mile limit it is frequently increased from 3 to $3\frac{1}{2}$ ins.



FIG. 18
THE CATCH ABOARD

Herring are caught with drift nets at night near the surface. In the daytime they frequent the sea bottom and can then be caught with a trawl net. Trawling for herrings was first practised by the fishermen of Milford Haven and Fleetwood in 1901. They used an ordinary otter trawl lined with a piece of herring net. A specially constructed herring trawl is now used, of which the cod end is made of $2\frac{1}{2}$ in. mesh instead of the usual $3\frac{1}{2}$ in.

When trawling for herrings the steamer goes at full speed, generally for two to four hours, unless a shoal is encountered, when half-an-hour is frequently sufficient.

Herrings are trawled in from 70 to 100 fathoms of water over a soft bottom. The main centre for trawled herrings is North-West of Ireland, other fisheries being carried on off the South-West of Ireland, the West of Scotland, and in the North Sea. In 1913 over 500,000 cwts. of herrings were taken with trawl nets in these areas.

This method of catching herrings aroused serious opposition among the drift net fishermen. They asserted that the trawl catches and destroys a high proportion of immature fish, and also destroys the herring eggs as it passes along the sea bottom. In 1913 the matter was investigated by a Parliamentary Committee, but any Government action was checked by the outbreak of war.

Since 1905 the trawling grounds frequented by British steam trawlers have been divided for statistical purposes into eighteen fishing areas. The names and areas of these regions are shown in the chart of the trawling grounds ([Fig. 19](#)).

[Table I](#) shows in hundredweights the average catch per day's absence from port in different areas.

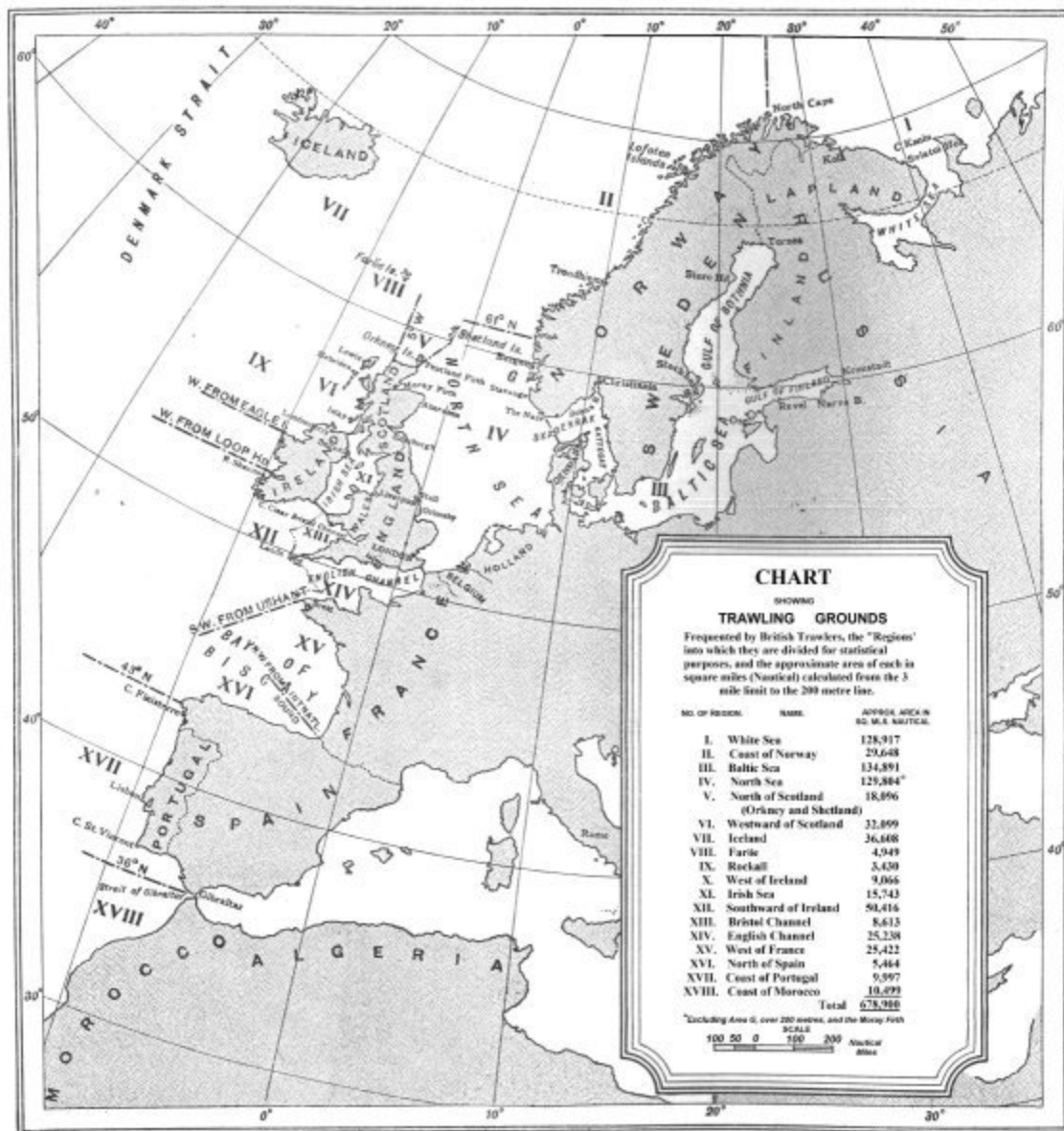


FIG. 19

CHART SHOWING TRAWLING GROUNDS

Frequented by British Trawlers, the "Regions" into which they are divided for statistical purposes, and the approximate area of each in square miles (Nautical) calculated from the 3 mile limit to the 200 metre line.

NO. OF REGION.

NAME.

APPROX. AREA IN
SQ. MLS. NAUTICAL

I.	White Sea	128,917
II.	Coast of Norway	29,648
III.	Baltic Sea	134,891
IV.	North Sea	129,804*
V.	North of Scotland (Orkney and Shetland)	18,096
VI.	Westward of Scotland	32,099
VII.	Iceland	36,608
VIII.	Farøe	4,949
IX.	Rockall	3,430
X.	West of Ireland	9,066
XI.	Irish Sea	15,743
XII.	Southward of Ireland	50,416
XIII.	Bristol Channel	8,613
XIV.	English Channel	25,238
XV.	West of France	25,422
XVI.	North of Spain	5,464
XVII.	Coast of Portugal	9,997
XVIII.	Coast of Morocco	10,499
Total		678,900

**Excluding Area G, over 200 metres, and the Moray Firth*

[Larger image](#) (383 kB)

TABLE I

	1906	1913	1920
White Sea	40·15	44·12	25·45
Iceland	44·22	46·10	58·54
Farøe	31·19	28·19	27·03
Rockall	38·98	39·27	49·53
North of Scotland	25·01	25·76	27·31
North Sea	17·60	14·08	24·94
English Channel	11·36	8·95	25·70
Irish Sea	15·66	11·94	18·79
Bristol Channel	13·15	13·98	26·38
West of Scotland	21·18	28·11	28·17
West of Ireland	21·48	30·22	25·87
South of Ireland	26·97	23·74	26·63
Biscay	15·98	13·22	18·73
Portugal and Morocco	6·55	13·81	19·29

In England and Wales more fish is landed by trawlers than by all other methods of fishing combined. Trawl-caught fish—soles, plaice, turbot, halibut, cod—are much more valuable than fish caught by drift nets, e.g.

herring and mackerel. In England and Wales, in 1913, the weight of pelagic fish caught amounted to 389,262 tons, and of demersal fish 418,038 tons. Although the quantity of the demersal fish was, therefore, only little larger than of the pelagic fish, its value was £7,463,003, compared with £2,531,979, the value of the pelagic fish.

CHAPTER VII

SHELLFISH

SHELLFISH are divided into two classes: Crustacea, including the lobster, crab, shrimp, prawn, and mollusca, including the oyster, mussel, cockle and periwinkle. Shellfish generally abound in comparatively shallow water near the shore.

Perhaps the most important members of the crustacea are the various minute, pelagic copepoda, of which incalculable myriads form an important constituent of the plankton in all seas. These copepoda live upon the diatoms and other microscopic, marine vegetable life floating at the surface of the sea. The most important edible members of the crustacea are the lobster and the shrimp.

The lobster is found along the coasts of the North Atlantic and Mediterranean, particularly along the European coasts from Norway to the Mediterranean, and off North America from Labrador to Cape Hatteras. The lobster lives in shallow water at about 12 fathoms depth, and frequents a rocky bottom. The lobster's eggs remain attached to the female until the larvae hatch out. From 10,000 to 12,000 eggs are carried in this way by a female lobster. She protects them from the ravages of fish that will otherwise consume them as food, and by keeping them constantly irrigated with fresh sea-water she promotes their healthy life and development. The eggs may take as long as twelve months to hatch, and although "berried" lobsters are seen in greatest numbers in the spring they are also captured at all seasons of the year.

When hatched the young lobster larvae leave their mother and float up to the surface water, where they develop for a time among the plankton. During the larval period the lobster is a free and active swimmer.

The young larvae are consumed in large quantities by fish such as herring, mackerel and sprat, especially during the summer months when they are most abundant. While developing into a complete lobster it passes through at least three distinct changes of form. When the larva has attained the length of about $\frac{3}{5}$ in. it already possesses many of the characteristic

features of the adult. Soon afterwards, it sinks to the sea bottom and gradually grows into a complete adult. During the growth of the lobster it frequently casts its shell and grows a new one. Growth only takes place when the shell is cast and while the new shell is hardening. During the first few weeks of its life the lobster casts its shell about once a week, but this casting happens less and less frequently as the lobster grows older. The new shell is formed beneath the old one, and although at first quite soft rapidly hardens when the old one has been cast off. Most adult lobsters cast their shells in July, August and September.

A lobster grows slowly, and when from 9 to 10 ins. long is probably from four to five years old. It becomes mature when about 6 ins. long—that is when about three years old.

The lobster is usually caught in creels or “pots” baited with portions of stale fish—generally flounder, skate, eels, etc. Lobster fisheries tend to deteriorate in value very rapidly. Owing to the lobsters’ keen sense of smell, the method of capture by means of creels or pots is very efficient, so that the lobsters are caught in great numbers, with the result that the fishery soon shows signs of exhaustion, the average size of the lobster caught becoming smaller. The lobster fishery is entirely confined to the shallow water near the shore, and can only be replenished and maintained by the young lobsters that hatch out in that neighbourhood. Large quantities of lobster spawn are destroyed every year when berried lobsters are caught. It is estimated that, on an average, 30 per cent of the lobsters caught are berried females. The fishermen either remove the spawn and throw it back into the sea—where, of course, it almost certainly becomes fish food—or sell it to be used in making certain special sauces.

Various attempts have been made by legislation in different countries to prevent the capture of berried females, and so protect the lobster spawn, but, since berried females are found all the year round and comprise about 30 per cent of all the lobsters captured, it is practically impossible to prohibit the capture of berried lobsters without seriously penalizing the fishermen.

A better policy would be to hatch lobster eggs in large numbers artificially, and when the young lobsters are well established add them to

the natural stock. This is actually done on a large scale and with excellent results in America and Norway.

In Europe lobsters are generally sent to market in a fresh state, but in America they form the basis of an extensive canning industry. In 1913 over 2,500,000 lobsters were captured round the coasts of Great Britain and Ireland, the total value of the fish being more than £110,000.

Shrimping is one of the most important methods of inshore fishing, and gives employment to a large number of fishermen round our coasts. The shrimp is found on sandy or muddy ground in shallow water near the coast. A female shrimp, like the lobster and the crab, carries its eggs under its tail.

Shrimps are caught with a fine-meshed trawl net, drawn by a boat or by horse and cart, or with push nets or hose nets. One great objection to shrimping is that the shallow, sandy areas on which it takes place are much frequented by young fish—particularly dabs, plaice, soles, whiting and codling. Owing to the small mesh of the shrimp trawl, these small fish are captured in large numbers and are generally dead or dying when discharged from the net. Generally, the shrimps are separated from the small fish by riddling, and the smaller shrimps are then separated from the larger ones by a second riddling process, and are returned to the sea. The shrimps are thrown into boiling salt water, rapidly stirred for a few seconds, and spread out on the deck to cool. From three to four hauls are made per day, a good day's fishing consisting of from 30 to 40 quarts of shrimps. Large numbers of shrimps are potted.

The other important group of shellfish is the mollusca. Molluscs, i.e. "soft creatures," are essentially soft, mobile animals, protected by shells. They are classed as bi-valves, for example oyster and mussel, and uni-valves, for example limpet and whelk. There is no real difference between a bi-valve and a uni-valve, for what appear to be the two shells of the bi-valve are really one shell divided into two parts by a line of soft, uncalcified material which forms a hinge between the two halves of the shell; this hinge tends to keep the shell open, but the muscular action of the living animal inside keeps it closed when required.

With the exception of the mussel, very few shellfish actually live on the shore between the tide marks. Most of the seashore shells are brought by the sea from animals that lived in from 10 to 20 fathoms of water. The

cockle lives buried in the sand, about an inch below the surface. The oyster lives on stones and shells below low-water mark.

All molluscs are attached tightly to the shell at one or two points, and cannot be removed from the shell alive. In the case of the bi-valves the animal is attached to the two shells by a muscle which draws the two valves of the bi-valve together. When this muscle is relaxed, for example in normal circumstances, when feeding at the bottom of the sea—the shell remains open. Some shellfish—notably the scallop—actually swim by opening and shutting the two valves of their shell.

The most important uni-valves are the periwinkle, the limpet and the whelk. Uni-valves possess a well-marked head and neck, a pair of eyes and a mouth. They are remarkable for the possession of a tongue, formed like a ribbon rasp, furnished on its upper surface with a large number of small teeth. The number and arrangement of these teeth differ in different species. With this ribbon rasp the uni-valve, for example a dog-whelk, can rasp a hole through the shell of an oyster and feed upon the contents.

Bi-valves do not possess a ribbon rasp, neither have they a projecting head, nor in most cases any eye. They possess a mouth, furnished with four flapper-like lips or gill plates. They feed on microscopic, floating plants that are drawn within their mouth by currents set up in the water by the rhythmic vibrations—from three to four hundred strokes per minute—of millions of hairs that hang down from soft plates supported under the protecting arch of the shell and called the “beard.” These currents of water not only bring food to the mouth of the bi-valve, but also irrigate the gill plates and so enable the animal to breathe. The oyster lies on the sea bottom with its muscle relaxed and its shell gaping.

A North European oyster acts alternately as female and male. It produces eggs—as many as a million in a season—and a fortnight after the eggs have been shed, the same oyster produces millions of spermatazoa, which form a cloud of fine dust in the water. These spermatazoa rapidly scatter in all directions, and, entering the tubular reproductive sacs of oysters that are producing eggs, fertilize them.

American and Portuguese oysters are definitely male and female, the eggs being discharged by the female and fertilized subsequently in the sea by the male.

The eggs remain attached to the parent's gill plates, and in a day or so develop into minute, shell-less oysters. The parent oyster is then said to be "white-sick." About two days later the young oysters have become dark-coloured and are found to have formed minute convex shells, rather like those of a cockle. The parent is then "black-sick." A week later the young oysters escape and rise in thousands to the surface water, swimming by means of fine hairs or cilia that are attached to the upper edge of the shells. They are carried far and wide by tides and surface currents. Many are eaten by young fish and shrimps. As they grow the shells become heavier, and after a time they sink to the sea bottom. This is known as the "fall of spat." If they fall on stony ground, where they will be well irrigated and nourished through the movement of the water, they will thrive. Many, however, fall on soft, unsuitable ground and perish.

The European oysters spawn in the summer (from May to September). They become mature in three years, are at their prime in from five to seven years, and rarely live longer than ten years.

Oysters are gathered from natural beds or from artificial grounds. The oyster breeders place movable tiles or frames for the spat to fall on. When the young have become affixed to these "stools" they are frequently carried away to develop in a different locality. The oysters are finally fattened in sea ponds or inlets that contain a large diatom population. At Marennes, on the west coast of France, the water in which the oysters are grown contains a particular blue diatom. After feeding upon these diatoms, the beard of the oyster becomes stained a bluish-green colour—the well-known "Marennes vertes" oysters.

A natural oyster bed is formed on stony ground free from mud and sand, so that the oyster, after becoming attached to a stone, is completely surrounded by clear sea-water. Oysters do not flourish in water containing less salt than ordinary sea-water. Thus, there are no oysters in the Baltic Sea.

The chief enemies of the oyster are the dog-whelk that bores through the shell, and the starfish that pulls the valves apart and attacks the oyster inside.

The oyster is widely distributed in tropical and temperate seas all over the world. The approximate value of the annual oyster crop of the world is

£4,000,000, representing a crop of 10 billion oysters.

In Europe up to 75 per cent of the oysters are reared from spat in artificial beds—not more than 7 per cent being “native.” In the United States, however, over 40 per cent are still obtained from natural beds.

The simplest form of oyster culture is the preservation of the natural bed. These beds are easily destroyed or made unproductive by over-dredging. Colonies are broken up. Other animals are admitted. Breeding oysters are covered up by stones and shells, and suffocated. Ridges suitable for the development of the spat are broken down.

After the beds have been properly protected and preserved the next stage is to extend the area of the natural beds. This involves a knowledge of the conditions of depth, temperature, salinity and character of bottom that are necessary to the successful growth of the oyster. Finally the productivity of an oyster “park” and the quality of its produce can be greatly improved by providing artificial “stools” for the reception and development of the spat. Many substances can be used for this purpose. The Romans used earthenware tiles, and similar tiles are used to this day in France. Brushwood, trees, stones and stakes, and old oyster shells (cultch) are also used.

The earthenware tiles used in France are hollowed on one side to receive the spat, and are coated with lime to facilitate the removal of the oysters when they are a year old. They are then from $\frac{1}{2}$ to 1 inch in diameter, and are picked off the stools and placed on stands where they are thinned out from time to time as they grow.

The chief oyster fisheries in Britain are at Whitstable, Colchester and Brightlingsea. Nearly 40,000,000 oysters were gathered on the coasts of England and Wales in 1920, and were sold for about £250,000.

Perhaps the next most important edible bi-valve is the mussel. Frequently, mussel beds are situated near the mouth of rivers, and consequently tend to be contaminated by sewage. It has been established by various investigators—notably Dr. Klein and Professor James Johnstone—that mussels are able to cleanse themselves of sewage pollution in a comparatively short time if they are re-laid in sterilized water. Experiments on a large scale have been carried out with the mussel beds at the mouth of

the Conway river since September, 1916. The mussels are gathered from the beds and placed about two deep on wooden grids in a large concrete cleansing tank of 40,000 gallons capacity. The mussels are first thoroughly hosed with water at high pressure to remove all adherent mud, etc. The tank is then filled with sterilized sea-water and the mussels are allowed to remain in it for 24 hours. During this period the mussels effectually free themselves from bacteria. The tank is then emptied, the mussels are hosed again, the tank is again filled with sterile water and after a further 24 hours is emptied. The mussels are once more flushed with the hose. After this treatment the mussels reach a high standard of purity. The scheme has proved to be a complete success, not only from a scientific point of view, but also as a commercial proposition. The sum of 1s. per bag of mussels (140 lbs.) is charged to the fishermen for this treatment.

CHAPTER VIII

FISHERIES FOR WHALES

WHALES are the most important members of a large family of land animals including also the seals, walrus, and porpoise, that have gradually become adapted to live in the sea. They have acquired an externally fish-like form, but in every other respect they retain the characteristic features of mammalian structure. They are warm-blooded, air-breathing quadrupeds, that suckle their young. In the whale, the fore-limbs have become simple five-fingered flippers, while only isolated, vestigial bones of the hind-limbs remain buried uselessly in the body. Unlike fishes, the tail is set horizontally, thus enabling the creature to rise easily to the surface to breathe. The warm-blooded body is kept warm by a layer of fat placed immediately beneath the skin, and varying in thickness from 8 to 20 ins., and known as the blubber. The nostrils, instead of being situated at the end of the snout, are placed far back at the apex of the head to form the blowhole.

Whales are divided into two well-marked groups, known as the whaleboned and the toothed whales respectively, according to the particular form of their dentition.

The most important of the whaleboned whales is the Greenland, or Arctic Right, whale. It attains a length of upwards of 45 to 50 ft., and is remarkable for the enormous extent of its head and mouth cavity. The head extends for a third of the length of the body, so that the mouth cavity may be as much as 18 ft. long, 12 ft. broad and 11 ft. in height, the dimensions of a small chapel! The upper jaw is narrower than the lower and arches backwards, thus increasing the actual height of the mouth cavity and providing ample room for the blades of whalebone with which the jaws are furnished in place of teeth. These blades of whalebone number about 380, and range in length from 8 ft. to, in exceptional cases, 12 ft. They are suspended in the mouth of the whale like stalactites, set fairly close together, and, since the edges of each blade are fringed with fine whalebone, the whole arrangement forms a very efficient strainer. This enables the whale to feed upon the plankton—or “krill,” as it is called by

the whalers—and small fish, e.g. herring and capelan. The whale fills his enormous cavern of a mouth with water containing the floating food particles, and then, by raising his tongue, slowly expels the water through the whalebone sieve. The food particles are retained by the whalebone, and are then licked off and swallowed.

The Greenland whale inhabits the Arctic seas north of latitude 54°N. A closely related variety, the Bowhead whale, forms the basis of a fishery in the Behring Sea.

The largest whales known are the so-called Rorqual whales. The name of these whales is derived from the large number of longitudinal folds or pleatings that form a characteristic feature of their throat. Rorqual whales attain a length of from 80 to 85 ft. The head is relatively small, and the long, slender body carries a distinct dorsal fin. The whalebone is coarse and short. The Rorqual whales are the most abundant and widely distributed of all whales. They are found in all open seas, with the exception of those in the extreme Arctic and Antarctic regions.

The Southern Right whale, or Black whale, is found in the temperate seas of both Northern and Southern hemispheres.

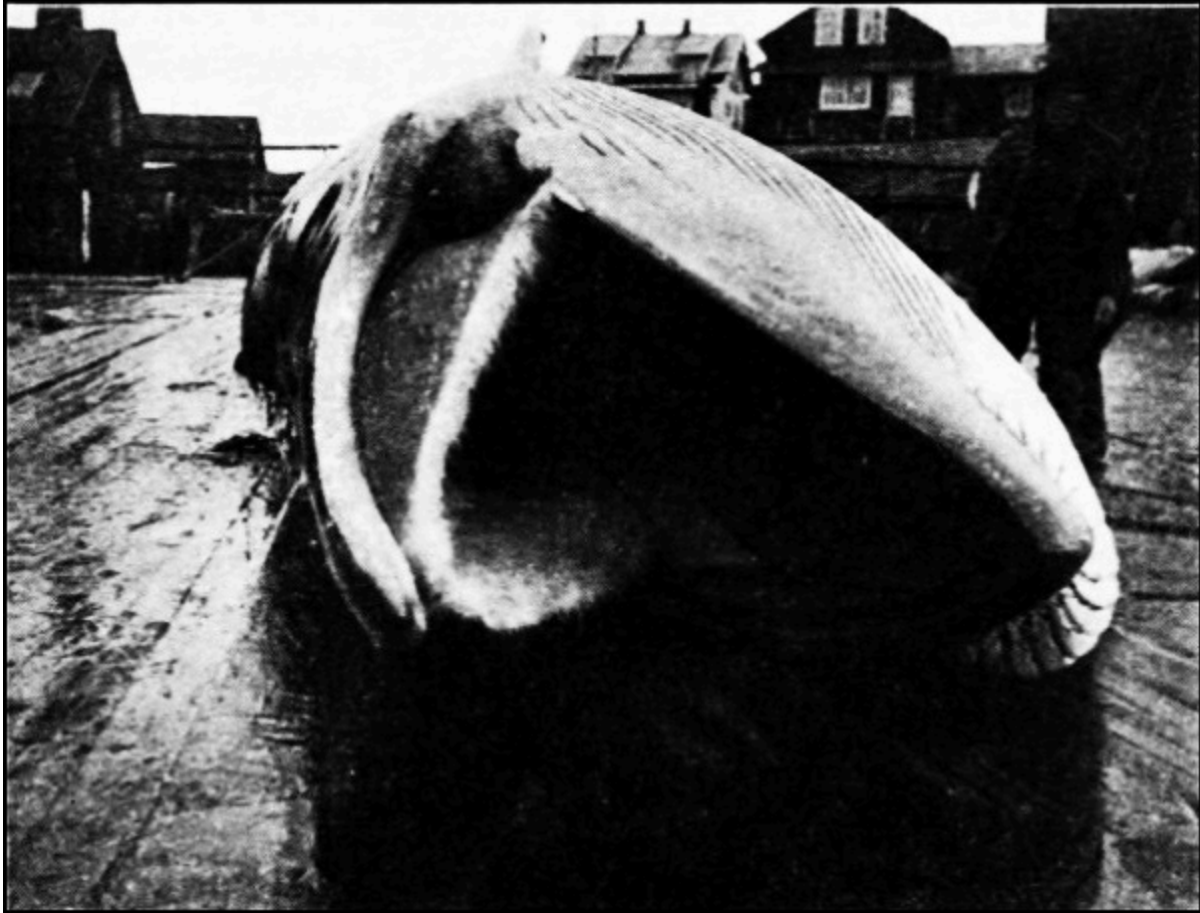


FIG. 20

A WHALE'S MOUTH

The carcass is ready for cutting up at a Shetland whaling station.

Of the toothed whales, the most important is the Cachalot or Sperm whale. It is chiefly captured in Southern seas, and is killed in large numbers for the sake of the spermaceti and sperm oil that occur in large quantities in its head cavity. Sperm and other toothed whales feed upon fish and cuttlefish.

The breeding habits and migrations of the different species of whales are at present little understood. During the summer, when the water in the Polar circles swarms with certain varieties of pelagic crustacea, the whales congregate in these regions and are then most profitably hunted. At the end of the summer they appear to migrate towards warmer water nearer the Equator. They bring forth their young in warm, shallow water, and return to

the whaling grounds in the spring. A young whale calf may be as much as 20 ft. long at birth.

Whales were captured by the Norwegians over 1,000 years ago. In the Middle Ages—from the ninth to the seventeenth centuries—the Basques hunted the Black whale in the Bay of Biscay, and supplied Europe with oil and whalebone. Towards the end of the sixteenth century, as the Biscay whales became rare and more difficult to find, the whalers ventured further afield, and in 1612 discovered the Greenland whale. The Black or Biscay whale is now almost extinct, and there is every likelihood that the Greenland Right whale will also soon be exterminated. The capture of Sperm and Rorqual whales, although equally important, is a comparatively modern development.

Modern whale fishing has become a very efficient art, owing largely to the invention of the shot-harpoon by a Norwegian, Sven Foyn, in 1870. This harpoon is discharged from a gun from the deck of a fast steamship. It penetrates the body of the whale in the vital region just behind the flipper. The invention of this weapon has made the killing of whales a matter of comparative ease and certainty. The inevitable consequence of this is that the whales are being killed in such large numbers that they are in danger of general extermination. Even before the introduction of the shot-harpoon, whales were being destroyed at an astonishing rate. Thus, during 40 years in the middle of the last century, over 300,000 whales were captured by the United States whale fisheries alone. The value of these whales was £65,000,000, so that each whale realized on an average £216. Of recent years—before 1914—a single large Greenland whale has realized as much as £900 for whalebone and £300 for oil. At the present time, over 20,000 whales are killed each year.

The old eighteenth century whaler of about 400 tons burden carried about 30 officers and men, and was equipped for a three years' voyage. Each whaler carried six whale boats. These whaleboats were about 27 ft. long and built sharp at each end. Each boat was furnished with mast and sails, and was provided with two 200-fathom whale lines. When a whale was sighted four of these boats, each manned by six men, started in pursuit. The boats ranged themselves alongside the whale and a harpoon was driven into it from each boat. The whale immediately dived to the bottom of the sea and remained there sometimes for as long as forty minutes. When he returned to

the surface to breathe, more harpoons were thrown and he dived again. Ultimately, owing to loss of blood, the whale kept near the surface and was then dispatched by a lance thrust behind the flipper into the vital parts.

The modern Greenland whaler is an iron vessel of about 500 tons. She is fitted with auxiliary engines of 75 horse-power. She carries from fifty to sixty hands and eight whaleboats. She is fitted with tanks for 250 tons of oil. Before the war she would cost about £17,500 to build and £500 a month to maintain. Each whaleboat carries a harpoon gun in order to make sure of the first harpoon getting a good hold.

In Rorqual fishing, off Newfoundland, the harpoon is tipped with a bomb and time fuse. This explosive harpoon is discharged into the whale from the deck of the whaler—a fast steamer—and explodes with fatal effect.

The chief whale fisheries are carried on off Greenland for the Greenland whale, off the coast of Newfoundland for Rorquals. There is the Norwegian bottlenosed-whale fishery around Iceland, and the American Bowhead-whale fishery in the Behring Sea. In Southern Seas the Humpback, Fin whale, and Blue whale (Sibbald’s Rorqual) constitute an overwhelming majority of the whales captured. The Right whale and the Sperm whale, although captured in relatively small numbers, are individually more valuable. Other smaller species, e.g. the Sei whale (Rudolph’s Rorqual), the lesser Rorqual and the Killer or Grampus, are also found in large numbers in the Antarctic.

When the whale has been killed it is either made fast alongside the whaler and cut up, or it is towed ashore to a “factory” to be cut up and stripped. The blubber is stripped off, cut up into small pieces, and boiled down with water to separate the oil. The yield of oil varies for different species, as shown in [Table II](#). The whalebone is removed and, if a Sperm whale, the oil is removed from the skull cavity with buckets. An average large Sperm whale will yield from 2½ to 3 tons of Sperm oil.

TABLE II

Average Yield of Oil in Barrels	
Species of Whale.	(6 Barrels = 1 Ton).
Right	60 to 70
Blue	70 „ 80
Fin	35 „ 50

Sei	10 „ 15
Humpback	25 „ 35
Sperm	60 „ 65

Whale oil is marketed in five grades: Nos. 0, 1, 2, 3, and 4. Nos. 0 and 1 are made entirely from blubber; No. 2 from tongues and kidney fat and from the residue of the blubber boilings; No. 3 is made from the flesh and bones, and No. 4 from refuse. The different grades contain progressively from $\frac{1}{2}$ to 1 per cent water and dirt, and from 2 to 30 per cent free fatty acid.

Grades 0, 1 and 2 of whale oil are used in the manufacture of soap, glycerine being obtained from it as a by-product. In its natural condition the oil is soft, and has to be “hardened” before it can be used for soap making. The hardened whale oil is white, odourless and tasteless, and is an excellent substitute for tallow. In this condition it is also used as a substitute for lard and, to a small extent, is used in making margarine.

Grades 3 and 4 are used in the manufacture of lubricating greases. Whale oil alone is used for shafting and machinery bearings. When mixed with mineral oil, it is used for looms, spindles and textile machinery. Whale oil is also used as an illuminant, for currying leather, and in making chamois leather, for batching flax and other vegetable fibres, and in oiling wool for combing.

In 1913, the world’s annual catch of whale oil had reached 800,000 barrels. During the war the supply was considerably less, for example in 1917 it was only 358,000 barrels.

Whalebone. Whalebone from the mouths of the Right or whaleboned whales is in considerable demand among dressmakers and milliners. Its principal use is in the brush trade, chiefly in making brushes for mechanical purposes. It is prepared for use by being boiled in water for about 12 hours until it is quite soft. It is then cut into strips or bristles or filaments, according to the use for which it is intended. It is light, flexible, tough and fibrous.

Sperm Oil. Sperm oil is really a liquid wax. It is an excellent lubricant—particularly for rapidly moving machinery, e.g. spinning spindles, or for delicate machinery such as watches. It does not become gummy or rancid, and retains its viscosity at high temperatures. It has no corrosive action.

When cooled to low temperatures, it deposits a solid wax—spermaceti—which is used in the manufacture of high grade candles. Sperm oil is also used for dressing leather, in oil tempering steel, and as an illuminant.

Ambergris. Ambergris is a solid, fatty, inflammable substance, dull grey in colour, which occurs as a concretion in the intestines of sperm whales. It is generally found floating in the sea or on the shore. It is used in the perfume industry mixed with other perfumes.

The development of the whaling industry in the south seas has led to the industrial development of previously uninhabited islands. On South Georgia, which was previously uninhabited, actual industrial villages have been established. A church has been erected, and there are three slips for cutting up the whales, two guano factories, reservoirs for the oil, and houses for the staff. This Antarctic island has a floating population of many hundreds of sailors and workmen. A doctor resides there during the whaling season and, since 1908, the British Government has established a post office in this polar land. In 1922 the eyes of all the world were turned to this far-away land, the Gate of the Antarctic, as the body of Sir Ernest Shackleton, the hero of the Antarctic, was laid to rest there.

CHAPTER IX

THE CURING AND PRESERVATION OF FISH

THE preservation of fishes for use as food long after they have been caught is a matter of constantly increasing importance to the prosperity of the fishing industry. In most other food supplying industries the produce can be kept fresh for the market comparatively easily. Dry grain will keep indefinitely; vegetables and fruits with proper care will generally remain “fresh” long enough to reach distant markets. Oxen, sheep and pigs may be transported to the market alive, and then slaughtered as required. But a fish as soon as it is taken from the water dies and speedily begins to decay.

Fish, like other foodstuffs, whether animal or vegetable, decays as a result of the growth in it and on it of certain micro-organisms (bacteria, moulds). These micro-organisms swarm in the air and on exposed surfaces all the world over. Generally speaking, they flourish best at ordinary temperatures and in a moist environment.

Foodstuffs can be preserved from decay only by preventing the growth and development of these decay organisms. They can be killed outright by any of the ordinary sterilizing processes such as exposure to sufficient extremes of heat or cold, or by treatment with disinfectant substances (germicides) such as carbolic acid or hypochlorites. Clearly, however, foodstuffs cannot be preserved indefinitely by the simple process of killing all the organisms that are resident on the foodstuff at the time of treatment, for, as soon as the foodstuff is exposed to the air, it will become infected afresh.

They can be preserved—

(1) By boiling, and packing immediately afterwards in air-free containers.

This process is, of course, the basis of the great meat packing industry. The meat is packed in a tin, the tin and its contents are heated in steam or boiling water until the meat is cooked and all the decay organisms are destroyed. The tin is then sealed, air-free and air-tight.

(2) By freezing.

Cold storage is a widely used method of preserving foodstuffs. The low temperature prevents the growth and development of decay organisms and, as long as the foodstuff is kept sufficiently cold, arrests decay.

Prehistoric animals long extinct are sometimes found firmly embedded in the Polar ice, as fresh as they were on the day they were drowned.

It is found that the stability and subsequent quality of frozen meat or fish depend directly upon the manner in which it has been frozen. It may be frozen in air, or when immersed in brine. Of these two methods the latter is much quicker, because brine is over twenty-five times as good a conductor of heat as air is. During the slower air-freezing process the quality of the flesh is impaired by the separation of the contained water into comparatively large crystals of ice. This leads to the displacement of the membrane and tissues of the meat, so that in thawing again the meat drips and becomes tough. When immersed in brine freezing occurs too rapidly for this separation of water to occur to any marked extent.

The keeping qualities of brine-frozen fish also are greater than those of air-frozen fish, owing to the protecting coating of ice which effectively prevents contact with bacteria or mould spores.

(3) By drying.

Primitive man preserved his meat by drying it in the sun, or in the smoke of a fire. To-day the preparation of fish, dried fruits, desiccated vegetables, etc., is a world-wide industry.

Generally speaking, decay organisms can only develop in a moist environment. All fresh foodstuffs contain a large proportion of water. The removal of this water effectively checks decay. Drying alone, however, does not always produce a permanent "cure," as the foodstuff is always liable to get moist again. For that reason it is customary to combine the drying process with treatment with an antiseptic substance such as salt. Smoke drying is better than sun drying, for although the ultra-violet rays of the direct sunlight effectively kill bacteria and mould spores wood smoke contains antiseptic substances with which the meat becomes impregnated, so that even the chance of any subsequent infection is greatly reduced.

(4) By treating with an antiseptic substance such as salt.

Antiseptic substances differ from disinfectant substances in that they do not kill micro-organisms, but only arrest their development.

As a rule, they are effective preserving agents, and do not make the food poisonous or unpalatable.

All these methods can be, and are, used for preserving fish, the method most commonly used being treatment with salt. Fish, however, are often kept in ice on board during a fishing trip and are then either packed in ice for transit under special storage conditions (if required fresh) or they are salted down.

Methods of Salting. Different methods of salting are used, according to the character of the fish and the locality. The fish are either cleaned (split and gutted) or salted “round” (whole). In general, the method used is one of the following—

(1) DRY-SALTING. The fish are cleaned, rolled in dry salt, and packed in layers in open casks. Each layer of fish is covered with a layer of salt.

(2) BRINE PICKLING. The fish are immersed in saturated brine, salt being added from day to day to restore the strength of the brine as it becomes weakened by the water which it extracts from the fish.

(3) KENCHING. The fish, either split or round, are piled in layers in the hold of the ship, or on the floor of the warehouse, each layer being covered in turn with a layer of salt. The brine, as it forms, is allowed to drain away.

Of all these three methods, the first is undoubtedly more effective, more economical, and requires less attention than the second. The third method is often used on board ship and sometimes on shore as a temporary expedient when the catch is too large for the number of containers available.

In the dry-salt method, the fish are packed tightly in the casks, and are not afterwards disturbed. When cured they possess a characteristic dry, shrunken appearance.

Fish pickled in brine need attention every day. The brine has to be closely watched so that it shall not become too weak. Fresh salt has to be added daily, and the fish stirred up with wooden paddles to ensure uniform pickling.

Fish cured in this way are softer and more plump than those cured by the dry-salting method.

When a fish is packed in salt the salt rapidly extracts water from the flesh and a strong brine results.

The salt dissolves in the remaining flesh juices of the fish, and rapidly diffuses throughout the fish, thoroughly permeating it. By this process, therefore, the fish is partially dried and becomes thoroughly impregnated with salt.

The gradual change in the composition of the flesh is reflected in the following analysis—

Sample.	% Water.	% NaCl.	% Fat.
Fresh herring, ungutted	67·33	0·63	13·78
Herring lightly salted, before gutting	66·33	1·27	12·11
Herring from rousing tub, gutted and salted, ready to pack into barrel	61·09	1·41	16·14
Herring, after 7 days salted in barrel	52·67	7·43	17·10
Herring, after 8 days salted in barrel	46·90	11·49	22·50

The efficiency of the cure and the appearance of the finished product will be influenced by the following factors—

- (a) The temperature—whether summer or winter;
- (b) The freshness of the fish;
- (c) The quality of the salt—its purity and grain;
- (d) The quantity of salt used;
- (e) The duration of the process.

(a) *The Temperature.* As soon as a fish is dead, it commences to decay.

In hot weather, decay proceeds more rapidly and the interior portion of the meat may become soured before the salt reaches it. Clearly, if the rate at which the salt penetrates the fish is retarded by the salt being impure, or of too fine a grain, or by the brine being too weak, the probability of the fish being spoilt is very much increased.

The dry salt method leads to a much quicker penetration of the fish than the brine method, and should always be used in warm weather.

(b) *The Freshness of the Fish.* The decay processes gather impetus day by day. It is clear, therefore, that in order to avoid the possibility of “souring,” the fish should be salted with the least possible delay.

(c) *The Quality of the Salt.* (1) *Its Purity.* The impurities commonly present in Fishery Salt are the sulphates and chlorides of calcium and magnesium.

The following analysis show the composition of typical samples of Fishery Salt.

Composition.	German Rock Salt.	Italian.	Spanish.	French.	English.
	%	%	%	%	%
Salt (Sodium chloride)	97·28	96·59	96·63	95·86	98·9
Calcium chloride	—	0·32	—	0·16	—
Magnesium chloride	0·25	1·19	0·96	0·35	0·08
Magnesium sulphate	—	1·75	0·73	—	—
Sodium sulphate	0·44	—	—	—	0·04
Sodium bicarbonate	0·01	—	—	—	—
Insoluble (Calcium sulphate, sand, etc.)	2·02	0·15	1·68	3·63	0·98
	100·00	100·00	100·00	100·00	100·00
Moisture	0·20	6·54	4·47	1·39	3·25

The Spanish and Italian salts are solar salts, obtained by evaporating sea-water by the heat of the sun. Solar salt nearly always contains more magnesium salts than brine salt does. This constitutes a serious disadvantage to the fish curer.

Of the calcium salts which occur as impurities in Fishery Salt, the sulphate is practically insoluble in brine, and is probably without action upon the salting process.

Calcium chloride, on the other hand, resembles magnesium chloride and is an undesirable constituent of Fishery Salt, for calcium chloride, and to a lesser extent magnesium chloride and magnesium sulphate, diminish the rate at which the salt penetrates the fish. Curing will, therefore, be delayed, and in warm weather (above 70°F.) this may result in the souring of the fish.

To obtain rapid and thorough curing, therefore, it is necessary—especially in warm weather—to use salt which contains as little calcium and magnesium salts as possible.

Pure salt, used dry, produces a soft, yellow-meated fish which is flexible in the hand. Salt containing calcium chloride or magnesium chloride produces a harder and stiffer fish with a markedly whiter colour.

Salted fish can only be stored satisfactorily in a dry place. Fish which has been cured with impure salt is hygroscopic and will run wet in the store.

This hygroscopic moisture weakens the preserving action of the salt. Fish that has been cured with a pure salt will keep much drier under ordinary storage conditions.

(2) *Its Grain.* The crystals of Fishery Salt should be coarse and hard. Coarse crystals dissolve slowly, and so produce a more gradual cure than fine-grained salt does. Fine-grained salt extracts the water so rapidly from the surface tissues that it coagulates them. This retards the further penetration of the salt into the fish, so that the fish has the appearance of being slack salted.

Round versus Cleaned Fish. The thoroughness with which a cut fish is cleaned and washed influences the temperature at which the fish can be salted successfully, and materially affects the quality and taste of the product. Tressler¹ has shown that the chief cause of fish spoiling when salted in hot weather is the decomposition of the blood which remains in the flesh. Even in cold weather, it is found that the extra washing and cleaning greatly improve the quality of the fish. As the presence of blood in the fish also leads to discolouration during the salting process, a thoroughly cleaned and washed fish is, after salting, much whiter in appearance and has a finer taste.

Many fish are skinned before they are salted. It has been observed that a skinned fish will cure almost twice as quickly as an unskinned fish. This is because salt penetrates the meat of the fish at approximately twice the rate at which it penetrates the skin. It is desirable, therefore, particularly in hot climates, to skin the fish before salting. This, of course, is only commercially practicable with certain large kinds of fish such as cod.

The Reddening of Salted Fish. Salted fish sometimes undergo a change, either during the salting process if improperly carried out, or more generally in the store, which is characterized by the development on the surface of the fish of irregular red and brown patches. This reddening occurs not only on

the fish, but also on the floors and walls of the curing factories, on the sides and decks of fishing boats, and even on the salt itself. It occurs most readily in warm weather.

The reddening has been shown to be due to the growth of a micro-organism (a micro-coccus). With this micro-coccus are generally associated a bacillus and a micro-fungus which produce the brown mould on the fish.

Fish become infected with these micro-organisms by contact with boats or docks or warehouses.

Every precaution should be taken to keep such places clean and properly disinfected.

The “rusting” of fatty fish, e.g. herring, is due to the oxidation of certain free, fatty acids split off from the fats by enzyme action.

CHAPTER X

THE FOOD VALUE OF FISH

WITH few exceptions, the different species of fishes that are caught industrially are important because of their food value.

Some fishes are unsuitable for food because they have an unattractive taste; others are directly poisonous. Thus, in the Japanese fish of the genus tetrodon, the roe is poisonous, although the remainder of the fish is edible. Some fishes are poisonous during the spawning season. Others are provided with a special poison gland connected with special spines or barbs. In edible fishes, given the suitable conditions, poisons may be formed by bacterial activity in the flesh of the fish. Poisons so formed give rise to the kind of fish poisoning known as botulism. Cases of botulism have resulted from eating canned salmon and sardines that have become spoiled. In some cases, bacteria present in a diseased fish may produce poisonous substances in the body of the fish. *Bacillus paratyphosus* has been isolated from some poisonous fish, and certain poison-producing bacteria have been found in others.²

Certain shellfish are notoriously liable to be poisonous. The exact nature of the microbes concerned in the production of poisonous substances in shellfish is at present unknown; it is clear, however, that such poisonous substances may be produced in shellfish in three ways—

(1) Microbes of various infectious diseases, such as typhoid fever, may be absorbed by the shellfish from sewage.

(2) The shellfish may be diseased, or be seriously contaminated, by living in dirty water.

(3) Decomposition may set in after the shellfish have been removed from the water—particularly if they have been kept too long in a warm place.

It has been found recently that shellfish that have been deliberately fattened on sewage can be effectively cleansed in such a way as to get rid of ingested sewage bacteria. This process has been carried out successfully on a commercial scale at Conway by the Ministry of Agriculture and Fisheries.

Danger from infected shellfish may also be safely avoided by boiling them. When shellfish are gathered at the right season of the year and from suitable localities, they are a perfectly safe and wholesome food.

Of the many species of edible fishes that are known and used, the number is by no means complete, and new species are added from time to time. Thus, in 1916, the United States Bureau of Fisheries introduced a new edible fish (*Lopholatilus chamaeleonticeps*), which they christened the tile fish. After this fishery had been in existence for twelve months, the known catch of tile fish amounted to over 10,000,000 lbs., valued at more than \$400,000. In 1917, the same Bureau introduced the dog-fish under a new name. As people were prejudiced against the name “dog fish,” the Bureau altered it to “gray fish,” “which is descriptive, not preoccupied, and altogether unobjectionable.” The fish is now caught in large numbers, and forms the basis of a very flourishing canning industry. Attempts have been made recently to utilize as food the edible portions of the shark (which is closely related to the dog fish) and the porpoise.

The food value of most fishes varies very much according to the condition of the fish when it is caught—that is whether it is spawning or not. Further, it may be considerably modified by the changes that take place subsequently in the composition of the flesh during the processes of curing, cooking or preserving.

Generally speaking, all marine fish annually pass through a well-marked series of seasonal changes, the stages of which appear to depend upon changes in the temperature, salinity and alkalinity of the sea. These changes are directly connected with the development of roe and milt, with the fluctuation in the percentage of oil and fat in the liver and body tissues, and also with the rate of growth. Thus the chemical composition of the fish, and hence its food value, varies greatly according to the season at which it is caught.

Norwegian brisling (“Skipper Sardines”) are caught in the summer just before spawning time. At this time the fat content is high; in winter the fat content is low, and the fish possesses small commercial value.

The gradual change in the composition and food value (in calories per pound) of the herring as spawning time approaches is well shown in [Table](#)

[III](#). (Prof. J. Johnstone, Trans., Liverpool Biolog. Soc., Vol. xxxiii (1919), p. 106.)

TABLE III

MANX SUMMER HERRINGS, 1916

COMPOSITION OF THE FLESH OF THE FISH: MONTHLY MEANS

Date.	Condition.	Water.	Oil and Fat.	Proteid.	Ash.	Total.	Food Value.
May	Empty	75·0	2·5	21·1	2·3	100·9	1,100
June	Filling	66·1	11·4	18·6	2·0	98·1	1,806
July	Filling	55·8	21·6	18·4	2·3	98·1	2,762
August	Half full	48·4	31·5	16·5	2·3	98·7	3,608
Sept.	Full	51·9	25·2	17·3	2·6	97·0	3,050

The herrings are caught in September when they assemble in shoals for the purpose of spawning. They are thus most easily caught at the time when their food value is at a maximum.

The flesh of clupeoid fish—herrings, sprats, pilchards, sardines—contains a quantity of oil disseminated throughout the flesh in the form of fine globules. From the above table it will be seen that the percentage of oil in the flesh of the herring may be as low as 2·5 per cent in May, and as much as 31·5 per cent in August. In summer the adipose tissue forms two distinct layers, one situated just below the skin, the other being parallel to the first, but separated from it by a layer of muscular tissue. In winter the oil content becomes so small that these layers of adipose tissue disappear. A comparatively small amount of oil is contained in the liver of the fish.

In gadoid fishes, e.g. cod, as well as in skates and rays, the oil is almost entirely confined to the liver. During the summer the liver grows larger and richer in oil, until sometimes the oil amounts to more than half the total weight of the liver. (When cod are caught the livers are removed and kept apart, to be treated subsequently for their oil.) The percentage of oil in the flesh of the cod varies from 0·1 per cent to 1·0 per cent. Unlike that of the herring, therefore, the food value of the flesh of the cod does not fluctuate markedly according to the season.

When fish are dry-salted a certain proportion of the proteins and mineral salts in the flesh is extracted by the brine pickle that is formed. In Russia

and Poland, where the greater proportion of salted herrings are consumed, the peasants eat them without further cooking, and also consume the pickle.

A great gain in food value per pound results from the removal of so much water from the flesh of the fish. Freshly caught cod flesh contains about 80 per cent water and 17 per cent protein; after being dry-salted for export it contains about 25 per cent of water and 55 per cent protein.

Thus, 1 lb. of dry cod is equal in food value to about 3 lbs. of fresh cod. The increased food value of salted fish will be seen from the following analyses—

THE EFFECT OF CURING AND DRYING UPON THE
FOOD VALUE OF DIFFERENT FISHES

Food.	Protein.	Fat.	Carbohydrate.	Ash.	Water.	Food Value. Cal. per lb.
Haddock (fresh)	12·0	0·2	—	0·9	51·6	232
„ (smoked)	14·9	0·2	—	3·4	57·4	286
Herring (fresh)	14·0	10·4	—	1·5	45·7	699
„ (salted)	21·2	15·4	—	7·7	30·9	944
„ (bloater)	15·7	9·6	—	1·5	52·0	697
„ (kipper)	14·1	11·1	—	3·4	46·9	730
Sprats (fresh)	12·6	10·7	—	1·3	49·4	686
„ (smoked)	21·2	14·9	—	3·2	39·4	1,023

Thus, the food value of salted sprats or herrings per pound is 50 per cent more than that of the same fish when fresh.

The original food value of a fish is generally diminished by the cooking process. The fish may be boiled or broiled for direct consumption, or it may be steam cooked in cans and sealed up for future consumption, as in the canning industry. When oily fishes, such as herrings, are cooked, the oil globules burst and some of the oil is lost, and the food value of the fish becomes correspondingly less. When salted fish is soaked in fresh water before being cooked, some of the gelatin and other coagulable proteins are extracted from the flesh. This loss of protein can be checked either by broiling the fish, when the protein near the surface becomes coagulated and so prevents the loss of protein from the interior of the fish, or by placing the fish that is to be boiled direct into boiling water, and not into the cold water before the heating has begun.

In addition to this diminution of the food content of the fish, the process of cooking, contrary to general expectation, also diminishes slightly its digestibility.

In the canning process the fish to be canned are cleaned (gutted) and boned, and packed into tins, together with the necessary sauce or seasoning. The tins are then closed, a small hole being left temporarily in the lid. The tins are placed on steam-heated racks, and the contents thoroughly cooked. In this way the contents are sterilized as well as cooked, and the air originally present in the tin is all driven out by the steam through the small hole in the lid. This hole is sealed with a spot of solder while the contents of the tin are still at boiling point. The tin and its contents are allowed to cool down, and are dispatched to the store-room. During storage the contents of the sealed tin gradually “mature.” This maturing process may last from six months to ten years. During this period the bones soften, the flesh becomes soft and pasty, and the taste becomes richer. The precise nature of the changes that take place during this maturing process is not fully understood; probably maturing is partly due to the action of certain enzymes in the flesh of the fish, and partly to the slow but continuous chemical action of the various juices present in the tin. Attempts to pickle herrings from the Zuyder Zee have been unsuccessful owing to a lack of the enzyme action that makes other herrings tender when pickled. The enzyme, although present, is apparently rendered inactive by the presence of an anti-enzyme.

The last, but by no means the least, important factor to be considered in estimating the food value of any particular fish is its retail price. The price of the different kinds of fishes is by no means proportional to their individual food values. It is determined primarily by the abundance or otherwise of the available supply of each individual species. Thus, the various pelagic fish—mackerel, herring, sprat—that are easily caught in enormous quantities at certain seasons of the year are by far the most valuable. Of trawl-caught fish, cod and whiting are more plentiful and are, therefore, cheaper than hake, although, again, the cheaper fish has the greater food value.

In some cases certain fish, although fairly abundant, are in poor demand owing to some prejudice on the part of the public, and are generally sold in poorer districts, or to the fried fish trade, at a disproportionately low price, for example skate, dog-fish, angler fish, john dory.

Taste and appearance also contribute to the popularity and, therefore, indirectly to the retail price of fish, such as the sole and the salmon.

In Table IV the present retail prices (Sept., 1921) and the food values of a number of different fishes are compared. From these figures, the actual food value per shillingworth of each fish has been calculated.

The cheapest fish, therefore, are also those possessing the greatest food value, e.g. the herring in all its forms, dried cod and ling, and mackerel. These compare favourably both in cost and food value with meat, such as beef and mutton.

TABLE IV
FOOD VALUE PER SHILLINGSWORTH OF DIFFERENT FISHES

Fish.	Food Value.	Retail Price Sept. 1921	Food Value per Shilling.
	Cals. per lb.	per lb. s. d.	
Halibut (cuts)	258	2 3	115
Sole	346	2 6	138
Turbot	270	1 6	180
Brill	327	1 8	196
Haddock	232	1 2	198
Hake	256	1 3	204
Smoked haddock	286	1 3	228
Plaice	367	1 6	244
Cod (section)	296	1 1	252
Whiting	215	– 10	258
Salmon (section)	847	3 –	282
Eels	799	1 10	436
Dried ling	560	1 –	560
Mackerel	515	– 10	618
Dried cod	750	1 –	750
Kipperd herring	730	– 9	972
Herring	709	– 8	1062
Bloaters	715	– 8	1072
Red herrings	1220	– 8	1830
Salt herrings	1129	– 5	2712

Finally, the popularity or otherwise of any foodstuff necessarily depends upon its flavour. Fishes differ greatly in this respect. In many cases the flavour of a fish can be seriously impaired by an unsuitable method of

cooking. A full-flavoured fish like the mackerel lends itself to a variety of methods of cooking, equally good results being obtained by baking, grilling, frying in fillets or boiling. The plaice, sole, ling, hake, mullet, and turbot are essentially fish for frying, while cod, haddock and whiting are best boiled. To prepare a fish for the table requires considerable skill, but it is an art that, once acquired, can be used to render even what are regarded as inferior varieties both wholesome and palatable. In this country, fishes have long been a neglected form of food. They have a high food value, they are easily digestible, and are cheap and plentiful.

It has been shown recently that edible fish contain vitamins. Vitamins are complex chemical compounds of hitherto unknown composition, and of little understood properties, that occur in minute quantities in a great variety of natural food stuffs. These vitamins appear to be essential to healthy animal existence. Without them, the body rapidly becomes attacked by certain diseases, e.g. rickets, beri-beri, scurvy, and unless this deficiency of the diet is corrected, death soon follows. Three different vitamins have been discovered, known as vitamins A, B, and C. Vitamin A is contained in the oily part of most fish, while Vitamin B is present in certain fish roes.

CHAPTER XI

FISH PRODUCTS

THE industrial value and importance of fishes is by no means limited to their use as food. They yield large quantities of valuable oil. The fish waste, or offal, chiefly heads, skins, bones and viscera—that is discarded by the fish curer, is worked up to yield fish glue, fertilizers and cattle food. The skins of certain large fishes, for example the shark, are tanned and manufactured into a valuable leather.

The story of the fishing industry would not be complete without a brief description of the methods by which these products are manufactured.

Fish Oils. The various kinds of oil that are obtained from different species of fish and other marine animals, such as whales and seals, may be divided into three classes, according to the part of the fish from which they are extracted.

(1) Fish oils proper are disseminated throughout the flesh of the fish in the form of fine globules. They are extracted from the entire fish, e.g. herring, sardine, sprat, menhaden.

(2) Liver oils are located in the fish liver, e.g. cod, shark.

(3) Blubber oils constitute a thick layer of adipose tissue just under the skin of the marine mammalia, e.g. whale, seal, dolphin, porpoise.

In oily fish, such as herrings and sprats, each minute globule of oil is enclosed within a thin skin. It is practically impossible to rupture this skin and liberate the oil simply by the application of pressure. When, however, these globules are heated the skin shrivels, the oil globules expand and burst the skin, and the liquid oil is liberated and can then be extracted from the flesh by pressure. To obtain the oil, therefore, the fish are boiled or steam heated in large vats until the oil is set free. The hot mass is then placed in a press and the oil squeezed out. The residue is made into cattle food and fertilizer.

In obtaining the best sorts of liver oils, e.g. codliver oil, the livers are taken from the fish as soon as they are caught, and are heated in steam-

jacketed vessels until the cell membranes burst and the oil exudes. The oil is then separated by pressure.

Inferior qualities of oil are obtained by treating putrid livers in the same way at the end of the voyage. These tainted liver oils are unfit for medicinal purposes, but are used in large quantities in the leather industry.

Blubber (which is from 8 to 20 ins. thick) is stripped from the whale as soon after capture as possible. Generally the dead whale is made fast alongside the whaler, a deep, spiral cut is made round its body, and the blubber is stripped off and hauled aboard. This is then cut into pieces, chopped up in mincing machines and fed into melting pans and heated with steam, often under pressure. The oil gradually exudes and collects upon the water, the cell membranes, etc.—the greaves—settling to the bottom. At the conclusion of the boil, the oil is drawn off from above the aqueous (gluey) layer, and is clarified by straining through sieves or filters. The “greaves” is placed in hair or woollen bags and submitted to hydraulic pressure, by which means a further quantity of oil is obtained.

Fish oils, unless specially purified for medicinal purposes, are dark-coloured liquids, with a characteristic, unpleasant, fishy smell, due to the presence of small quantities of fishy decomposition products, for example trimethylamine.

When cooled, many samples of fish oil deposit solid masses of fish tallow (fish stearine).

Fish oils, and, to a less extent, the marine animal oils, e.g. whale, seal, porpoise, are drying oils like linseed oil, that is they possess to a very marked degree a capacity for absorbing oxygen from the air, and so become thickened and viscous. This thickening is generally induced by blowing air through the warm oil. Oils that have been thickened in this way are known as “blown” oils.

Blown fish oils are mixed with mineral oils for use as lubricants for heavy machinery. They have been used as vehicles for paints in place of linseed oil, but with somewhat disappointing results. They are used successfully in place of linseed oil in the manufacture of printers’ ink, and in making paints for painting smoke stacks. Such paints resist successfully the action of heat and light.

More particularly, they are used in the leather industry. Fish oils are used chiefly in the manufacture of chamois leather. Ordinary chamois or wash-leather is made from the flesh-splits of sheep skins. The skin is well washed and softened, and freed from hair by treatment with lime. It is then split, and the loose and fatty middle layer removed by a sharp knife. The lime is removed by a short bran-drench and the superfluous moisture is pressed out. The skin is thus rendered porous and easily able to absorb the oil. It is stretched on a table and oiled with fish or whale oil. The oiled skin is folded up and worked for two or three hours in the faller stocks and then shaken out and hung up for a short time to cool and partially dry. The process is repeated a number of times, until all the water originally present in the skin has been replaced by oil. The oiled skins are then piled in a warm place. The oil gradually oxidizes—probably owing to some fermentation process—and the skins become yellow and very hot. From time to time the skins are strewn on the floor to cool and then re-piled, the process being repeated until the oxidation of the oil is complete. In France the freshly-oiled skins are hung in hot stoves, and the oxidation of the oil is completed in one operation.

The skins are then dipped in water and passed through hydraulic presses, by which the surplus oil is removed. This surplus thick, oxidized oil is known as “degras” or “moellon,” and is used for stuffing leathers that have already been tanned. Stuffed leathers are supple and impervious to water, and are used for harness, belting, etc. A further quantity of oil may be removed from the “chamoised” leather by treating it with potash or carbonate of soda, “sod” oil being recovered from the extract by neutralization with sulphuric acid. The value of sod oil for oiling dressed leather is due to a resinous acid of unknown composition, that is soluble in alkali but insoluble in petroleum ether.

Enamel or patent leather is generally coated, after tanning, with a linseed oil varnish, boiled with prussian blue, and dried in a steam heated chest at 70° to 80°C., the process being repeated until a sufficiently thick coat is produced. Fish oils are now used successfully in place of linseed oil. The enamel leather produced, although not quite so glossy as that made with linseed oil, is said to be more pliable.

Fish oils are also employed in the manufacture of such closely-related, although happily diverse, substances as soap and margarine. All animal and

vegetable fats and oils are essentially compounds of glycerine, with one or other of three acids: palmitic, stearic and oleic. Palmitic and stearic acids and their compounds are solids at the ordinary temperature, whereas oleic acid and its compounds are liquid. This difference appears to be connected in some way with the molecular structure of these substances. When oleic acid is heated with hydrogen gas under pressure, in the presence of finely-divided nickel, it absorbs hydrogen and is transformed into stearic acid. Oleic acid, therefore, is said to be unsaturated with respect to hydrogen, whereas stearic acid is called a saturated acid. This process, whereby a liquid oil is transformed into a solid fat, is called hydrogenation, or hardening.

Both the margarine industry and the soap industry require large quantities of hard fats. Originally the soap industry absorbed the available supplies of hard animal fats such as beef suet, hog's lard, and mutton suet. The margarine industry depended upon these same supplies of animal fats, and the rapid growth in the production of margarine during recent years has seriously diminished the supply of hard fats necessary for the manufacture of soap.

The hydrogenation of whale oil and various fish oils has now made it possible to supply this demand, and has also made possible the industrial utilization of substances, such as fish oils, for which formerly comparatively little use could be found.

Hardened whale oil melts at 40° to 50°C., and is a white solid entirely devoid of taste or smell. It is used for making soap, and as a lard substitute for cooking purposes.

Fish Glue. Fish glue is the most important liquid glue on the market. The bulk of the fish glue manufactured to-day is made from the waste and offal that are discarded by the curers. This waste consists of heads, bones, viscera and skins. The best glue is obtained from the skins of non-oily, demersal fish, for example cod, haddock, soles, plaice and hake.

The waste is washed in running water to free it from salt. Sometimes the waste—particularly the heads—is decomposed with hydrochloric acid and afterwards neutralized with lime. It is then charged into a cooker provided with a perforated, false bottom. The stock is covered with water and heated with steam. The glue is extracted and gradually concentrates in the water.

When this glue liquor is sufficiently concentrated (from 5 to 6 per cent), it is run off (the first run) and more water is added to the waste and the cooking continued. After about 10 hours cooking, nearly all the glue has been extracted and the liquor is again run off (the second run). The cooked waste is then withdrawn, and any remaining glue liquor is pressed out of it and added to the second run. From 2 to 4 per cent of phenol or boric acid are added to prevent decomposition by bacteria.

The glue liquor is evaporated down to a concentration of 32 per cent in open vats or closed evaporators, and is bleached with sulphurous acid. A small amount of some essential oil, e.g. cassia, clove, wintergreen, is added to check mould growth and mask the fishy odour. Glue is also made in a similar way from the “greaves” obtained from whale blubber.

Fish glue is manufactured in three grades.

GRADE I is made from skins, only the first run being used. It is used for photo-engraving work, for the production of half-tone plates.

GRADE II is made from second run skin liquors and fish waste. It is sold in small cans and bottles for general repair work.

GRADE III is prepared from fish heads, and is sold in large cans and barrels for sizing, box making, cabinet making, and general joiner work.

The glue is sometimes made more flexible by the addition of glycerine and glucose. The flexibility of fish glue makes it useful for the manufacture of court plaster, labels, stamps, and in book-binding.

The residue from the press is dried and sold as chicken feed or fertilizer. For the latter purpose it is frequently mixed with Carnallite.

Fish Gelatine. Fish gelatine or isinglass is obtained from the swimming bladder of the sturgeon and also of the cod. The bladders are exported, either opened (pipe isinglass) or washed, split open and dried (purse, lump or leaf isinglass).

Isinglass is the purified and dried inner skin of the bladder. It has but feeble adhesive power. It is used for clarifying wines, ciders and beers, and for making jellies and plasters.

Fertilizers. In many places near the sea, fish are employed whole as manure. Sprats particularly are caught in large numbers and distributed over

the fields, and left to decompose. Fresh sprats contain 63·7 per cent of water, 1·94 per cent nitrogen, 2·1 per cent ash (0·43 potash and 0·90 phosphoric acid).

Fish guano or fish manure is generally prepared from the fish waste discarded by the curer. An average sample of this manufactured fish manure will contain 12 per cent water, 60 per cent organic matter, yielding 10 per cent ammonia, 16 per cent of calcium phosphate, and a residue of salt, sand, magnesia and potash, the amount of potash being inconsiderable. Fish guano is mainly valuable as a source of ammonia, the ammonia content ranging from 6 to 11 per cent, according to the kind of fish used and its previous history, e.g. whether fresh or salted.

In many places, such as London, the fish offal from the shops and restaurants is collected, dried and ground up for use as manure. In Germany in 1918 herrings' heads were removed by the curers to be utilized for the production of oil, albumen, and phosphate of lime. The herring meal contained up to 50 per cent of albumen and calcium phosphate, the latter being obtained from the bones and heads. The albumen was extracted chemically and prepared for human consumption. The oil was extracted with benzol or other solvents, and, after hardening, was used in the manufacture of butter substitutes. Fish waste or offal is fed into a continuous cooker. This cooker consists essentially of a long, cylindrical vessel, through which runs a hollow steel shaft on which are mounted perforated radial vanes in such a way that the whole arrangement forms a spiral conveyor. By means of the hollow shaft and vanes, steam is blown into the mass of fish waste as it travels slowly through the vessel, so that it is completely cooked and disintegrated by the time that it emerges at the other end.

The cooked mass is then fed into a press in which a screw conveyor urges it through a gradually tapering cylinder with perforated sides. In this way the oil is extracted from it, and it is then dried and disintegrated by a rotary drier.

There is always a little residual oil in fish manure that tends to delay its decomposition in the soil. It is important, therefore, that the oil be removed as completely as possible.

Dry fish manure requires careful storing, as the presence of this small amount of oxidizable oil tends to promote spontaneous combustion.

In addition to its value as a fertilizer, the high content of protein (albumen)—namely, 50 per cent—makes fish meal a suitable food for livestock and poultry.

The commercial importance of this industry will be realized when we remember that practically half of the total catch of fish in the world is discarded by the curers as waste.

Fish Leather. The hides of such marine mammals as the walrus and the seal have long formed the basis of a regular tanning industry.

Of recent years, however, particularly in America, successful attempts have been made to tan the skins of certain fish, notably the shark. The skins are treated with alkali to remove fat and oil, the alkali is then neutralized with acid, after which the skins are washed and tanned. The leather is said to be soft and pliable, and well adapted for many uses.

Shark skins are also tanned hard, and used to print a grain on imitation pigskin.

Shark fishing was commenced off the American coast in October, 1918. The fish are hunted from fast, powerful motor boats, with specially constructed nets. A small shark 5 ft. long will yield a hide 10 sq. ft. in area.

Shark skin is naturally very tough and durable, and in its untanned condition is used by jewellers as a natural emery paper for grinding and polishing metal surfaces. It is also used as an abrasive in working hard woods and ivory.

A method has been devised by which a shark skin can be split into three. The first split, after tanning, is strong and thick, and suitable for high grade, heavy shoes. The second furnishes leather suitable for second grade footwear, and the third resembles suede and is used in making fancy articles. In addition to the shark's skin, the fins, blood, teeth, flesh, and oil of the fish are also utilized commercially and yield a satisfactory profit.

FOOTNOTES:

[1](#) U.S. Bureau of Fisheries, Document No. 884 (1920).

[2](#) Marshall, *Microbiology*.

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Printed in Bath, England, by Sir Isaac Pitman & Sons, Ltd.

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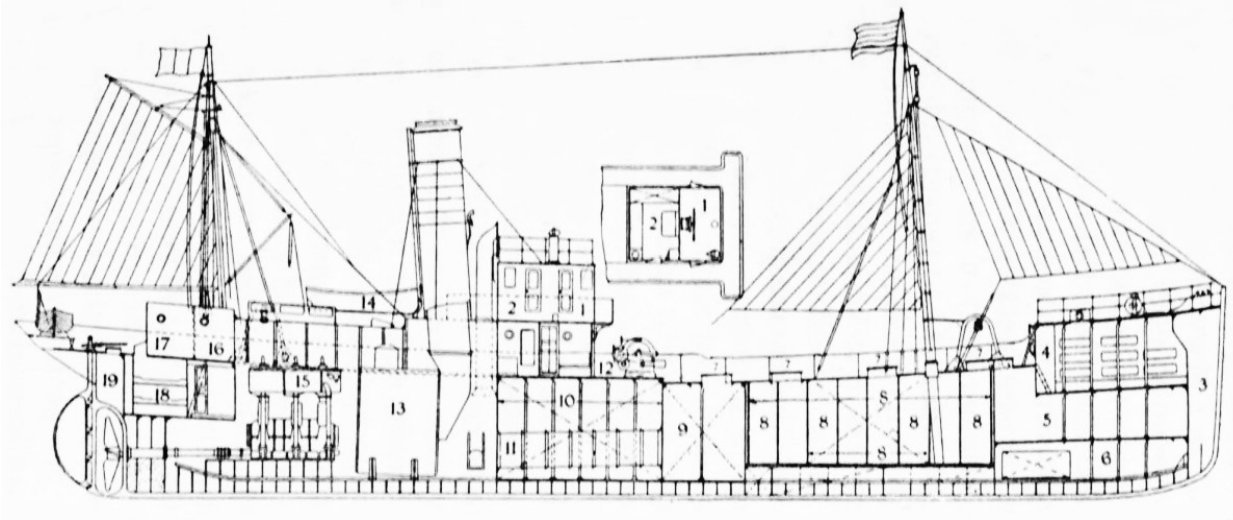
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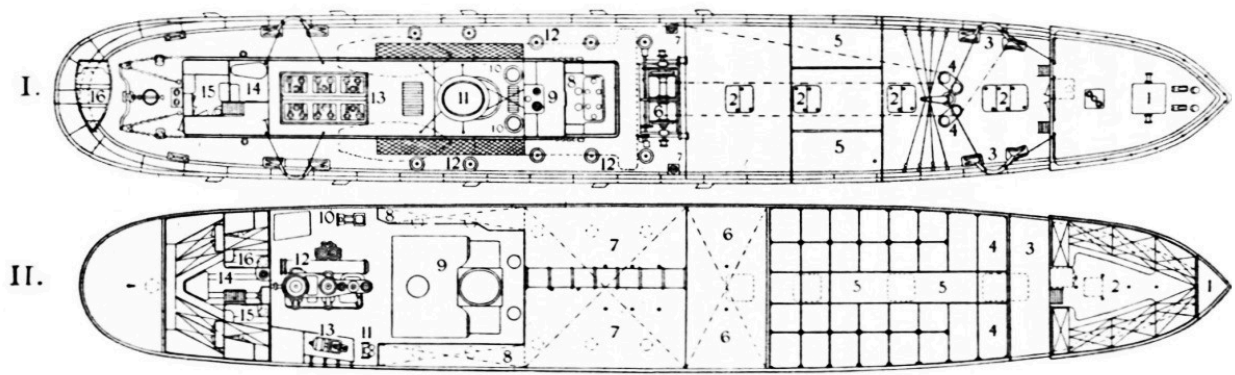
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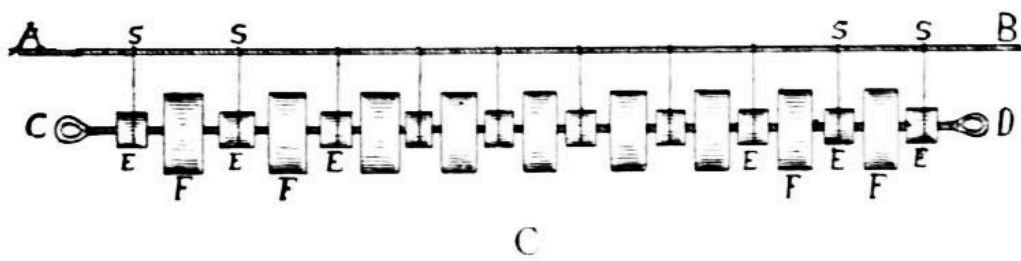
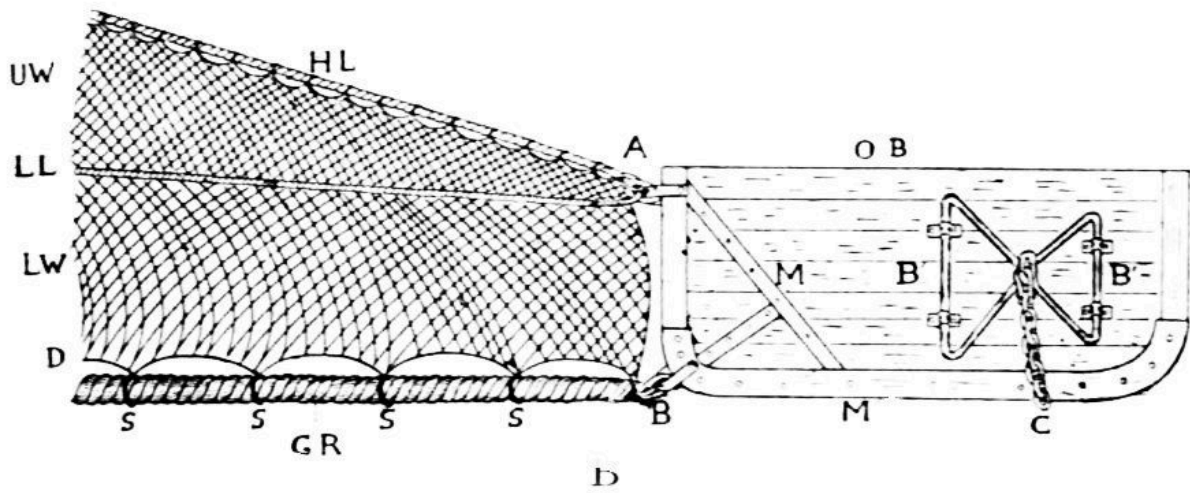
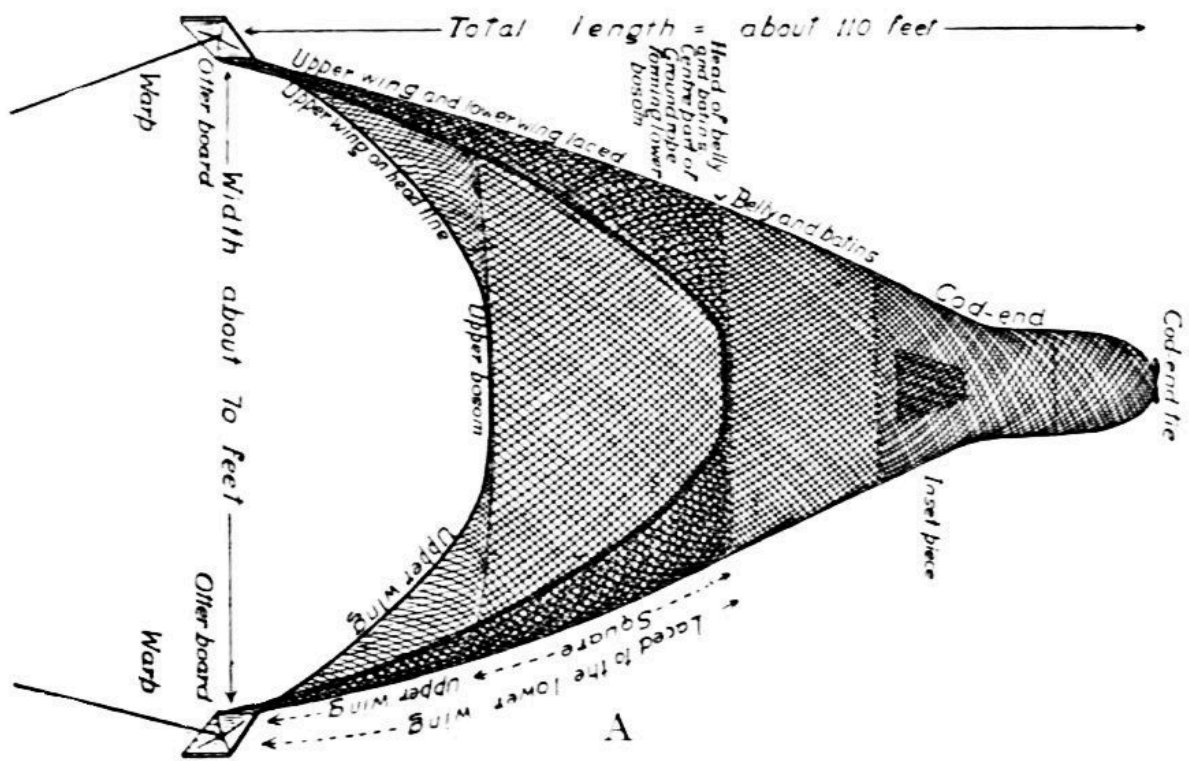
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